Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

SEMI E30-0125
SPECIFICATION FOR THE GENERIC MODEL FOR COMMUNICATIONS AND CONTROL OF MANUFACTURING EQUIPMENT (GEM)

This Standard was technically approved by the Information & Control Global Technical Committee. This edition was approved for publication by the global Audits and Reviews Subcommittee on October 31, 2024 and December 5, 2024. Available at www.semiviews.org and www.semi.org in January 2025; originally published in 1992; previously published October 2024.

NOTICE: The section symbol (§) is used to reference a numbered section, paragraph, or header and all subordinate paragraphs, headers, and embedded material (i.e., EXCEPTIONs, lists) therein. The paragraph symbol (¶) is used to reference a particular numbered paragraph and embedded material therein. When duplicated (i.e., §§ and ¶¶) the symbols are used to reference more than one section or paragraph, respectively.

NOTICE: Paragraphs entitled 'NOTE:' are not an official part of this Standard or Safety Guideline and are not intended to modify or supersede the official Standard or Safety Guideline. These have been supplied by the global technical committee to enhance the usage of the Standard or Safety Guideline.

Table of Contents

1 Purpose 3
2 Scope 4
3 Limitations 4
4 Referenced Standards and Documents 4
5 Terminology 5
6 State Models 6
6.3 *State Model Methodology* 7
6.4 *Communications State Model* 8
6.5 *Control State Model* 12
6.6 *Equipment Processing States* 16
7 Equipment Capabilities and Scenarios 18
7.2 *Establish Communications* 19
7.3 *Data Collection* 21
7.4 *Alarm Management* 37
7.5 *Remote Control* 41
7.6 *Equipment Constants* 43
7.7 *Process Program Management* 44
7.8 *Material Movement* 67
7.9 *Equipment Terminal Services* 67
7.10 *Error Messages* 70
7.11 *Clock* 73
7.12 *Spooling* 75
7.13 *Control* 82
7.14 *Documentation File Transfer* 85
7.15 *SEDD File Transfer* 87
8 Data Items 89
8.3 *Data Item Restrictions* 89

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

8.4 *Variable Item List* 89
8.5 *Data Collection Item Properties* 91
9 Collection Events....91
10 SECS-II Message Subset 93
11 GEM Compliance 96
11.2 *Fundamental GEM Requirements*....96
11.3 *GEM Compliance* 97
11.4 *Documentation*....97
Appendix 1 Alternative Scenarios for Large Formatted Process Programs....100
Appendix 2 Alternative Scenarios for Large Unformatted Process Programs....104
Appendix 3 Alternative Scenarios for SEMI E42 Recipes 108
Appendix 4 Additional Items....117
Related Information 1 119

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

1 Purpose

1.1 GEM defines a standard implementation of SECS-II for all semiconductor manufacturing equipment. The GEM Standard defines a common set of equipment behavior and communications capabilities that provide the functionality and flexibility to support the manufacturing automation programs of semiconductor device manufacturers. Equipment suppliers may provide additional SECS-II functionality not included in GEM as long as the additional functionality does not conflict with any of the behavior or capabilities defined in GEM. Such additions may include SECS-II messages, collection events, alarms, remote commands, processing states, variable data items (data variables, status variables or equipment constants), or other functionality that is unique to a class (etchers, steppers, etc.) or specific instance of equipment.

1.1.1 GEM is intended to produce economic benefits for both device manufacturers and equipment suppliers. Equipment suppliers benefit from the ability to develop and market a single SECS-II interface that satisfies most customers. Device manufacturers benefit from the increased functionality and standardization of the SECS-II interface across all manufacturing equipment. This standardization reduces the cost of software development for both equipment suppliers and device manufacturers. By reducing costs and increasing functionality, device manufacturers can automate semiconductor factories more quickly and effectively. The flexibility provided by the GEM Standard also enables device manufacturers to implement unique automation solutions within a common industry framework.

1.1.2 The GEM Standard is intended to specify the following:

- A model of the behavior to be exhibited by semiconductor manufacturing equipment in a SECS-II communication environment,
- A description of information and control functions needed in a semiconductor manufacturing environment,
- A definition of the basic SECS-II communications capabilities of semiconductor manufacturing equipment,
- A single consistent means of accomplishing an action when SECS-II provides multiple possible methods, and
- Standard message dialogues necessary to achieve useful communications capabilities.

1.1.3 The GEM Standard contains two types of requirements:

- fundamental GEM requirements and
- requirements of additional GEM capabilities.

1.1.4 The fundamental GEM requirements form the foundation of the GEM Standard. The additional GEM capabilities provide functionality required for some types of factory automation or functionality applicable to specific types of equipment. A detailed list of the fundamental GEM requirements and additional GEM capabilities can be found in § 11, GEM Compliance. Figure 1 illustrates the components of the GEM Standard.

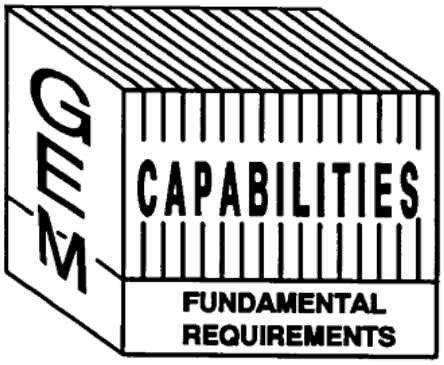

Figure 1
GEM Components

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

1.1.5 Equipment suppliers should work with their customers to determine which additional GEM capabilities should be implemented for a specific type of equipment. Because the capabilities defined in the GEM Standard were specifically developed to meet the factory automation requirements of semiconductor manufacturers, it is anticipated that most device manufacturers will require most of the GEM capabilities that apply to a particular type of equipment. Some device manufacturers may not require all the GEM capabilities due to differences in their factory automation strategies.

2 Scope

2.1 The scope of the GEM Standard is limited to defining the behavior of semiconductor equipment as viewed through a communications link. The SEMI E5 (SECS-II) Standard provides the definition of messages and related data items exchanged between host and equipment. The GEM Standard defines which SECS-II messages should be used, in what situations, and what the resulting activity should be. Figure 2 illustrates the relationship of GEM, SECS-II, and other communications alternatives.

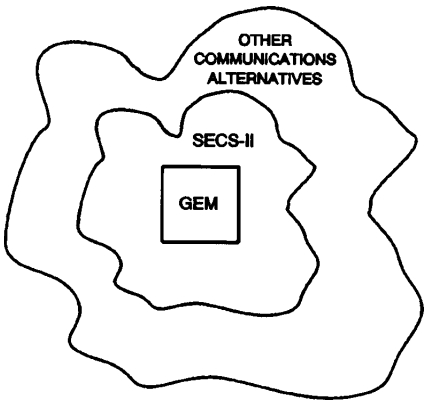

Figure 2
GEM Scope

NOTICE: SEMI Standards and Safety Guidelines do not purport to address all safety issues associated with their use. It is the responsibility of the users of the Documents to establish appropriate safety and health practices, and determine the applicability of regulatory or other limitations prior to use.

3 Limitations

3.1 The GEM Standard does not attempt to define the behavior of the host computer in the communications link.

3.2 The capabilities described in this Standard are specifically designed to be independent of lower-level communications protocols and connection schemes (e.g., SECS-I, SMS, point-to-point, connection-oriented or connectionless). Use of those types of standards is not required or precluded by this Standard.

4 Referenced Standards and Documents

4.1 *SEMI Standards and Safety Guidelines*

4.1.1 The following SEMI Standards are related to the GEM Standard. The specific portions of these Standards referenced by GEM constitute provisions of the GEM Standard.

SEMI E4 — Specification for SEMI Equipment Communications Standard 1 Message Transfer (SECS-I)

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E37 — Specification for High-Speed SECS Message Services (HSMS) Generic Services

SEMI E42 — Recipe Management Standard: Concepts, Behavior, and Message Services

SEMI E139 — Specification for Recipe and Parameter Management (RaP)

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

SEMI E172 — Specification for SECS Equipment Data Dictionary (SEDD)

SEMI E173 — Specification for XML SECS-II Message Notation (SMN)

4.2 *IETF® Documents*[1]

RFC 4180[2] — Common Format and MIME Type for Comma-Separated Values (CSV) Files

4.3 *ISO® Standards*[3]

ISO 32000 — Document Management — Portable Document Format (PDF)

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

5 Terminology

5.1 *Abbreviations and Acronyms*

5.1.1 *GEM* — generic equipment model

5.1.2 *PDF* — portable document format

5.1.3 *SMN* — SECS-II message notation

5.2 *Definitions*

5.2.1 *alarm* — related to any abnormal situation on the equipment that may endanger people, equipment, or material being processed. Such abnormal situations are defined by the equipment manufacturer based on physical safety limitations.

5.2.1.1 *Discussion* — Note that exceeding control limits associated with process tolerance does not constitute an alarm nor do normal equipment events such as the start or completion of processing.

5.2.2 *capabilities* — operations performed by semiconductor manufacturing equipment. These operations are initiated through the communications interface using sequences of SECS-II messages (or scenarios). An example of a capability is the setting and clearing of alarms.

5.2.3 *collection event* — an occurrence (or grouping of related occurrences) on the equipment that might be significant to the host.

5.2.4 *communication failure* — occurs when an established communications link is broken. Such failures are protocol specific. Refer to the appropriate protocol Standard (e.g., SEMI E4 or SEMI E37) for a protocol-specific definition of communication failure.

5.2.5 *communication fault* — occurs when the equipment does not receive an expected message, or when either a transaction timer or a conversation timer expires.

5.2.6 *control* — to exercise directing influence.

5.2.7 *equipment model* — a definition based on capabilities, scenarios, and SECS-II messages that manufacturing equipment should perform to support an automated manufacturing environment (see also Generic Equipment Model.).

5.2.8 *event* — a detectable occurrence significant to the equipment.

5.2.9 *GEM Compliance* — defined with respect to individual GEM capabilities to indicate adherence to the GEM Standard for a specific capability. § 9 includes more detail on GEM Compliance.

5.2.10 *Generic Equipment Model* — a reference model for any type of equipment. It contains functionality that can apply to most equipment, but does not address unique requirements of specific equipment.

5.2.11 *host* — the intelligent system that communicates with the equipment. [SEMI E4, SEMI E5]

---

[1] Internet Engineering Task Force, IETF Administration LLC, 1000 N West Street, Suite 1200, Wilmington, DE 19801, USA; http://www.ietf.org. IETF trademark is owned by the IETF Trust.

[2] https://tools.ietf.org/html/rfc4180

[3] International Organization for Standardization, ISO Central Secretariat, BIBC II, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland; Telephone: +41.22.749.01.11, http://www.iso.org. ISO trademark is owned by International Organization for Standardization.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

5.2.12 *message fault* — occurs when the equipment receives a message that it cannot process because of a defect in the message.

5.2.13 *operational script* — a collection of scenarios arranged in a sequence typical of actual factory operations. Example sequences are system initialization power up, machine setup, and processing.

5.2.14 *operator* — a human who operates the equipment to perform its intended function (e.g., processing) and typically interacts with the equipment via the equipment supplied console.

5.2.15 *process unit* — refers to the material that is typically processed as a unit via single run command, process program, etc. Examples include wafers, cassettes, magazines, and boats.

5.2.16 *processing cycle* — a sequence wherein all of the material contained in a typical process unit is processed. This is often used as a measure of action or time.

5.2.17 *scenario* — a group of SECS-II messages arranged in a sequence to perform a capability. Other information may also be included for clarity. Note that in this Specification, SECS-II messages are identified using an abbreviation 'S<n>F<m>' where '<n>' is the stream number, '<m>' is the function number, and neither a comma nor space is used between the stream number and the 'F'. For example, S1F1 is message Stream 1, Function 1.

5.2.18 *state model* — a collection of states and state transitions that combine to describe the behavior of a system. It includes a definition of the conditions that delineate a state, the actions/reactions possible within a state, the events that trigger transitions to other states, and the process of transitioning between states.

5.2.19 *system default* — refers to state(s) in the equipment behavioral model that are expected to be active at the end of system initialization. It also refers to the value(s) that specified equipment variables are expected to contain at the end of system initialization.

5.2.20 *system initialization* — the process that an equipment performs at power-up, system activation, and/or system reset. This process is expected to prepare the equipment to operate properly and according to the equipment behavioral models.

5.2.21 *user* — a human or humans who represent the factory and enforce the factory operation model and is considered to be responsible for many setup and configuration activities that cause the equipment to best conform to factory operations practices.

5.2.22 *variable data (V)* — a data item containing a status variable (SV), data variable (DV), or equipment constant (EC) value.

5.2.23 *variable data ID (VID)* — a unique identifier of a variable data item. The set of VIDs is the union of all SVIDs, ECIDs, and DVIDs.

6 State Models

6.1 The following sections contain state models for semiconductor manufacturing equipment. These state models describe the behavior of the equipment from a host perspective in a compact and easy to understand format. State models for different equipment will be identical in some areas (e.g., communications), but may vary in other areas (e.g., processing). It is desirable to divide the equipment into parallel components that can be modeled separately and then combined. An example of a component overview of an equipment is provided as Figure 3.

6.2 Equipment manufacturers shall document the operational behavior of their equipment using state model methodology. State models are discussed in § 6.3 and Related Information 1, § R1-5, and in a referenced article. Documentation of a state model shall include the following three elements:

- A *state diagram* showing the possible states of the system or components of a system and all of the possible transitions from one state to another. The states and transitions shall each be labeled. Use of the Harel notation (refer to Related Information 1, § R1-5) is recommended.
- A *transition table* listing each transition, the beginning and end states, what stimulus triggers the transition, and any actions taken as a result of the transition.
- A *definition of each state* specifying system behavior when that state is active.

6.2.1 Examples of the above elements are provided in Related Information 1, § R1-5.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

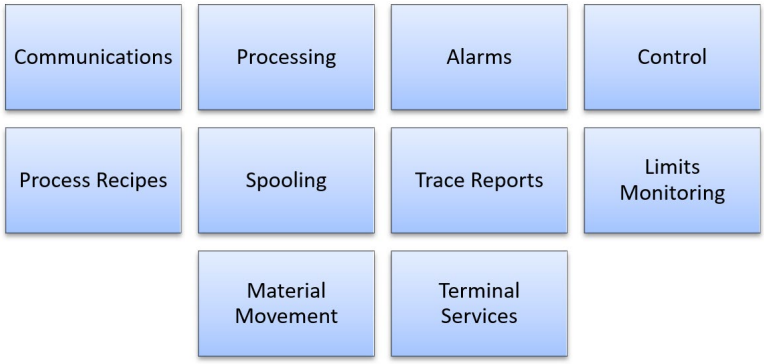

Figure 3
Example Equipment Component Overview

6.2.2 The benefits of providing state models are:

1. State machine models are a useful specification tool,
2. A host system can anticipate machine behavior based upon the state model,
3. End-users and equipment programmers have a common description of machine behavior from which to work,
4. 'Legal' operations can be defined pertaining to any machine state,
5. External event notifications can be related to internal state transitions,
6. External commands can be related to state transitions,
7. State model components describing different aspects of machine control can be related to one another (example: processing state model with material transport state model; processing state model with internal machine safety systems).

6.3 *State Model Methodology* — To document the expected functionality of the various capabilities described in this Standard, the 'Statechart'[4] notation developed by David Harel has been adopted. An article by Harel listed in § 4 is recommended reading for a full understanding of the notation. The convention used in this and following sections is to describe the dynamic functionality of a capability with three items: a textual description of each state or substate defined, a table that describes the possible transitions from one state to another, and a graphical figure that uses the symbols defined by Harel to illustrate the relationships of the states and transitions. The combination of these items define the state model for a system or component. A summary of the Harel notation and a more detailed description of the text, table, and figure used to define behavior with this methodology is contained in Related Information 1, § R1-5.

6.3.1 The basic unit of a state model is the state. A state is a static set of conditions. If the conditions are met, the state is current. These conditions might involve sensor readings, switch positions, time of day, etc. Also, part of a state definition is a description of reactions to specific stimuli (e.g., if message SxFy is received, generate reply message SxFy + 1). Stimuli may be quite varied but for semiconductor equipment would include received SECS messages, expired timers, operator input at an equipment terminal, and changes in sensor readings.

6.3.2 To help clarify the interpretation of this Standard and the state models described herein, it is useful to distinguish between a state and an event and the relationship of one to the other. An event is dynamic rather than static. It represents a change in conditions, or more specifically, the awareness of such a change. An event might involve a sensor reading exceeding a limit, a switch changing position, or a time limit exceeded.

6.3.3 A change to a new active state (state transition) shall always be prompted by a change in conditions, and thus an event. In addition, a state transition may itself be termed an event. In fact, there are many events that may occur on an equipment, so it is important to classify events based on whether they can be detected and whether they are of

[4] Harel, D. "Statecharts: A Visual Formalism for Complex Systems." Science of Computer Programming 8 (1987): pp. 231–274.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

interest. In this Standard, the term event has been more narrowly defined as a detectable occurrence that is significant to the equipment.

6.3.4 A further narrowing of the definition of event is represented by the term 'collection event', which is an event (or group of related events) on the equipment that is considered significant to the host. It is these events that (if enabled) are reported to the host. By this definition, the list of collection events for an equipment would typically be only a subset of total events. The state models in this Standard are intended to be limited to the level of detail in which the host is interested. Thus, all state transitions defined in this Standard, unless otherwise specified, shall correspond to collection events.

6.4 *Communications State Model* — The Communications State Model defines the behavior of the equipment in relation to the existence or absence of a communications link with the host. § 7.2 expands on this section by defining the Establish Communications capability. This model pertains to a logical connection between equipment and host rather than a physical connection.

6.4.1 *Terminology* — The terms communication failure, connection transaction failure, and communication link are defined for use within this Standard only and should not be confused with the same or similar terms used elsewhere.

- Refer to SEMI E4 (SECS-I) or SEMI E37 (HSMS) for a protocol specific definitions of communications failure.
- A connection transaction failure occurs when attempting to establish communications and is caused by:
  - a communication failure,
  - the failure to receive an S1F14 reply within a reply timeout limit, or
  - receipt of S1F14 that has been improperly formatted or with COMMACK not set to 0. Refer to SEMI E5 for further definition of Data Item COMMACK, Establish Communications Acknowledge Code.
- A reply timeout period begins after the successful transmission of a complete primary message for which a reply is expected (refer to SEMI E4 (SECS-I) or SEMI E37 (HSMS) for a protocol-specific definition of reply timeout).
- A communication link is established following the first successful completion of any one S1F13/F14 transaction with an acknowledgement of 'accept'. The establishment of this link is logical rather than physical.
- Implementations may have mechanisms which allow outgoing messages to be stored temporarily prior to being sent. The noun queue is used to cover such stored messages. They are queued when placed within the queue and are dequeued by removing them from this storage.
- Send includes 'queue to send' or 'begin the process of attempting to send' a message. It does not imply the successful completion of sending a message.
- The host may attempt to establish communications with equipment at any time due to the initialization of the host or by independent detection of a communications failure by the host. Thus, the host may initiate an S1F13/F14 transaction at any time.

6.4.2 *CommDelay Timer* — The CommDelay timer represents an internal timer used to measure the interval between attempts to send S1F13. The length of this interval is equal to the value in the EstablishCommunicationsTimeout. The CommDelay timer is not directly visible to the host.

6.4.2.1 EstablishCommunicationsTimeout is the user-configurable equipment constant that defines the delay, in seconds, between attempts to send S1F13. This value is used to initialize the CommDelay timer.

6.4.2.2 The CommDelay timer is initialized to begin timing. The CommDelay timer is initialized only when the state WAIT DELAY is entered.

6.4.2.3 The CommDelay timer is expired when it 'times out', and the time remaining in the interval between attempts to send is zero. When the timer expires during the state WAIT DELAY, it triggers a new attempt to send S1F13 and the transition to the state WAIT CRA. Note that CRA is the mnemonic defined for Establish Communications Request Acknowledge (S1F14).

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

6.4.3 *Conventions*

- The attempt to send S1F13 is made only upon transit into the state WAIT CRA. The CommDelay Timer should be set to 'expired' at this time.
- The CommDelay timer is initialized only upon transit into the state WAIT DELAY. A next attempt to send S1F13 shall occur only upon a transit to the state WAIT CRA.

6.4.4 *Communication States* — There are two major states of SECS communication, DISABLED and ENABLED. The system default state shall be user-configurable at the equipment (e.g., via a jumper setting or nonvolatile memory variable).

6.4.4.1 Once system initialization has been achieved, the operator shall be able to change the communication state selection at any time via equipment terminal functions or momentary switch. A two-position type switch shall not be used due to possible conflict with the system default.

6.4.4.2 The ENABLED state has two substates, NOT COMMUNICATING and COMMUNICATING, described below. The equipment shall inform the operator of the current communication state via continuous display at the equipment, including the NOT COMMUNICATING and COMMUNICATING substates.

6.4.4.3 In the event of a connection transaction failure, a user-configurable equipment constant EstablishCommunicationsTimeout is used to establish the interval between attempts to send an S1F13 (Establish Communications Request) while in the NOT COMMUNICATING substate.

6.4.4.4 Figure 4 shows the relationship between the superstates and substates of the Communications State Model. A description of the events triggering state transitions and the actions taken is given in Table 1.

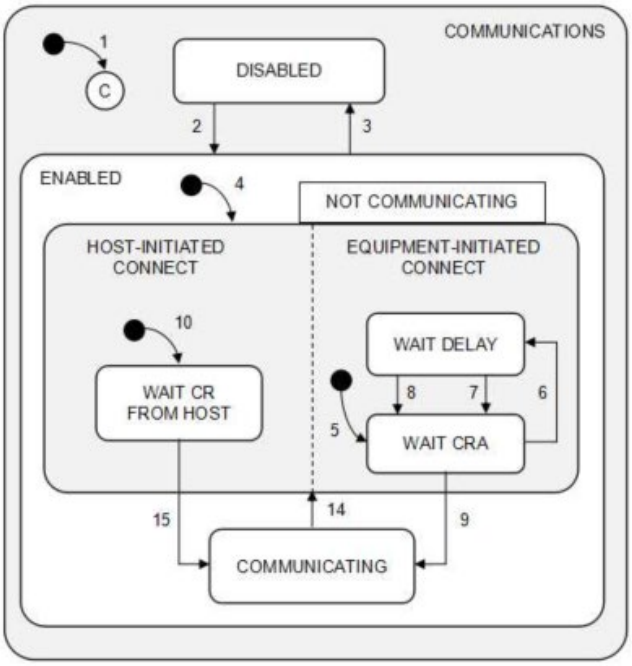

Figure 4
Communications State Model Diagram

6.4.4.5 The states of the Communications State Model are defined as follows:

6.4.4.5.1 *DISABLED*

6.4.4.5.1.1 In this state SECS-II communication with a host computer is nonexistent. If the operator switches from ENABLED to DISABLED, all SECS-II communications shall cease immediately. Any messages queued to send shall

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

be discarded, and all further action on any open transactions and conversations shall be halted. Refer to SEMI E5, for definitions of SECS-II transaction and conversation protocols. Handling of messages currently being transmitted is an issue for lower level message transfer protocols and is not addressed in this Standard.

6.4.4.5.1.2 The DISABLED state is a possible system default.

6.4.4.5.2 *ENABLED*

6.4.4.5.2.1 ENABLED has two substates, COMMUNICATING and NOT COMMUNICATING. Whenever communications are enabled, either during system initialization or through operator selection, the substate of NOT COMMUNICATING is active until communications are formally established. Lower-level protocols (such as SECS-I) are assumed to be functioning normally in that they are capable of supporting the communication of SECS-II syntax.

6.4.4.5.2.2 The ENABLED state is a possible system default.

6.4.4.5.3 *ENABLED/NOT COMMUNICATING*

6.4.4.5.3.1 No messages other than S1F13, S1F14, and S9Fx shall be sent while this substate is active. The equipment shall discard any messages received from the host other than S1F13 or S1F14 (Establish Communications Acknowledge). It shall also periodically attempt to establish communication with a host computer by issuing an S1F13 until communications are successfully established. However, only one equipment-initiated S1F13 transaction may be open at any time.

6.4.4.5.3.2 The NOT COMMUNICATING state has two AND substates, HOST-INITIATED CONNECT and EQUIPMENT-INITIATED CONNECT, both of which are active whenever the equipment is NOT COMMUNICATING. These two substates clarify the behavior of the equipment in the event that both the equipment and the host attempt to establish communications during the same period of time. Note that in the Harel notation, an exit from any AND substate is an exit from the parent state and thus from all other AND substates of that parent substate.

6.4.4.5.4 *NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT*

6.4.4.5.4.1 This state has two substates, WAIT CRA and WAIT DELAY. Upon any entry to the NOT COMMUNICATING state, whenever EQUIPMENT-INITIATED CONNECT first becomes active, a transition to WAIT CRA occurs, the CommDelay timer is set to ‘expired’, and an immediate attempt to send S1F13 is made.

6.4.4.5.5 *NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT/WAIT CRA*

6.4.4.5.5.1 An Establish Communications Request has been sent. The equipment waits for the host to acknowledge the request.

6.4.4.5.6 *NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT/WAIT DELAY*

6.4.4.5.6.1 A connection transaction failure has occurred. The CommDelay timer has been initialized. The equipment waits for the timer to expire.

6.4.4.5.7 *NOT COMMUNICATING/HOST-INITIATED CONNECT*

6.4.4.5.7.1 This state describes the behavior of the equipment in response to a host-initiated S1F13 while NOT COMMUNICATING is active.

6.4.4.5.8 *NOT COMMUNICATING/HOST-INITIATED CONNECT/WAIT CR FROM HOST*

6.4.4.5.8.1 The equipment waits for an S1F13 from the host. If an S1F13 is received, the equipment attempts to send an S1F14 with COMMACK = 0.

6.4.4.5.9 *ENABLED/COMMUNICATING*

6.4.4.5.9.1 Communications have been established. The equipment may receive any message from the host, including S1F13. When the equipment is COMMUNICATING, SECS communications with a host computer shall be maintained. This state remains active until communications are disabled or a communication failure occurs. If the equipment receives S1F13 from the host while in the COMMUNICATING substate, it should respond with S1F14 with COMMACK set to zero. If the equipment receives S1F14 from a previously sent S1F13, no action is required.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

6.4.4.5.9.2 In the event of communication failure, the equipment shall return to the NOT COMMUNICATING substate and attempt to re-establish communications with the host.

6.4.4.5.9.3 It is possible that the equipment may be waiting for an S1F14 from the host in EQUIPMENT-INITIATED CONNECT/WAIT CRA at the time an S1F13 is received from the host in HOST-INITIATED CONNECT/WAIT CR FROM HOST. When this situation occurs, both equipment and host have an open S1F13/S1F14 transaction. Since communications are successfully established on the successful completion of any S1F13/S1F14 transaction, either of these two transactions may be the first to complete successfully and to cause the transition from NOT COMMUNICATING state to COMMUNICATING. In this event, the other transaction shall remain open regardless of the transition to COMMUNICATING until it is closed in a normal manner.

6.4.4.5.9.4 If the equipment has not yet sent an S1F14 to a previously received S1F13 at the time when COMMUNICATING becomes active, the S1F14 response shall be sent in a normal manner. This includes transmissions that may have started but not yet successfully completed at the time that the transition to COMMUNICATING occurs. A failure to send the S1F14 is then treated as any other communication failure.

6.4.4.5.9.5 If the equipment-initiated S1F13/S1F14 is still open when the transition to COMMUNICATING occurs, subsequent failure to receive a reply from the host is considered a communication fault by equipment. An S9F9 should be sent when a transaction timer timeout occurs (refer to § 7.10 for definitions of communication faults and message faults, as well as detail on Stream 9 Error Messages). The existence of a transaction timer is not a requirement in some protocols.

6.4.5 *State Transitions* — Table 1 contains a full description of the state transitions depicted in Figure 4.

6.4.5.1 When the operator switches from the DISABLED state to the ENABLED state, no collection event shall occur, since no messages can be sent until communications have been established. The process of establishing communications serves to notify the host that communications are ENABLED. No other collection events are defined for the Communications State Model.

Table 1 Communications State Transition Table

#	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	(Entry to COMMUNICATIONS)	System initialization.	System Default	None.	The system default may be set to DISABLED or ENABLED.
2	DISABLED	Operator switches from DISABLED to ENABLED.	ENABLED	None.	SECS-II communications are enabled.
3	ENABLED	Operator switches from ENABLED to DISABLED.	DISABLED	None.	SECS-II communications are prohibited.
4	(Entry to ENABLED)	Any entry to ENABLED state.	NOT COMMUNICATING	None.	May enter from system initialization to ENABLED or through operator switch to ENABLED.
5	(Entry to EQUIPMENT-INITIATED CONNECT)	(Any entry to NOT COMMUNICATING)	WAIT CRA	Initialize communications. Set CommDelay timer ‘expired’. Send S1F13.	Begin the attempt to establish communications.
6	WAIT CRA	Connection transaction failure (including a protocol-specific communication failure).	WAIT DELAY	Initialize CommDelay timer. Dequeue all messages queued to send.	If appropriate, dequeued messages shall be placed in spool buffer in the order generated. Wait for timer to expire.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

#	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
7	WAIT DELAY	CommDelay timer expired.	WAIT CRA	Send S1F13	Wait for S1F14. May receive S1F13 from Host.
8	WAIT DELAY	Received a message other than S1F13.	WAIT CRA	Discard message. No reply. Set CommDelay timer 'expired'. Send S1F13.	Indicates opportunity to establish communications.
9	WAIT CRA	Received expected S1F14 with COMMACK = 0.	COMMUNICATING	None.	Communications are established.
10	(Entry to HOST-INITIATED CONNECT)	(Any entry to NOT COMMUNICATING)	WAIT CR FROM HOST	None.	Wait for S1F13 from Host.
14	COMMUNICATING	Communication failure. (Refer to SEMI E4 or SEMI E37 for a protocol-specific definition of communication failure.)	NOT COMMUNICATING	Dequeue all messages queued to send.	Dequeued messages may be placed in spool buffer as appropriate.
15	WAIT CR FROM HOST	Received S1F13.	COMMUNICATING	Send S1F14 with COMMACK = 0.	Communications are established.

6.5 *Control State Model* — The CONTROL state model defines the level of cooperation between the host and equipment. It also specifies how the operator may interact at the different levels of host control. While the COMMUNICATIONS state model addresses the ability for the host and equipment to exchange messages, the CONTROL model addresses the equipment's responsibility to act upon messages that it receives.

6.5.1 The CONTROL model provides the host with three basic levels of control. In the highest level (REMOTE), the host may control the equipment to the full extent possible. The middle level (LOCAL) allows the host full access to information, but places some limits on how the host can affect equipment operation. In the lowest level (OFF-LINE), the equipment allows no host control and only very limited information. The host may establish communications. This does not affect equipment operation and for that reason is not termed a control operation. The host may determine the equipment identification via the S1F13/F14 transaction.

6.5.2 The control model and communications model (when implemented) do not interact directly. That is, no action or state of one model directly causes a change in behavior of the other. It is true, however, that when the communication state is NOT COMMUNICATING then most message transaction are not functional. When messages cannot be transmitted, the control capabilities and all other GEM capabilities are affected.

6.5.3 Refer to Figure 5 as the CONTROL substates and state transitions are defined.

6.5.4 *OFF-LINE*

6.5.4.1 When the OFF-LINE state is active, operation of the equipment is performed by the operator at the operator console. While the equipment is OFF-LINE, message transfer is possible. However, the use of messaging for any automation purpose is severely restricted. While the OFF-LINE state is active, the equipment will only respond to those messages used for the establishment of communications or a host request to activate the ON-LINE state.

6.5.4.2 While OFF-LINE, the equipment will respond with an SxF0 to any primary message from the host other than S1F13 or S1F17. It will process and respond to S1F13 and S1F17. S1F17 is used by the host to request the equipment to transition to the ON-LINE state. The equipment will accept this request and send a positive response only when the HOST OFF-LINE state is active (refer to transition 11 definition below).

6.5.4.3 While the OFF-LINE state is active, the equipment shall attempt to send no primary message other than S1F13, S9Fx, and S1F1 (refer to ATTEMPT ON-LINE substate). If the equipment receives a reply message from the host other than S1F14 or S1F2, this message is discarded. Sending of S1F13 is based upon the COMMUNICATIONS state

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

model. S9Fx messages may be issued only in response to the messages to which the equipment will normally respond while OFF-LINE (i.e., S1F13 and S1F17).

6.5.4.4 No messages enter the spool when the system is OFF-LINE. Spooling may be active when the Communications State of NOT COMMUNICATING is active. This might occur during OFF-LINE, but since the equipment will not attempt to send messages except as mentioned in the previous paragraph, no messages will enter the spool. The equipment may send S1F1 or S1F13, but since Stream 1 messages are not eligible for spooling, they will not enter the spool either.

6.5.5 OFF-LINE has three substates: EQUIPMENT OFF-LINE, ATTEMPT ON-LINE, and HOST OFF-LINE.

6.5.6 *OFF-LINE/EQUIPMENT OFF-LINE*

6.5.6.1 While this state is active, the system maintains the OFF-LINE state. It awaits operator instructions to attempt to go ON-LINE.

6.5.7 *OFF-LINE/ATTEMPT ON-LINE*

6.5.7.1 While the ATTEMPT ON-LINE state is active, the equipment has responded to an operator instruction to attempt to go to the ON-LINE state. Upon activating this state, the equipment attempts to send an S1F1 to the host.

6.5.7.2 Note that when this state is active, the system does not respond to operator actuation of either the ON-LINE or the OFF-LINE switch.

6.5.8 *OFF-LINE/HOST OFF-LINE*

6.5.8.1 While the HOST OFF-LINE state is active, the operator's intent is that the equipment be ON-LINE. However, the host has not agreed. Entry to this state may be due to a failed attempt to go ON-LINE or to the host's request that the equipment go OFF-LINE from ON-LINE (refer to the transition table for more detail). While this state is active, the equipment shall positively respond to any host's request to go ON-LINE (S1F17). Such a request shall be denied when the HOST OFF-LINE state is not active.

6.5.9 *ON-LINE*

6.5.9.1 While the ON-LINE state is active, SECS-II messages may be exchanged and acted upon. Capabilities that may be available to the host should be similar to those available from the operator console wherever practical.

6.5.9.2 The use of SxF0 messages is not required while the ON-LINE state is active. Their use is discouraged in this case. The only allowed use is to close open transactions in conjunction with message faults.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

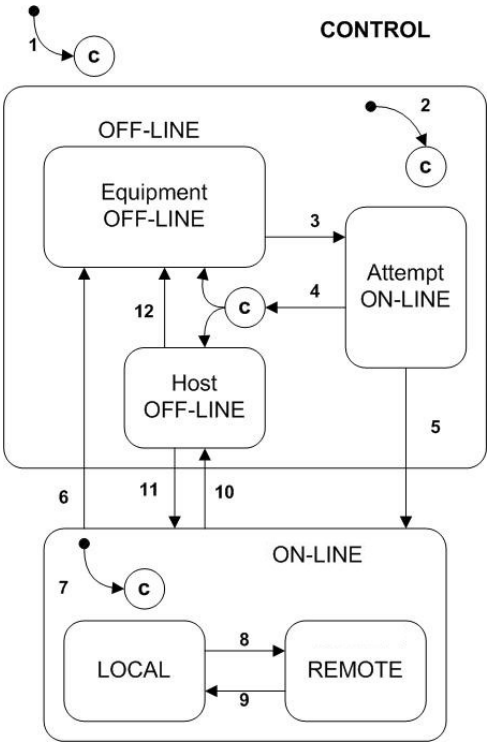

Figure 5
CONTROL State Model Diagram

6.5.10 *ON-LINE/LOCAL*

6.5.10.1 Operation of the equipment is implemented by direct action of an operator. All operation commands shall be available for input at the local operator console of the equipment.

6.5.10.2 The following capability requirements apply when the LOCAL state is active:

6.5.10.2.1 The equipment shall prohibit remote commands that cause physical movement, initiate processing, or affect the process in progress.

6.5.10.2.2 The equipment shall prohibit modifying any equipment constants by the host that affect the process. Other equipment constants shall be changeable by the host's request during the processing. All available equipment constants shall be modifiable by the host's request when no processing is in progress.

6.5.10.2.3 The equipment shall allow the host to request and send process programs. When the action does not affect any currently executing process program, the equipment can allow the host to select process programs for execution.

6.5.10.2.4 The equipment shall allow the host to configure data reporting capabilities including alarm, collection event, and trace data reporting. The equipment shall send these reports to the host at appropriate times.

6.5.10.2.5 The equipment shall provide data to the host as requested including status data, equipment constants, collection event reports, process program directories, and alarm reports.

6.5.10.2.6 The equipment shall allow Terminal Services as defined in this Specification.

6.5.10.3 As long as the LOCAL state is active, the equipment shall allow the host access to any other implemented equipment capabilities that are not specifically restricted in the above section.

NOTE 1: Capabilities mentioned above which are not implemented on a specific equipment may be ignored in this context.

6.5.11 *ON-LINE/REMOTE*

6.5.11.1 For equipment which supports the GEM capability of remote control (refer to § 7.5), while the REMOTE state is active, the host shall have access, through the communications interface, to the necessary commands to operate

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

the equipment through the full process cycle in an automated manner. The equipment does not restrict any host capabilities when REMOTE is active. The degree of control executed by the host may vary from factory to factory. In some cases, the operator may be required to interact during remotely controlled processes. This interaction may involve set-up operations, operator assist situations, and others. This state is intended to be flexible enough to accommodate these different situations.

6.5.11.2 To support the different factory automation policies and procedures, it shall be possible to configure the equipment to restrict the operator in specific nonemergency procedures. These restrictions shall be configurable so that the equipment may be set up to allow the operator to perform necessary functions without contention with the host. The categories for configuration shall include (but are not limited to):

- change equipment constants (process-related),
- change equipment constants (non-process-related),
- request Process Program from the host,
- select Process Program,
- start Process Program,
- pause/resume Process Program,
- operator assist,
- material movement to/from equipment, and
- equipment-specific commands (on a command-by-command basis if needed).

NOTE 2: Capabilities mentioned above which are not implemented on a specific equipment may be ignored in this context.

6.5.11.3 No capabilities that are available to the operator when the LOCAL state is active should be unconditionally restricted when the REMOTE state is active. The supplier may provide for configurable restriction of operator capabilities not included in the list above if desired. No configurability is necessary for any operator functions not available to the host.

6.5.11.4 The control functions shall be shared to some degree between the host and the local operator. At the very least, the operator shall have the capability to change the CONTROL state, actuate an Emergency Stop, and interrupt processing (e.g., STOP, ABORT, or PAUSE). All of these capabilities except Emergency Stop may be access-limited. Definition of the method of limiting operator access (password, key, etc.) to a capability is not within the scope of this Standard.

6.5.11.5 The host software should be designed to be compatible with the capabilities allotted to the operator.

Table 2 CONTROL State Transition Table

#	*Current State*	*Trigger*	*New State*	*Action*	*Comments*
1	(Undefined)	Entry into CONTROL state (system initialization).	CONTROL (Substate conditional on configuration).	None	Equipment may be configured to default to ON-LINE or OFF-LINE. The configuration mentioned for transitions 1 and 2 should be a single setting. This would provide the user with a choice of entering the EQUIPMENT OFF-LINE, ATTEMPT ON-LINE, HOST OFF-LINE, or ON-LINE states.
2	(Undefined)	Entry into OFF-LINE state.	OFF-LINE (Substate conditional on configuration.)	None	Equipment may be configured to default to any substate of OFF-LINE.
3	EQUIPMENT OFF-LINE	Operator actuates ON-LINE switch.	ATTEMPT ON-LINE	None	Note that an S1F1 is sent whenever ATTEMPT ON-LINE is activated.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

#	*Current State*	*Trigger*	*New State*	*Action*	*Comments*
4	ATTEMPT ON-LINE	S1F0.	New state conditional on configuration.	None	This may be due to a communication failure, reply timeout, or receipt of S1F0. Configuration may be set to EQUIPMENT OFF-LINE or HOST OFF-LINE. Communication failures are protocol specific. Refer to the appropriate protocol Standard (e.g., SEMI E4 or SEMI E37) for a protocol-specific definition of communication failure.
5	ATTEMPT ON-LINE	Equipment receives expected S1F2 message from the host.	ON-LINE	None	Host is notified of transition to ON-LINE at transition 7.
6	ON-LINE	Operator actuates OFF-LINE switch.	EQUIPMENT OFF-LINE	None	'Equipment OFF-LINE' event occurs. Event reply will be discarded while OFF-LINE is active. Any host initiated transaction open at the equipment shall be completed. This may happen either by sending the appropriate reply to the host prior to sending the event message or by sending an SxF0 message following the event message (i.e., after the transaction).
7	(Undefined)	Entry to ON-LINE state.	ON-LINE (Substate conditional on REMOTE/LOCAL switch setting.)	None	'Control State LOCAL' or 'Control State REMOTE' event occurs. Event reported based on actual ON-LINE substate activated.
8	LOCAL	Operator sets front panel switch to REMOTE.	REMOTE	None	'Control State REMOTE' event occurs.
9	REMOTE	Operator sets front panel switch to LOCAL.	LOCAL	None	'Control State LOCAL' event occurs.
10	ON-LINE	Equipment accepts Request OFF-LINE message from host (S1F15).	HOST OFF-LINE	None	'Equipment OFF-LINE' event occurs.
11	HOST OFF-LINE	Equipment accepts host Request ON-LINE message from host (S1F17).	ON-LINE	None	Host is notified to transition to ON-LINE at transition 7.
12	HOST OFF-LINE	Operator actuates OFF-LINE switch.	EQUIPMENT OFF-LINE	None	'Equipment OFF-LINE' event occurs.

6.6 *Equipment Processing States* — The behavior of the equipment in the performance of its intended function shall be documented. This processing state model is highly dependent on the equipment process, technology, and style. However, there are expected to be common aspects to these models.

6.6.1 The Processing State Model Diagram, Figure 6, is provided as an example of an implementation model. This model demonstrates the expected nature of the processing state model documentation. There is no requirement that these specific states be implemented.

6.6.2 The equipment shall generate collection events for each processing state transition, as well as provide status variables (ProcessState, PreviousProcessState) whose values are the current processing state and the previous processing state.

6.6.3 In referring to the Processing State Model Diagram, note that the initialization state INIT is not an actual processing state. It is shown here simply to indicate that the IDLE processing state is entered upon completion of

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

equipment system initialization. On the following pages detailed descriptions are provided for the equipment processing states and state transitions (numbered) as shown in the diagram.

6.6.4 *Description of Equipment Processing States*

6.6.4.1 *IDLE*

6.6.4.1.1 In this state the equipment is awaiting instructions.

6.6.4.2 *PROCESSING ACTIVE*

6.6.4.2.1 This state is the parent of all substates where the context of Process Program execution exists.

6.6.4.3 *PROCESS*

6.6.4.3.1 This state is the parent of those substates that refer to the active preparation and execution of a process program.

6.6.4.4 *SETUP*

6.6.4.4.1 In this state all external conditions necessary for process execution are satisfied, such as ensuring material is present at the equipment, input/output ports are in the proper state, parameters such as temperature and pressure values are within limits, etc. If all setup operations are already complete, then this becomes a fall through state and a transition to the next state takes place.

6.6.4.5 *READY*

6.6.4.5.1 In this state the equipment is ready for process execution and is awaiting a START command from the operator or the host.

6.6.4.6 *EXECUTING*

6.6.4.6.1 Executing is the state in which the equipment is executing a Process Program automatically and can continue to do so without external intervention.

6.6.4.7 *PAUSE*

6.6.4.7.1 In this state processing is suspended and the equipment is awaiting a command.

6.6.4.7.2 Each state transition is defined in the following table. Note that all transitions in this table should be considered collection events.

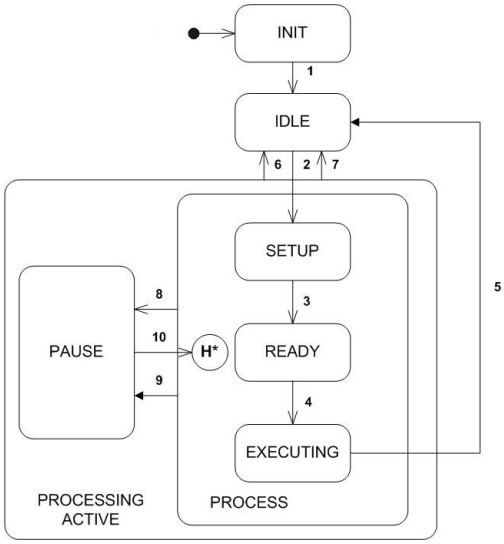

Figure 6
Processing State Model Diagram

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Table 3 Processing State Transition Table

#	*Current State*	*Trigger*	*New State*	*Action*	*Comments*
1	INIT	Equipment initialization complete.	IDLE	None.	None.
2	IDLE	Commit has been made to set up.	SETUP	None.	None.
3	SETUP	All setup activity has completed and the equipment is ready to receive a START command.	READY	This activity is equipment-specific.	None.
4	READY	Equipment has received a START command from the host or operator console.	EXECUTING	This activity is equipment-specific.	None.
5	EXECUTING	The processing task has been completed.	IDLE	None.	None.
6	PROCESSING ACTIVE	Equipment has received a STOP command from host or operator console.	IDLE	None.	None.
7	PROCESSING ACTIVE	Equipment has received an ABORT command from host or operator console.	IDLE	This activity is equipment-specific.	None.
8	PROCESS	The equipment decides to PAUSE due to a condition such as alarm.	PAUSE	This activity is equipment-specific.	For this type of problem, an operator assist is usually required.
9	PROCESS	Equipment has received a PAUSE command from host or operator console.	PAUSE	This activity is equipment-specific.	None.
10	PAUSE	Equipment has received a RESUME command from host or operator console.	Previous PROCESS substate	This activity is equipment-specific.	None.

7 Equipment Capabilities and Scenarios

7.1 This section describes the details of the capabilities required by GEM and provides scenarios for their use. Capabilities are operations performed by semiconductor manufacturing equipment. These operations are initiated through the communications interface using SECS-II messages. A scenario is a group of SECS-II messages arranged in a sequence to perform a capability. Other information may be included with the scenario for clarity. For each capability, the reader is provided with a statement of purpose, pertinent definitions, a detailed description, requirements, and scenarios.

7.1.1 The following capabilities are discussed:

- Establish Communications
- Event Notification
- Dynamic Event Report Configuration
- Variable Data Collection
- Trace Data Collection
- Limits Monitoring
- Status Data Collection
- On-Line Identification
- Alarm Management

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

- Remote Control
- Equipment Constants
- Process Program Management
- Material Movement
- Equipment Terminal Services
- Error Messages
- Clock
- Spooling
- Control

7.1.2 Because the GEM Standard does not attempt to define the behavior of the host computer, the host computer may initiate any GEM message scenario at any time and the equipment shall respond as described in the GEM Standard. When a GEM message scenario is initiated by either the host or equipment, the equipment shall behave in the manner described in the GEM Standard when the host uses the appropriate GEM messages.

7.2 *Establish Communications* — The Establish Communications capability provides a means of formally establishing communications following system initialization or any loss of communications between communicating partners, and thus of notifying the communication partner that a period of noncommunication has occurred.

7.2.1 *Purpose* — Communications between host and equipment are formally established through use of the Establish Communications Request/Establish Communications Acknowledge transaction.

7.2.1.1 The use of S1F1/F2 for this purpose is ambiguous since the transaction can be used for other purposes and may occur at any time.

7.2.1.2 The S1F13/F14 transaction, used in conjunction with the Communications State Model, provides a means for equipment to notify the host, or the host to notify the equipment, that there has been a period of inability to communicate. The successful completion of this transaction also signals a possible need for synchronization activities between host and equipment.

7.2.2 *Definitions*

7.2.2.1 *COMMACK*— Acknowledge code returned in the Establish Communications Acknowledge message. Refer to SEMI E5 for a full definition of this data item.

7.2.2.2 *EstablishCommunicationsTimeout* — An equipment constant used to initialize the interval between attempts to resend an Establish Communications Request. This value specifies the number of seconds for the interval. Refer to SEMI E5 for a full definition of this variable data item.

7.2.3 *Description* — There are potential problems when one side of the communications link fails and the other side does not detect it. From the point of view of the host, a loss of communications has many possible causes. In some cases, host-controlled settings on the equipment may need to be reset. In other cases, the equipment may have continued an automatic processing sequence during the period of no communication and may have changed states. The definition of a formal protocol for establishing communications alerts the host to the need to synchronize itself with the equipment's current status.

7.2.3.1 Equipment shall consider communications as formally established whenever either of the following conditions have been satisfied. Satisfaction of either of these conditions shall result in a transition to the COMMUNICATING substate. Refer to § 6.4 for further detail.

- Communications Request has been sent to the host and an Establish Communications Acknowledge has been received within the transaction timeout period and with an acknowledge code of Accept, or
- Communications Request has been received from the host, and an Establish Communications Acknowledge response has been successfully sent with an acknowledge code of Accept.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.2.3.2 When the equipment sends an Establish Communications Request to the host, this notifies the host of the possible need to synchronize itself with the equipment.

7.2.3.3 When the equipment is attempting to establish communications, an Establish Communications Request shall be sent periodically until communications have been formally established as described above. The interval between attempts shall be user-configurable and begins as soon as a connection transaction failure is detected (refer to § 6.4).

7.2.3.4 Attempting to establish communications is not a low-level connectivity issue, but rather a logical application issue used by either party to notify its partner that the host may need to perform synchronization activities with the equipment.

7.2.4 *Requirements*

- Equipment shall support the Communication State Model (refer to § 6.4).
- Equipment shall provide the EstablishCommunicationsTimeout equipment constant described above.

7.2.5 *Scenarios*

7.2.5.1 *Host Attempts to Establish Communications*

```
COMMENT                    HOST                EQUIPMENT              COMMENT

                                                                      Communications state is
                                                                      Enabled (any substate)

Establish                        S1F13-->
Communications

                                         <--S1F14                     Reply COMMACK = Accept
                                                                      and
                                                                      Communications state =
                                                                      COMMUNICATING
```

7.2.5.2 *Equipment Attempts to Establish Communications and Host Acknowledges*

```
COMMENT                    HOST                EQUIPMENT              COMMENT

                                                                      Communications State =
                                                                      NOT COMMUNICATING

                                                                      [LOOP]

                                                                      Set Communication state
                                                                      to WAIT_CRA

                                         <--S1F13                     Send Establish
                                                                      Communications Request

Establish                        S1F14-->                             [IF] S1F14 received
Communications                                                        without communication
Acknowledge                                                           failure or connection
                                                                      transaction failure

                                                                        [IF COMMACK =
                                                                      Accepted]

                                                                      Set Communication state
                                                                      to COMMUNICATING

                                                                      Exit LOOP

                                                                        [END IF]

                                                                      [ENDIF]

                                                                      Set Communication state
                                                                      to WAIT_DELAY

                                                                      Initialize CommDelay
                                                                      timer. When the
                                                                      CommDelay timer expires
                                                                      or a message other than
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                                  S1F13 is received, then
                                                                  continue in the LOOP.
                                                                  [ENDLOOP]
```

7.2.5.3 *Simultaneous Attempts to Establish Communications* — For equipment that supports interleaving, it is possible that either the host or equipment could send an Establish Communications Request before receiving the request from its partner. As communications are established by the successful acceptance of any one Establish Communications Request, it is immaterial who sends the request first. The roles of host and equipment may be reversed.

7.2.5.3.1 Equipment Receives S1F14 From Host Before Sending S1F14:

7.2.5.3.1.1 Note that Communications are established at the successful completion of the S1F13/14 transaction where COMMACK is set to zero. Communications are established on the successful transmission of S1F14, even if there is an open S1F13.

```
COMMENT                    HOST                EQUIPMENT              COMMENT

                                                                      Communications State =
                                                                      NOT COMMUNICATING

                                       <--S1F13                       Establish
                                                                      Communications Request

Establish                        S1F13-->
Communications Request

Reply COMMACK = Accept           S1F14-->                             S1F14 received from
                                                                      Host
                                                                      Communications
                                                                      established and
                                                                      Communications state =
                                                                      COMMUNICATING

                                       <--S1F14                       Reply COMMACK = Accept
```

7.2.5.3.2 Equipment Sends S1F14 To Host Before Receiving S1F14:

```
COMMENT                    HOST                EQUIPMENT              COMMENT

                                                                      Communications State =
                                                                      NOT COMMUNICATING

                                       <--S1F13                       Establish
                                                                      Communications Request

Establish                        S1F13-->
Communications Request

                                       <--S1F14                       Reply COMMACK = Accept
                                                                      Communications
                                                                      established and
                                                                      Communications state =
                                                                      COMMUNICATING

Reply COMMACK = Accept           S1F14-->                             S1F14 received from
                                                                      Host
```

7.3 *Data Collection* — Data collection allows the host to monitor equipment activity via event reporting, trace data reporting, limits monitoring, and query of selected status or other variable data.

7.3.1 *Event Data Collection* — Event data collection provides a dynamic and flexible method for the user to tailor the equipment to meet individual needs with respect to data representation and presentation to the host. The event-

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

based approach to data collection provides automatic notification to the host of equipment activities and is useful in monitoring the equipment and in maintaining synchronization with the equipment.

7.3.1.1 Event data collection may be broken into two logical parts: host notification when an event occurs, and dynamic configuration of the data attached to the event notification.

7.3.1.2 *Event Notification* — This section describes the method of notifying the host when equipment collection events occur.

7.3.1.2.1 *Purpose* — This capability provides data to the host at specified points in equipment operation. This asynchronous reporting eliminates the need for the host to poll the equipment for this information. Events on the equipment may trigger activity on the part of the host. Also, knowledge of the occurrence of events related to the equipment state models allows the host to track the equipment state. An equipment's behavior is related to its current state. Thus, this capability helps the host understand how an equipment will behave and how it will react to host behavior.

7.3.1.2.2 *Definitions*

7.3.1.2.2.1 *Collection Event* — An event (or grouping of related events) on the equipment that is considered significant to the host.

7.3.1.2.2.2 *Collection Event ID (CEID)* — A unique identifier of a collection event. Refer to SEMI E5 for a full definition of this data item.

7.3.1.2.2.3 *Event* — A detectable occurrence significant to the equipment.

7.3.1.2.2.4 *Report* — A set of variables predefined by the equipment or defined by the host via S2F33/F34.

7.3.1.2.3 *Detailed Description* — The equipment supplier shall provide a set of predefined collection events. Specific collection events are required by individual capabilities and state models. Examples of collection events include:

- The completion of each action initiated by a host requested command,
- Selected processing and material handling activities,
- Operator action detected by the equipment,
- A state transition,
- The setting or clearing of an alarm condition on the equipment, and
- Exception conditions not considered alarms.

7.3.1.2.3.1 Refer to § 9 for a list of required collection events.

7.3.1.2.3.2 The reporting of a collection event may be disabled per event by the user to eliminate unwanted messages. An event report message shall be sent to the host upon the occurrence of a particular collection event if the collection event (CEID) has been enabled. Attached to each event message is one or more event reports which contain variable data. § 7.3.1.3 describes the capability which allows for the dynamic customization of event reports. The values of any data contained in an event report message shall be current upon the occurrence of the event. This implies that event reports be built at the time of the event occurrence.

7.3.1.2.3.3 The equipment shall also provide the S6F15/F16 transaction to allow the host to request the data from a specific event report.

7.3.1.2.4 *Requirements*

1. The equipment supplier shall provide documentation of all collection events defined on the equipment and the conditions for each event to occur.
2. The equipment supplier shall provide unique CEIDs for each of the various collection events that are available for reporting.
3. The equipment supplier shall provide a method for enabling and disabling the reporting of each event. This method shall either be available via the host interface (refer to § 7.3.1.3) or the equipment's operator console.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

4. For each event, the equipment supplier shall provide either:
   - a default set of report(s) linked to the event which contain data pertinent to that event, or
   - the ability for the user to configure the data linked to that event via the equipment's operator console or host interface (refer to § 7.3.1.3).

7.3.1.2.5 *Scenarios*

7.3.1.2.5.1 Collection Event Occurs on the Equipment:

```
COMMENT                     HOST                    EQUIPMENT                COMMENT

                                                                             [IF] Event Report is
                                                                             Multi-block

                                        <--S6F5                              [THEN] send Multi-block
                                                                             inquire

Multi-block grant                        S6F6-->

                                                                             [ENDIF]

                                        <--S6F11                             Equipment sends Event
                                                                             Report

Host acknowledges Event                 S6F12-->
Report
```

7.3.1.2.5.2 Host Requests Event Report:

```
COMMENT                     HOST                    EQUIPMENT                COMMENT

Host requests an event                  S6F15-->
report

                                        <--S6F16                             Equipment sends event
                                                                             report
```

7.3.1.3 *Dynamic Event Report Configuration* — This section describes a capability which allows the host to dynamically modify the equipment collection event reporting setup.

7.3.1.3.1 *Purpose* — This capability is defined to provide the data reporting flexibility required in some manufacturing environments. It allows the host to increase or decrease the data flow according to need. For example, if the performance of an equipment degrades, the data flow from that equipment may be increased to help diagnose the problem.

7.3.1.3.2 *Definitions*

7.3.1.3.2.1 *EventsEnabled* — A status variable that lists the identifiers of collection events (CEIDs) enabled for reporting. Refer to SEMI E5 for a full definition of this variable data item.

7.3.1.3.2.2 *Report ID (RPTID)* — A unique identifier of a specific report. Refer to SEMI E5 for a full definition of this data item.

7.3.1.3.3 *Detailed Description* — The equipment shall support the following event report configuration functionality through the SECS-II interface:

- Host definition/deletion of custom reports,
- Host linking/unlinking of defined reports to specified collection events, and
- Host enabling/disabling the reporting of specified collection events.

NOTE 3: The equipment may also supply alternative means for defining reports and linking reports to events (e.g., via the operator console). Implementation of alternate means is not required.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.1.3.3.1 The equipment can be instructed by the host to enable or disable reporting of collection events on an individual or collective basis. A status variable (SV) shall be available that consists of a list of enabled collection events (refer to § 8.4, Variable Item List, EventsEnabled variable).

7.3.1.3.3.2 Reports may be attached to an event report message (S6F11). These reports are specifically linked to the desired event and typically contain variable data relating to that collection event. The reports may be provided by the equipment supplier or created by the user. The user shall be able to create reports and link them to collection events via the SECS-II interface.

7.3.1.3.3.3 The data reported in the event report messages may consist of Status Variables (SVs), Equipment Constants (ECs), and/or Data Variables (DVs). Note that data variable values shall be valid and current on certain collection events and certain states and might not be current at other times. The implementer shall document when a data variable value will be current and available for reporting.

7.3.1.3.4 *Requirements*

- The equipment manufacturer shall provide documentation of all variable data available from the equipment. This is to include variable name, variable type or class (SV, EC, DV), units, format codes, possible range of values, and a description of the meaning and use of this variable.
- The equipment manufacturer shall provide unique VIDs for the various variable data (V) available for data collection in the equipment. For example, this means that no SV shall have a VID which is the same as the VID of any EC or DV.
- All variable data shall be available for report definition and event data collection. Refer to § 8.4, Variable Item List, for a list of required variable data.
- All report definitions, report-to-event links, and enable/disable status of event reports shall be retained in nonvolatile storage.

7.3.1.3.5 *Scenario*

7.3.1.3.5.1 Collection Event Reporting Set-Up:

7.3.1.3.5.1.1 Note that DRACK is an acronym for Define Report Acknowledge Code, refer to SEMI E5 for full definition of this Data Item.

COMMENT	HOST		EQUIPMENT	COMMENT
[IF] Define Report is Multi-block		S2F39-->		
[THEN] send Multi-block inquire				
			<--S2F40	Multi-block grant.
[ENDIF]				
Send report definitions		S2F33-->		
			<--S2F34	DRACK = 0 the reports are OK
[IF] Link Event Report is Multi-block		S2F39-->		
[THEN] send Multi-block inquire				
			<--S2F40	Multi-block grant.
[ENDIF]				
Link reports to collection events		S2F35-->		CEIDs and the corresponding RPTIDs are received.
			<--S2F36	LRACK = 0 the collection event

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                                    linkages are
                                                                    acceptable.
Enable specific                    S2F37-->                         Enable/disable codes
collection events                                                   (CEEDs) collection
                                                                    events
                                                                    and the respective
                                                                    collection event
                                                                    reporting CEIDs
                                                                    received.
                                           <--S2F38                 ERACK = 0 OK, will
                                                                    generate the specified
                                                                    reports when the
                                                                    appropriate collection
                                                                    events happen
```

7.3.1.4 *Data Variable and Collection Event Namelist Request* — This section describes a method for the host to query valid event data collection configuration from the equipment.

7.3.1.4.1 *Purpose* — Data variable and collection event namelist requests allow the host to query the equipment for existing data variables and collection events as well as their dependencies in order to create valid event reports.

7.3.1.4.2 *Detailed Description* — The equipment shall support the following data variable and collection event namelist requests through the SECS-II interface:

- Data Variable Namelist Request (DVNR)
- Collection Event Namelist Request (CENR)
  - The CENR includes reporting of the dependencies between data variables and collection events.

7.3.1.4.3 *Requirements*

1. The equipment shall support the message S1F21 Data Variable Namelist Request (DVNR) to enable the host to query data variables from the equipment.
2. The equipment shall support the message S1F23 Collection Event Namelist Request (CENR) to enable the host to query collection events including their associated data variables from the equipment.

7.3.1.4.4 *Scenarios*

7.3.1.4.4.1 Host Initiates: Data Variable Namelist Request (DVNR):

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Host initiates DVNR              S1F21-->
                                           <--S1F22                 Data Variable Namelist
                                                                    (DVN)
```

7.3.1.4.4.2 Host Initiates: Collection Event Namelist Request (CENR):

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Host initiates CENR              S1F23-->
                                           <--S1F24                 Collection Event
                                                                    Namelist (CEN)
```

7.3.1.5 *Event Report Information* — This section describes a method for the host to query the list of defined report identifiers, report definitions, event report links and the list of enabled collection identifiers from the equipment.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.1.5.1 *Purpose* — Event Report Information requests allow the host to discover details about the report and event report link configuration. For example, when a host first connects to an equipment, the host can discover if previously configured report and event report setup was persisted by the equipment. Without these features, the host cannot know whether or not report and event report setup was changed. Additionally, these Event Report Information requests can be used when testing a GEM interface to confirm that other requests are working.

7.3.1.5.2 *Detailed Description* — The equipment shall support the following requests through the SECS-II interface:

- Request List of Report Identifiers (RQRPTID)
- Request List of Report Definitions (RQLRD)
- Request Event Report Links (RQERL)
- Request List of Enabled Events (RQEE)

7.3.1.5.3 *Requirements*

1. The equipment shall support the message S2F51 Request List of Report Identifiers (RQRPTID) to allow the host to query report identifiers from the equipment.
2. The equipment shall support the message S2F53 Request List of Report Definitions (RQLRD) to allow the host to query report definitions from the equipment.
3. The equipment shall support the message S2F55 Request Event Report Links (RQERL) to allow the host to query event report links from the equipment.
4. The equipment shall support the message S2F57 Request List of Enabled Events (RQEE) to allow the host to query the list of identifiers for enabled collection events from the equipment. Note that as long as no collection events are enabled or disabled between queries, the returned list shall be the same as the value for status variable EventsEnabled.

7.3.1.5.4 *Scenarios*

7.3.1.5.4.1 Host Initiates: Request List of Report Identifiers (RQRPTID):

```
        COMMENT                     HOST                    EQUIPMENT                COMMENT

Host initiates RQRPTID                    S2F51-->
                                                  <--S2F52                Return List of Report
                                                                          Identifiers (RTRPTID)
```

7.3.1.5.4.2 Host Initiates: Request List of Report Definitions (RQLRD):

```
        COMMENT                     HOST                    EQUIPMENT                COMMENT

Host initiates RQLRD                      S2F53-->
                                                  <--S2F54                Return List of Report
                                                                          Definitions (RTLRD)
```

7.3.1.5.4.3 Host Initiates: Request Event Report Links (RQERL):

```
        COMMENT                     HOST                    EQUIPMENT                COMMENT

Host initiates RQERL                      S2F55-->
                                                  <--S2F56                Return Event Report
                                                                          Links (RTERL)
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.1.5.4.4 Host Initiates: Request List of Enabled Events (RQEE):

```
COMMENT                    HOST                EQUIPMENT              COMMENT

Host initiates RQEE              S2F57-->

                                          <--S2F58              Return List of Enabled
                                                                Events (RTEE)
```

7.3.2 *Variable Data Collection*

7.3.2.1 *Purpose* — This capability allows the host to query for the equipment data variables and is useful during initialization and synchronization.

7.3.2.2 *Definitions*

7.3.2.2.1 *Report ID (RPTID)* — A unique identifier of a specific report. Refer to SEMI E5 for a full definition of this data item.

7.3.2.2.2 *Variable Data (V)* — A variable data item containing status, discrete, or constant data. Refer to SEMI E5 for a full definition of this data item.

7.3.2.3 *Detailed Description* — The host may request a report containing data variables from the equipment by specifying the RPTID. It is assumed that the report has been previously defined (e.g., using the Define Report S2F33 transaction [refer to § 7.3.1]). The values of any status variables (SVs) and equipment constants (ECs) contained within the report shall be current. Discrete data variable values (DVVALs) are only guaranteed to be valid upon the occurrence of a specific collection event. If a DVVAL cannot be specified in equipment due to some restrictions depend on hardware and/or software conditions, the zero-length item is reported.

7.3.2.4 *Requirements*

- Variable descriptions including the associated units of measure shall be provided by the equipment manufacturer.

7.3.2.5 *Scenario*

7.3.2.5.1.1 Host Requests Report:

```
COMMENT                    HOST                EQUIPMENT              COMMENT

Host requests data               S6F19-->
variables contained in
report RPTID

                                          <--S6F20              Equipment responds with
                                                                a list of variable data
                                                                for the given RPTID.
```

7.3.3 *Trace Data Collection*

7.3.3.1 *Purpose* — Trace data collection provides a method of sampling data on a periodic basis. The time-based approach to data collection is useful in tracking trends or repeated applications within a time window, or monitoring of continuous data.

7.3.3.2 *Definitions*

7.3.3.2.1 *Data Sample Period (DSPER)* — The time delay between samples. Refer to SEMI E5 for a full definition of this data item.

7.3.3.2.2 *Reporting Group Size (REPGSZ)* — The number of samples included per trace report transmitted to the host. See SEMI E5 for a full definition of this data item.

7.3.3.2.3 *Status Variable (SV)* — Status data item (included in trace report). Refer to SEMI E5 for a full definition of this data item.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.3.2.4 *Status Variable ID (SVID)* — A unique identifier of a status variable. Refer to SEMI E5 for a full definition of this data item.

7.3.3.2.5 *Total Samples* (TOTSMP) — Number of samples to be taken during a complete trace period. Refer to SEMI E5 for a full definition of this data item.

7.3.3.2.6 *Trace Request ID (TRID)* — An identifier associated with a trace request definition. Refer to SEMI E5 for a full definition of this data item.

7.3.3.3 *Detailed Description* — The equipment shall establish a trace report as instructed by the host (S2F23). For a trace report (S6F1), the host designates the following conditions for the trace:

- TRID – a name for the trace report
- DSPER – a time interval for data sampling
- TOTSMP – the total number of samples to be taken
- REPGSZ – the number of samples per trace report
- List of SVID – a list identifying data that will be sent in the trace report

The number of trace reports sent to the host is determined by total samples divided by reporting group size (TOTSMP/REPGSZ).

7.3.3.4 The equipment shall sample the specified data (SVs) at the interval designated by the host (DSPER) and shall send a predefined trace report to the host for the specified reporting group size (REPGSZ). The trace report definition shall be automatically deleted from the equipment after the last trace report has been sent.

7.3.3.4.1 The host may modify or re-initiate a trace function currently in progress by specifying the same TRID in a trace request definition, at which point the old trace shall be halted and the new trace shall be initiated, or the host can instruct the equipment to halt a trace report prior to its completion by specifying TOTSMP = 0 for that TRID, at which point the trace report definition shall be deleted.

7.3.3.4.2 A detailed example is included as Related Information 1, § R1-4.

7.3.3.5 *Requirements*

- The equipment shall have a local mechanism (e.g., internal clock) for triggering the periodic sampling and transmission of trace reports to the host.
- A minimum of four concurrent traces shall be supported by the equipment. The same SVID may be collected in multiple traces simultaneously.
- All SVIDs available at the equipment shall be supported for trace data collection. The exception to this is any SV that will not fit into a single block.

NOTE 4: SEMI E5 provides for SVs to be of a list format. Since this may in practice be a variable list, there is a potential problem with such an SV supported by the Trace Data Collection capability. This is a problem with the SEMI E5 Standard. Care should be exercised in the use of SVs using the list format.

7.3.3.6 *Scenario*

7.3.3.6.1 Host Initiates Trace Report:

COMMENT	HOST	EQUIPMENT	COMMENT
Trace Data initialization requested	S2F23-->		
		<--S2F24	Acknowledge, trace initiated
			[DO] TOTSMP/REPGSZ times
			[DO] REPGSZ many times:

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                                    collect SVID1,...SVIDn
                                                                  data,
                                                                    delay time by DSPER.
                                                                  [END_DO]
                                          <--S6F1                 Send SV1,...SVn
Acknowledge receipt                 S6F2-->
                                                                  [END_DO]
Optional:                          S2F23-->
Request trace
termination prior to
completion (TOTSMP = 0)
                                          <--S2F24                Acknowledge premature
                                                                  termination
```

7.3.3.7 *Trace Information* — This section describes a method for the host to query the list of trace identifiers and to query trace definitions.

7.3.3.7.1 *Purpose* —Trace Information requests allows the host to discover the current trace setup including trace identifiers and trace definitions. For example, these Trace Information requests can be used when testing a GEM interface to confirm that host requests to create traces were correctly received.

7.3.3.7.2 *Detailed Description* — The equipment shall support the following requests through the SECS-II interface:

- Request List of Trace Identifiers (RQTRID)
- Request Trace Definitions (RQTD)

7.3.3.7.3 *Requirements*

1. The equipment shall support the message S2F61 Request List of Trace Identifiers (RQTRID) to allow the host to query trace identifiers from the equipment.
2. The equipment shall support the message S2F63 Request Trace Definitions (RQTD) to allow the host to query trace definitions from the equipment.

7.3.3.7.4 *Scenarios*

7.3.3.7.4.1 Host Initiates: Request List of Trace Identifiers (RQTRID):

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Host initiates RQTRID             S2F61-->
                                          <--S2F62                Return List of Trace
                                                                  Identifiers (RTTRID)
```

7.3.3.7.4.2 Host Initiates: Request Trace Definitions (RQTD):

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Host initiates RQTD               S2F63-->
                                          <--S2F64                Return Trace Definition
                                                                  (RTTD)
```

7.3.4 *Limits Monitoring* — This capability relates to the monitoring of selected equipment variables and has three primary aspects:

- Defines a standard set of monitoring zones and limits.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

- Provides for reporting to the host when selected equipment variables transition between monitoring zones.
- Empowers the host to modify the values of the variable limit attributes for these same selected equipment variables.

7.3.4.1 *Purpose* — The limits monitoring capability provides the host a means of monitoring equipment conditions by a flexible, efficient, and asynchronous method which is consistent across equipment. It eliminates the need for constant polling of equipment by the host for current status variable values. Further, this capability allows the host to implement changes in the monitoring range as needed. This capability has application to both production operation and diagnostic/testing scenarios, and it also has applicability to statistical process control.

7.3.4.2 *Definitions*

7.3.4.2.1 *LimitVariable* — DV containing the VID of a specific equipment variable for which a zone transition collection event has been generated.

7.3.4.2.2 *EventLimit* — DV containing the LIMITID of the limit crossed by LimitVariable.

7.3.4.2.3 *TransitionType* — DV which defines the direction of the zone transition which has occurred: 0 = transition from lower to upper zone, 1 = transition from upper to lower zone.

7.3.4.2.4 *Limit* — Used in this section to represent the set of variable limit attributes that completely describe a variable monitoring 'barrier'. The attributes include VID, Units, UPPERDB, LOWERDB, LIMITMAX, and LIMITMIN. In some contexts, it may be interpreted more narrowly as the combination of UPPERDB and LOWERDB.

7.3.4.2.5 *$LIMITID_n$* — Refers to the identifier of a specific limit (as defined by UPPERDB and LOWERDB) among the set of limits for a monitored equipment variable. LIMITIDs are consecutively numbered, beginning at one through the number of limits possible (seven minimum).

7.3.4.2.6 *Monitoring Zone* — A subset of the possible range of values for a variable of interest to the host. A single limit divides the range into two zones. Multiple limits may be combined to divide the range even further.

7.3.4.2.7 *Zone Transition* — The movement of a variable value from one monitoring zone to another. This transition is a collection event and has a corresponding CEID.

7.3.4.2.8 *Deadband* — An overlap of two zones implemented to prevent constant zone transitions by a variable sitting on or near a limit (i.e., 'chattering').

7.3.4.2.9 *UPPERDB* — A variable limit attribute that defines the upper boundary of the deadband of a limit. The format and units shall be the same as the format of the variable being monitored. The value applies to a single limit (LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

7.3.4.2.10 *LOWERDB* — A variable limit attribute that defines the lower boundary of the deadband of a limit. The value applies to a single limit (LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

7.3.4.2.11 *UPPER ZONE* — The range of values lying above a limit.

7.3.4.2.12 *LOWERZONE* — The range of values lying below a limit.

7.3.4.2.13 *LIMITMAX* — The maximum value for any limits of a specific equipment variable. This value is set by the equipment manufacturer and typically coincides with the maximum value allowed for the monitored variable.

7.3.4.2.14 *LIMITMIN* — The minimum value for any limits of a specific equipment variable. This value is set by the equipment manufacturer and typically coincides with the minimum value allowed for the monitored variable.

7.3.4.2.15 *Undefined* — When used in reference to variable limits, it indicates that monitoring/reporting of zone transitions involving that particular limit are disabled.

7.3.4.3 *Description* — The limits monitoring capability provides the host with a minimum of seven configurable limits or barriers that may be applied to selected equipment status variables (SVs) of the types floating point, integer, and Boolean. When one of these barriers is crossed, a collection event is generated to alert the host to a change in monitoring zone or state of the monitored variable. These seven limits may be combined in a variety of ways to match the needs of the host system. Note that while at least seven limits per variable are available from the equipment, the host need not use all seven. An illustration of a combination of five of the limits to provide one type of variable

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

monitoring is shown in Figure 7. This illustration shows the reading which might be available to the equipment, not the limit excursions reported to the host. Reporting is covered later in the section. This section describes the key aspects of limits monitoring. Detailed implementation examples of limits monitoring are provided as Related Information 1, § R1-7.

NOTE 5: While the SEMI E5 Standard allows SVs to be lists, such variable lists are not allowed under this capability.

7.3.4.3.1 *Monitoring Limit Characteristics* — A limit is defined by a set of attributes that include the variable (VID) to which the limit corresponds, the units of that variable, the maximum and minimum possible values of the limit (LIMITMAX and LIMITMIN) and the specific borders of the limit (UPPERDB and LOWERDB). Refer to Figure 8. There is a limitation to the values of UPPERDB and LOWERDB which may be stated as:

- LIMITMAX≥UPPERDB≥LOWERDB≥LIMITMIN

7.3.4.3.1.1 A limit divides the possible range of variable values into two parts, the upper zone and the lower zone. At any time, the monitored variable is considered to be in one and only one of these zones. However, as Figure 8 shows, these two zones have an area of overlap. This is called the deadband.

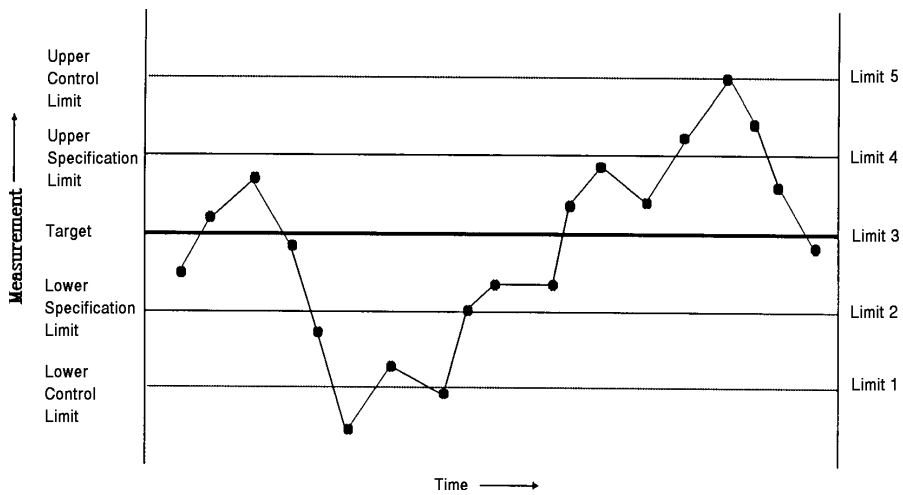

Figure 7
Limit Combination Illustration: Control Application

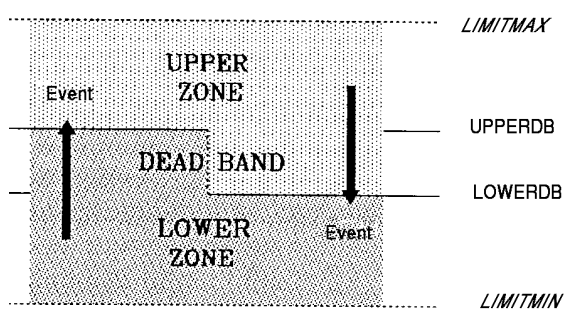

Figure 8
Elements of One Limit

7.3.4.3.1.2 The deadband is a key concept of limits monitoring, especially for floating point variables. Its purpose is to prevent a phenomenon known as chattering—the repeated changing of zones due to small, rapid fluctuations in variable value while near the zone boundary. In practice, the value of a variable shall reach the opposite boundary of the deadband before a zone transition can occur. Thus, if a variable's value reaches the UPPERDB and transitions into the upper zone, it will not return to the lower zone until it falls back to the LOWERDB. The difference between

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

UPPERDB and LOWERDB should always be greater than the typical amplitude of those fluctuations deemed to be insignificant. In some cases, the width of the deadband may set to zero (i.e., UPPERDB = LOWERDB). At first glance, this would seem to make indeterminate the current zone when an integer value sits on the limit. This is not the case, however, when movement of the value is considered. To illustrate, an example is given, assuming that UPPERDB = LOWERDB = 100. The list shows consecutive readings of the variable and the resultant zone:

- 99 Lower Zone (Initial Reading)
- 101 Upper Zone (Zone Transition)
- 100 Lower Zone (Zone Transition)
- 100 Lower Zone
- 99 Lower Zone
- 100 Upper Zone (Zone Transition)

7.3.4.3.1.3 Transition from one zone into another generates a collection event, as might be reported via S6F11. The host has the option of receiving notification by enabling event reporting for the event. For each variable that has monitoring capability, one CEID is reserved to indicate zone transitions for that variable. To aid in the determination of the nature of a transition event, three DVs have been defined:

- *LimitVariable* — The VID of the monitored variable to which the collection event refers.
- *EventLimit* — Contains the LIMITID of the limit reached or crossed by LimitVariable.
- *TransitionType* — Defines the direction of the zone transition which has occurred: 0 = transition from lower to upper zone, 1 = transition from upper to lower zone.

7.3.4.3.1.4 Sampling frequency is an important element of limits monitoring and should be considered during equipment specification. If changes in variable value are relatively fast compared to sampling frequency, it is possible for some zone transitions to be missed or for multiple zone transitions to occur between readings. Since it is possible for zone transitions to occur 'simultaneously' or for limits to be identically defined, the DV EventLimit has been defined to allow for a list of multiple zone transitions of a variable to be reported with a single collection event.

7.3.4.3.1.5 It also should be emphasized that a single CEID is used to report transitions in both directions across a limit. Thus, reporting for one direction but not the other cannot be configured.

7.3.4.3.1.6 The functionality of each limit for each variable can be described with the state model shown in Figure 9. Below, the three states are described more fully, followed by a table defining the transitions.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

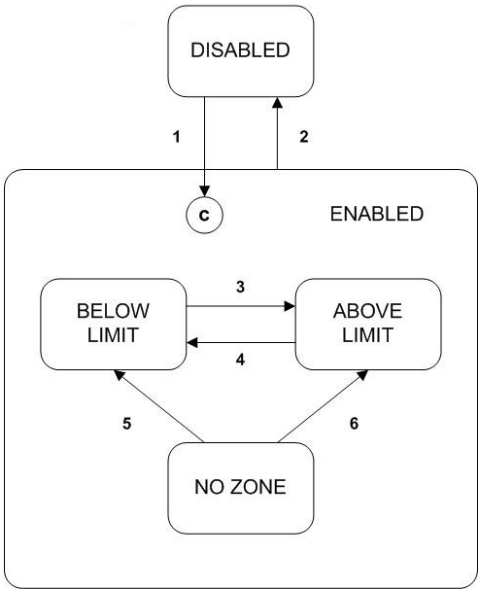

Figure 9
Limit State Model Diagram

7.3.4.3.1.7 *ABOVE LIMIT*

7.3.4.3.1.7.1 A variable is considered to be above a limit when its value increases to equal or exceed the upper boundary of the deadband, UPPERDB. The significance attached to this state is a function of the host's usage.

7.3.4.3.1.8 *BELOW LIMIT*

7.3.4.3.1.8.1 A variable is considered to be below a limit when its value decreases to equal or fall below the lower boundary of the deadband, LOWERDB. The significance attached to this state is a function of the host's usage.

7.3.4.3.1.9 *NO ZONE*

7.3.4.3.1.9.1 In some circumstances it is possible for the variable value to be in neither the upper zone nor the lower zone. This may occur upon definition of a new limit or upon equipment startup when the value of the variable lies in the deadband. In this case, the active state of the limit is considered to be NO ZONE. The limit shall remain in this state until the variable value reaches either boundary of the deadband.

7.3.4.3.2 *Modification of Limit Values* — Values for the monitoring limits on any monitored variable may be modified by the host using the message transaction S2F45/F46 (Define Variable Limit Attributes). The equipment shall reject any S2F45 message which contains limit information which conflicts with the following rules:

- LIMITMAX ≥ UPPERDB ≥ LOWERDB ≥ LIMIT-MIN
- If either UPPERDB or LOWERDB is defined, both shall be defined; if either UPPERDB or LOWERDB is undefined, both shall be undefined.

7.3.4.3.2.1 The first rule is defined and graphically depicted in Figure 8. The second rule refers to the host's ability to turn any limit 'on' or 'off'. While a minimum of seven limits shall be available for each monitored variable, it will be common for the host application to require less than seven or even none of the limits be used. The limits not needed can be disabled by leaving the values for UPPERDB and LOWERDB 'undefined'. Limits may be disabled for a VID or for all monitored VIDs by using zero-length lists in the S2F45 message.

7.3.4.3.2.2 All monitored variables shall be one of three types: integer, floating point, or Boolean. This may be accomplished by using the following formats: 11, 20, 3(), 4(), 5(). The binary format is not allowed. If the ASCII format is used, the equipment shall perform a conversion into one of the numeric types before performing any value comparisons, both for limit validations and zone transitions.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.4.3.3 *Limit Values Request* — The host may request the current limit values for a specified VID using the message transaction S2F47/F48 (Variable Limit Attribute Request).

7.3.4.4 *Requirements*

- A minimum of seven limits per monitored variable shall be available.
- One CEID per monitored variable shall be supplied for zone transition reporting.
- All limit definitions shall be kept in nonvolatile storage.
- The equipment shall enforce the limit validation rules defined above.
- The specification and documentation of which variables may be monitored with this capability is the responsibility of the equipment manufacturer based on the specific instance of equipment. This subject also may be addressed by equipment models of classes of semiconductor equipment.

Table 4 Limit State Transition Table

#	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	DISABLED	Limit attributed defined w/S2F45.	ENABLED	None	The substate of ENABLED is determined by the current value of the monitored variable.
2	ENABLED	Limit attributes set to undefined w/S2F45.	DISABLED	None	None.
3	BELOW LIMIT	Variable Increased to be ≥UPPERDB	ABOVE LIMIT	None	Zone Transition.
4	ABOVE LIMIT	Variable decreases to be ≤LOWERDB	BELOW LIMIT	None	Zone Transition.
5	NO ZONE	Variable decreases to be ≤LOWERDB	BELOW LIMIT	None	Zone Transition.
6	NO ZONE	Variable increases to be ≥UPPERDB	ABOVE LIMIT	None	Zone Transition.

7.3.4.5 *Scenarios*

7.3.4.5.1 Zone Transition Event Occurs in Equipment:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

                                                   <--S6F11v                Send enabled event
                                                                            report to host

Acknowledge                           S6F12-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.4.5.2 Host Defines Limit Attributes:

```
COMMENT                    HOST                EQUIPMENT                COMMENT
[IF] S2F45 is Multi-
block                            S2F39-->

                                          <--S2F40                Multi-block grant.

[END IF]

Host defines new                 S2F45-->
variable limit
attributes.

                                          <--S2F46                Equipment acknowledges
                                                                  host request.
```

7.3.4.5.3 Host Queries Equipment for Current Limits:

```
COMMENT                    HOST                EQUIPMENT                COMMENT
Host queries equipment           S2F47-->
for current variable
limit attribute
definitions.

                                          <--S2F48                Equipment returns
                                                                  report containing
                                                                  requested variable
                                                                  limit attribute values.
```

7.3.5 *Status Data Collection*

7.3.5.1 *Purpose* — This capability allows the host to query the equipment for selected status information and is useful in synchronizing with equipment status.

7.3.5.2 *Definitions*

7.3.5.2.1 *Status variable value (SV)* — A data item containing the value of a status variable. Refer to SEMI E5 for a full definition of this data item.

7.3.5.2.2 *Status variable ID (SVID)* — A unique identifier of a status variable. Refer to SEMI E5 for a full definition of this data item.

7.3.5.3 *Detailed Description* — The host may query equipment status by specifying the desired SVIDs. Upon such a request, the equipment sends the host the value of the selected status variables. The host also may request the description (name and units) of any or all available status variables.

7.3.5.4 *Requirements*

- The equipment manufacturer shall provide unique SVIDs for the various status variables (SV) available for data collection in the equipment.
- All status data is available for status data collection. Refer to § 8.4, Variable Item List, for a list of status variables.
- All SVs shall contain valid data whenever provided to the host.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.5.5 *Scenarios*

7.3.5.5.1 Request Equipment Status Report:

```
COMMENT                    HOST                EQUIPMENT                COMMENT

Host requests report of          S1F3-->
selected status
variable values

                                            <--S1F4                 Equipment responds with
                                                                    the requested status
                                                                    variable data.
```

7.3.5.5.2 Request Equipment Status Variable Name list:

```
COMMENT                    HOST                EQUIPMENT                COMMENT

Host requests equipment         S1F11-->
to identify selected
status variables.

                                            <--S1F12                Equipment responds with
                                                                    the requested status
                                                                    variable descriptions.
```

7.3.6 *On-Line Identification*

7.3.6.1 *Purpose* — Implementation of SEMI E5 (a GEM Fundamental Requirement) requires the equipment to accept the S1F1 from the Host at any time while it is ONLINE and COMMUNICATING, and respond with S1F2. The On-line Identification capability describes the host-initiated scenario. The equipment-initiated scenario is used for a different purpose and is defined in § 6.5 and § 7.13 describing the GEM 'Control Model'.

7.3.6.2 *Definitions*

7.3.6.2.1 *Equipment Model Type (MDLN)* — ASCII string containing the equipment model. Refer to SEMI E5 for a full definition of this data item.

7.3.6.2.2 *Equipment Software Revision Code (SOFTREV)* — ASCII string containing the equipment software revision. Refer to SEMI E5 for a full definition of this data item.

7.3.6.3 *Detailed Description* — On-line Identification allows the host to verify the presence and identity of the equipment.

7.3.6.4 *Requirements* — (Host-Initiated) an S1F2 response from the equipment shall provide MDLN and SOFTREV information which reflects the hardware and software configuration of the equipment.

7.3.6.4.1 SOFTREV shall uniquely identify different releases of equipment software. Any change in equipment software shall result in a corresponding change to SOFTREV.

7.3.6.4.2 The equipment-initiated S1F1 is not required except as described in § 6.5 and § 7.13 describing the GEM 'Control Model'.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.3.6.5 *Scenario*

7.3.6.5.1 Host Initiated:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

Are you there?                          S1F1-->

                                                  <--S1F2                 Equipment replies with
                                                                          MDLN and SOFTREV.
```

7.3.7 *Unique Identification*

7.3.7.1 *Purpose* — Unique Identification allows each specific equipment to be uniquely identified to the host. Additionally, it allows the host or operator to assign an ID to the equipment.

7.3.7.1.1 Unique Identification allows for different equipment identification than the GEM feature On-Line Identification which only identifies an equipment's model number (MDLN) and software revision (SOFTREV). Using On-Line Identification only, two equipment of the same model number and software revision cannot be easily distinguished.

7.3.7.2 *Requirements*

7.3.7.2.1 The equipment shall provide a status variable called 'EqpSerialNum' as defined in SEMI E5.

7.3.7.2.2 The value of status variable EqpSerialNum shall be persisted for the lifetime of the specific equipment.

7.3.7.2.3 The equipment shall provide a status variable called 'E30EquipmentSupplier' with SECS-II format 20 and a maximum length of 128 bytes.

7.3.7.2.4 The value of status variable E30EquipmentSupplier shall identify the supplier of the equipment. This specification does not define any requirements related to subsequent changes to the supplier's identification such as acquisition or company name changes.

7.3.7.2.5 Each equipment from a supplier is expected to have a unique serial number. The name of each equipment supplier is expected to be unique. Therefore, the combination of status variables E30EquipmentSupplier and EqpSerialNum should establish an equipment instances's unique identity.

7.3.7.2.6 The equipment shall include an equipment constant called 'EqpName' as defined in SEMI E5.

7.3.7.2.7 The value of equipment constant EqpName can be specified by the host or operator to identify the equipment. As examples, the value might reflect the location of the equipment, the order of an equipment in a line, a nickname, or the function of the equipment.

7.4 *Alarm Management* — The alarm management capability provides for host notification and management of alarm conditions occurring on the equipment.

7.4.1 *Purpose* — Historically, a precise definition of an equipment alarm has been absent. Consequently, differing interpretations have resulted in inconsistent implementations. This is addressed by providing a more rigorous definition (refer to definition in § 7.4.2 below) of an alarm.

7.4.1.1 In addition, it is often important for equipment to report more extensive information to the host than has been available in the S5F1/F2 (Alarm Report Send/Acknowledge) transaction. The data required in such cases is very dependent on equipment type, host information requirements, and alarm situation. This issue is addressed by providing event reporting methods that are tied to alarm state changes.

- Lastly, the alarm management capability provides mechanisms for:
- Reporting the time of an alarm state change,
- Uploading a list of alarm texts,
- Enabling and disabling the notification of specific alarms, and
- Host query of alarms set and enabled status on the equipment.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.4.2 *Definitions*

7.4.2.1 *Alarm* — An alarm is related to any abnormal situation on the equipment that may endanger people, equipment, or material being processed. Such abnormal situations are defined by the equipment manufacturer based on physical safety limitations. Equipment activities potentially impacted by the presence of an alarm shall be inhibited.

7.4.2.1.1 Note that exceeding control limits associated with process tolerance does not constitute an alarm nor do normal equipment events such as the start or completion of processing.

7.4.2.2 *AlarmsEnabled* — Status variable consisting of a list of the alarm IDs currently enabled for reporting to the host. See SEMI E5 for a full definition of this variable data item.

7.4.2.3 *AlarmsSet* — Status variable consisting of a list of the alarm IDs currently in the ALARM SET (or unsafe) state. See SEMI E5 for a full definition of this variable data item.

7.4.2.4 *ALCD* — Alarm code data item used in the S5F1 (Alarm Report Send) and S5F6 (List Alarm Data) messages. This code is divided into two parts, the alarm set/cleared bit and the 7 bit alarm category code. Only the set/cleared bit is used—bit 8 = 1 means alarm set, = 0 means alarm cleared. The alarm category code is not used. Refer to SEMI E5 for a full definition of this data item.

7.4.2.5 *ALID* — Alarm identifier. Refer to SEMI E5 for a full definition of this data item.

7.4.2.6 *ALTX* — Data item contained in the S5F1 and S5F6 messages containing a brief textual description of an alarm. Refer to SEMI E5 for a full definition of this data item.

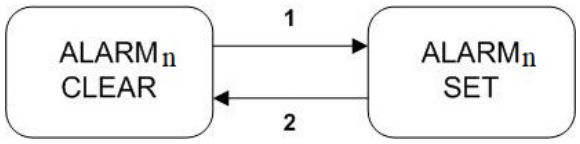

Figure 10
Alarm State Model Diagram for Alarm ALIDn

7.4.3 *Detailed Description* — Two alarm notification mechanisms are defined to achieve the flexibility necessary for the reporting required by host systems. First, stream 5 alarm reporting enables a brief, yet fixed, method for notification of alarm occurrences using the S5F1/F2 transaction. Second, to address the host's potential need for more extensive and flexible data reporting, two collection events ('alarm set' and 'alarm cleared') are defined for each possible alarm condition on the equipment to allow the use of event data collection mechanisms. In the latter case, reports are sent by the equipment using the Event Report/Acknowledge transaction (refer to § 7.3 on event data collection).

7.4.3.1 The $alarm_n$ detected and cleared events are derived from the state model for an alarm (refer to Figure 10 and Table 5). In this model each of n alarms can be in one of two possible states, either ALARM CLEAR or ALARM SET. The transition from the ALARM CLEAR to the ALARM SET state is defined as the collection event '$Alarm_n$ detected' (transition 1). Conversely, the transition from ALARM SET to ALARM CLEAR is defined as the collection event '$Alarm_n$ cleared' (transition 2).

NOTE 6: The alarm capability is intended as an addition to standard safety alarms (e.g., lights, horns). There is no intent to replace direct operator notification of such problems, nor is there the expectation that the host can prevent or directly address such alarms.

Table 5 Alarm State Transition Table

#	*Current*	*Trigger*	*New State*	*Action*	*Comment*
1	$ALARM_n$ CLEAR	$Alarm_n$ is detected on the equipment.	$ALARM_n$ SET	Initiate local actions (if any) to ensure safety. Update AlarmsSet and $ALCD_n$ values. Generate and issue alarm message if enabled.	Inhibited activities require operator or host intervention prior to resuming.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

#	*Current*	*Trigger*	*New State*	*Action*	*Comment*
2	$ALARM_n$ SET	$Alarm_n$ is no longer detected on the equipment.	$ALARM_n$ CLEAR	Update AlarmsSet and $ALCD_n$ values. Generate and issue alarm message if enabled.	Inhibited activities require operator or host intervention prior to resuming.

7.4.3.2 The equipment manufacturer is responsible for identifying all alarms on their equipment by:

- Applying the above alarm definition,
- Consulting Related Information 1, § R1-3 for examples of alarms for various equipment configurations,
- Noting that the presence of an alarm typically requires some action or intervention before resuming safe operation of the equipment, and by
- Referring to Table 6 below which delineates the differences between collection events and alarms.

Table 6 Collection Event and Alarm Comparison

COLLECTION EVENT	*ALARM*
Any occurrence detectable by the equipment.	Related to only those occurrences that are abnormal, undesirable, AND endanger people, equipment, or physical material being processed.
Certain collection events may trigger a state transition(s).	Each alarm has an associated two-state state model: ALARM SET (or unsafe) and ALARM CLEAR (or safe).
Equipment activities are not necessarily inhibited by the occurrence of a collection event (unless it is associated with an alarm or intentional inhibit).	The presence of an alarm inhibits equipment activities to ensure safe operation until the alarm condition is cleared.
Certain collection events may occur in an expected sequence.	Alarms may occur at any time.

7.4.3.3 *Enable/Disable Alarms* — Upon request from the host, the equipment shall enable or disable reporting of certain alarms. Enabling or disabling a given alarm shall impact the communication of both the alarm set and clear messages equally (i.e., turn them both on or both off). This is not the case for enabling/disabling of the associated collection events, where the alarm-set and alarm-cleared events can be enabled and disabled separately. The current enable/disable settings shall be stored in nonvolatile memory. Changes to the enable/disable settings shall be reflected in the AlarmsEnabled status variable.

NOTE 7: The alarm itself is not being enabled or disabled, but the reporting of the alarm through SECS-II messages is being enabled or disabled.

7.4.3.4 *Send Alarm Report* — Upon detecting a change in the status of a given alarm, an associated alarm state model shall transition to the opposite state. Following initiation of local actions necessary to ensure safety, the equipment shall update the AlarmsSet and associated ALCD values and send an alarm message and/or an event message to the host assuming one or both are enabled. Alarm messages shall be sent before the corresponding event messages if both are enabled.

NOTE 8: The ALCD is divided into two parts, the alarm set/cleared bit and the 7 bit alarm category code. Only the set/cleared bit is used—bit 8 = 1 means alarm set, = 0 means alarm cleared. The alarm category code is not used.

7.4.3.5 *List Alarm Text* — Upon request from the host, the equipment sends values of alarm text associated with a specified list of alarm IDs using the S5F5/F6 (List Alarms Request/Data) transaction.

7.4.3.6 *List Currently Set Alarms* — The host may obtain a listing of currently set alarms by employing any data collection method specified by GEM and referencing the variable data item called ‘AlarmsSet’ (e.g., including " AlarmsSet in a report definition). When reported, AlarmsSet shall contain a list of all currently set alarms. Each alarm set or clear occurrence shall cause a change to the AlarmsSet status variable prior to reporting it to the host.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.4.3.7 *List Currently Enabled Alarms* — The host may obtain a listing of currently enabled alarms by employing any data collection method specified by GEM and referencing the variable data item called 'AlarmsEnabled' (e.g., including AlarmsEnabled in a report definition). When reported, AlarmsEnabled shall contain a list of all alarm ID's currently enabled for reporting. The equipment shall update the value of AlarmsEnabled upon corresponding change(s) to the enable/disable settings. The host can also query the list of enabled alarms using message S5F7.

7.4.4 *Requirements*

- A set of alarms relating to the physical safety limits of operator, equipment, or material being processed shall be defined for the equipment by the equipment manufacturer.
- The equipment shall maintain all enable/disable states and report definitions for alarms and collection events in nonvolatile memory.
- Each alarm defined on the equipment shall have a brief description of its meaning, an associated unique alarm identifier (ALID), alarm text (ALTX), an alarm status (ALCD) and two unique collection event identifiers (CEIDs), one for set and one for cleared.
- Enabled alarm reports shall be sent prior to corresponding enabled event reports.
- The list of enabled alarms reported using SV AlarmsEnabled shall match the list of enabled alarms reported in message S5F8 when no change(s) are made to the enable/disable settings between reports.

7.4.5 *Scenarios*

NOTE 9: Consult event reporting sections of this Standard for descriptions of enabling, disabling, and sending collection event reports.

7.4.5.1 Enable/Disable Alarms:

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Enable/disable Alarm              S5F3-->
                                          <--S5F4                Acknowledge
```

7.4.5.2 Upload Alarm Information:

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Request alarm data/text           S5F5-->
                                          <--S5F6                Send alarm data/text
```

7.4.5.3 Send Alarm Report:

```
COMMENT                    HOST                EQUIPMENT              COMMENT
                                                                     Alarm occurrence
                                                                     detected by the
                                                                     equipment.
                                          <--S5F1                Send alarm report (if
                                                                     enabled).
Acknowledge                       S5F2-->
                                          <--S6F11               Send event report (if
                                                                     enabled).
Acknowledge                      S6F12-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.4.5.4 Upload List of Enabled Alarms:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

Request list of alarms                  S5F7-->
enabled for reporting

                                                  <--S5F8                 Send list of alarms
                                                                          enabled for reporting.
```

7.5 *Remote Control*

7.5.1 *Purpose* — This capability provides the host with a level of control over equipment operations.

7.5.2 *Definitions*

7.5.2.1 *Host Command Parameter (CPNAME/CPVAL/CEPVAL)* — A parameter name/value associated with a particular host command (S2F41/S2F49). The equipment manufacturer shall provide unique names (CPNAMEs) for any supported command parameters. Command parameters are not specified in this Standard but are left to equipment manufacturers to define. Equipment models of specific classes of semiconductor equipment also may address this issue. Note that if there are no associated parameters a zero-length list is sent. The data item CEPVAL, which can be defined as a list, allows grouping of related parameters within a main parameter. If the CEPVAL is defined as a single (non-list) item, then it is the equivalent of a CPVAL.

7.5.2.1.1 The uses of OBJSPEC in the header structure of the S2F49 Enhanced Remote Command allows the equipment supplier to define a set of unique identifiers for different objects within the equipment such as: equipment subsystems, subsystem components, processing stations, ports, and exchange stations.

7.5.3 *Description* — The equipment responds to host commands that provide the following functions relative to individual equipment implementations:

- Start processing
- Select a process program
- Stop processing
- Temporarily suspend processing
- Resume processing
- Abort processing

7.5.3.1 Additional commands may be implemented by the equipment manufacturer (e.g., vent chamber, clear material, open door).

7.5.3.2 Remote commands shall be interpreted as ‘request action be initiated’ rather than ‘do action’. The equipment may then respond via S2F42/S2F50 with HCACK = 4 if the command ‘is going to be performed’. This alleviates any transaction timeouts for commands that may take a long time to perform. The completion of the action initiated by the remote command (i.e., HCACK = 0 or 4) shall result in either a state transition or other action that generates a collection event upon normal/abnormal completion.

7.5.3.3 The format for all remote commands is ASCII, with a maximum length of 20 characters. The character set is restricted to the printable characters (hexadecimal 21–7E). Note that spaces are not allowed.

7.5.3.4 The following remote commands (RCMDs), if implemented on the equipment, shall be supported as described below (refer to § 6.6 for a description of Equipment Processing States).

NOTE 10: The terms ‘current cycle’ and ‘safe break point’ used below are to be defined by the supplier or within the models of classes of semiconductor equipment.

7.5.3.5 *START* — This command is available to the host when a process program has been selected and the equipment is in the ‘ready’ processing state. The START command instructs the equipment to initiate processing. Variable parameter settings may be included as name/value command parameters CPNAME/CPVAL/CEPVAL.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.5.3.6 *PP-SELECT* — This command instructs the equipment to make the requested process program (s) available in the execution area. The process programs (PPIDs) are specified via the command parameter list. A status variable (PPExecName) contains the PPID of the process program(s) currently selected. PP-SELECT and PPExecName shall be implemented if process program is the Process Program type implemented.

7.5.3.7 *STOP* — Command to complete the current cycle, stop in a safe condition and return to the 'idle' processing state. Stop has the intent of stopping the process. The equipment is not required to support the continuation of processing. Stop leaves material either fully processed or partially processed so that the processing can be later completed. For example, for a single wafer process tool, five wafers have been processed while the remaining wafers remain unprocessed.

7.5.3.8 *PAUSE* — Command to suspend processing temporarily at the next safe break point. Pause has the intent of resuming the process at the same point where it was paused. The process may be RESUMED, STOPPED, or ABORTED while in a PAUSED condition. RESUME shall be able to continue the process from the same point where it was paused.

7.5.3.9 *RESUME* — Command to resume processing from the point where the process was paused.

7.5.3.10 *ABORT* — Command to halt the current cycle prior to its completion. Abort has the intent of immediately stopping the process and is used because of abnormal conditions. Abort makes no guarantee about the subsequent condition of material. In the above example, the wafers being processed at the time of the abort may not be completely processed. Other AbortLevels >1 may be defined by the manufacturer or addressed by models of specific classes of semiconductor equipment.

7.5.3.11 CPNAME = AbortLevel, CPVAL = 1 means halt the current cycle at the next 'safe break point', retrieve all material, stop in a safe condition and return to the idle state in the processing state machine.

7.5.4 *Requirements*

- The following Remote Commands, as defined under Descriptions, shall be implemented on equipment to satisfy minimum requirements for this capability:
  - START
  - STOP
- The RCMD value for all commands supported on the equipment shall be recognized if sent with all uppercase characters (e.g., 'STOP', 'START', 'PP-SELECT', 'PAUSE', etc.). In addition to accepting strings with all uppercase characters, the equipment can optionally accept strings with all lowercase characters or mixed-case strings. The equipment documentation should describe whether or not the optional lowercase or mixed-case strings are supported.
- Stream 2 currently provides for Host Command Send and Enhanced Remote Command. The equipment shall support one or both methods, based on appropriateness.
- The Enhanced Remote Command is used to address size, complexity, or the need to target a specific subsystem within the equipment, (i.e., processing station, port, exchange station, material handler, chamber).

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.5.5 *Scenarios*

7.5.5.1 Host Command Send Scenario:

```
COMMENT                    HOST                EQUIPMENT              COMMENT

Host Command Send                S2F41-->

                                         <--S2F42               Host Command
                                                                Acknowledge

                                                                [IF] Command Accepted
                                                                (HCACK = 0 or 4)

                                         <--S6F11               [THEN] Event Report-
                                                                state change or other
                                                                collection event
                                                                occurrence.

Event Report                     S6F12-->
Acknowledge
```

7.5.5.2 Enhanced Remote Command Scenario:

```
COMMENT                    HOST                EQUIPMENT              COMMENT

Enhanced Remote Command          S2F49-->

                                         <--S2F50               EnhancedRemote Command
                                                                Acknowledge

                                                                [IF] Command Accepted
                                                                (HCACK = 0 or 4)

                                         <--S6F11               [THEN] Event Report-
                                                                state change or other
                                                                collection event
                                                                occurrence.

Event Report                     S6F12-->
Acknowledge
```

7.6 *Equipment Constants*

7.6.1 *Purpose* — This capability provides a method for the host to read and to change the value of selected equipment constants on the equipment.

7.6.2 *Definitions* — None.

7.6.3 *Description* — This capability allows the host to reconfigure equipment constants to support a variety of situations. The following functions are included:

7.6.3.1 *Host Sends Equipment Constants* — Allows the host to change the value of one or more equipment constants.

7.6.3.2 *Host Equipment Constant Request* — Allows the host to determine the current value of equipment constants.

7.6.3.3 *Host Equipment Constant Namelist Request* — Allows the host to retrieve basic information about the equipment constants available at the equipment.

7.6.4 *Requirements*

- Equipment constants shall be stored in nonvolatile memory.
- The equipment shall be in a ‘safe’ condition to accept new constant(s) settings as defined by the equipment manufacturer.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

- The equipment shall provide a collection event to alert the host whenever an equipment constant is changed by the operator. Information indicating which constant was changed shall be available for the event report.

7.6.5 *Scenarios*

7.6.5.1 Host Sends Equipment Constants:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

Host sends equipment              S2F15-->
constants

                                          <--S2F16                EAC = 0 equipment sets
                                                                  constants
```

7.6.5.2 Host Requests Equipment Constants:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

Equipment constant                S2F13-->
request

                                          <--S2F14                Equipment constant data
```

NOTE 11: This capability also can be accomplished using S6F19 and S6F20. Refer to § 7.3.

7.6.5.3 Host Requests Equipment Constant Namelist:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

Equipment constant                S2F29-->
namelist request

                                          <--S2F30                Equipment constant
                                                                  namelist
```

7.6.5.4 Operator Changes Equipment Constant:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

                                                                  Operator changes
                                                                  equipment constant at
                                                                  equipment operator
                                                                  console.

                                          <--S6F11                Equipment reports
                                                                  equipment constant
                                                                  change.

Host acknowledges event           S6F12-->
```

7.7 *Process Program Management* — Instructions for equipment processing (e.g., process programs) shall be manageable through interaction between the equipment and the host system.

7.7.1 *Purpose* — Process Program Management provides a means to transfer equipment instructions (process programs) between the host and equipment. Additionally, it allows for shared management of these equipment instructions on the host and the equipment.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.1.1 An equipment shall comply with Process Program Management by implementing one or more of the following options:

- Stream 7 Unformatted
- Stream 21 Single Message Unformatted
- Stream 21 Multiple Message Unformatted
- Stream 7 Formatted

7.7.1.2 Each process program management option can include the following scenarios:

- Host queries the list of process programs on the equipment
- Host requests a process program from the equipment
- Host sends a process program to the equipment
- Equipment queries the list of available process programs on the host (limited to Stream 21 options)
- Equipment sends a process program to the host
- Equipment requests a process program from the host
- Host delete one or more process programs on the equipment

7.7.2 *Definitions*

7.7.2.1 *Process Program* — A process program is the pre-planned and reusable portion of the set of instructions, settings, and parameters under control of the equipment that determine the processing environment seen by the manufactured object and that may be subject to change between runs or processing cycles.

7.7.2.2 *Process Program Identifier* — A text string (PPID) used to identify a process program.

7.7.2.3 *Formatted Process Program* — A process program that is presented as an ordered sequence of process commands with their associated parameters as dictated by S7F23 and S7F26. The process commands and associated parameters are listed as separate data items in the SECS-II message when the process program is transferred. Where formatted process programs are supported, equipment shall also provide sufficient information to allow a user at the host to create, display, modify, and partially verify their contents.

7.7.2.4 *Unformatted Process Program* — An unformatted process program is considered a single block of ASCII or binary data when considered for transfer. The process program might be transferred as a single data item in a SECS-II message such as when using Stream 7 Unformatted and when the process program is 16777215 bytes or less and using Stream 21. Alternatively, the process program may be split into multiple data items when larger than 16777215 bytes and using Stream 21. The structure of the ASCII or binary data is determined by the equipment supplier. It is referred to as 'unformatted' because it does not use a SECS-II formatted structure when transferred like Formatted Process Programs. The term 'unformatted' is used to emphasize how the process program is represented by a single block of text or binary data. Although referred to as 'Unformatted', an unformatted process program shall conform to a structure defined by the equipment supplier.

7.7.2.5 *Process Program Change Event* — The collection event associated with the occurrence of the creation, modification, or deletion of a process program by the operator.

7.7.2.6 *PPChangeName* — A data variable containing the PPID of the process program affected by the event Process Program Change Event. This data variable is valid for the collection event Process Program Change Event. Refer to SEMI E5 for a full definition of this variable data item.

7.7.2.7 *PPChangeStatus* — A data variable that identifies the action taken on the process program named by PPChangeName. This data variable is valid for the collection event Process Program Change Event. Refer to SEMI E5 for a definition of this data variable.

7.7.2.8 *PPError* — A data variable to report error information about a process program that failed verification or t validation. Refer to SEMI E5 for a definition of this data variable.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.2.9 *PPErrorPPID* — A data variable with format code 20 to identify a process program when reporting a process program verification failure or a validation error.

7.7.2.10 *PPExecName* — The status variable containing the PPID (s) of the currently selected process program(s). Refer to SEMI E5 for a full definition of this variable data item.

7.7.2.11 *PP-SELECT* — The remote command used to select one or more process programs for execution. The process programs are specified by PPID via the command parameter list.

7.7.2.12 *Process Program Verification* — Verification is syntax checking of a process program. Verification ensures only that a process program is structured correctly. It does not ensure that the program has the correct parameters to run a particular process or product.

7.7.2.13 *Process Program Verification Failed Event* — The collection event to report when process program verification failure detected outside of scenarios already associated with S7F27.

7.7.2.14 *Process Program Validation* — Validation is type-and-range checking of parameters in a process program and is performed after verification.

7.7.2.15 *Process Program Validation Failed Event* — The collection event to report when process program validation failure detected.

7.7.3 *Process Program Description*

7.7.3.1 Process programs allow the equipment's process, and/or the parameters used by that process, to be set and modified by the engineer to achieve different results. Different process programs may be required for different products, while often the same process program will be used for all lots of a given product. Through the equipment's operator console, the engineer shall be able to create such process programs, to modify current process programs, and to delete process programs from equipment storage.

7.7.3.2 For the host to ensure that the proper process programs are in place at the equipment, process program management defines a means of transferring them from equipment to host and from host to equipment. The host also may need to delete process programs from the equipment's storage to make room for a new process program.

7.7.3.3 Process program management also allows the host to be notified when a local process program on the equipment is created, edited, or deleted.

7.7.3.4 Both formatted and unformatted process programs may be transferred between the host and equipment. This capability provides for both host-initiated and equipment-initiated transfers. The equipment-initiated transfer may be used at the request of the process engineer or operator at the equipment.

7.7.4 *Requirements*

7.7.4.1 The equipment manufacturer shall provide a method to create, modify, and delete process programs. This method shall exist on either the equipment or on a separate computing system.

7.7.4.2 If the equipment manufacturer has implemented process programs, then the equipment shall define a collection event to report the creation, the deletion, or the modification (completion of an editing session) of a process program (Process Program Change Event).

7.7.4.3 Process programs on the equipment shall be stored in nonvolatile storage.

7.7.4.4 The equipment shall be able to store in nonvolatile memory the number of process programs sufficient to execute three unique process cycles. For example, if a wire-bonder requires both an 'ALIGN' process program and a 'BOND' process program for a full process cycle, then it shall provide nonvolatile storage for at least three pairs of process programs. These stored process programs may not be modified in any way by the execution process, nor may the execution process be affected by the modification of any process program in storage. This includes requesting a process program from the host, local editing or by deletion while that process program is being executed.

7.7.4.5 The equipment supplier shall document any restrictions on the length or text format of PPID. The maximum length allowed by equipment may be less than that allowed by SECS-II.

7.7.4.6 The PPID in the SECS-II message shall identify the process program in the equipment's nonvolatile storage. When transferring a process program to the equipment, if there is already a process program in the equipment's

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

nonvolatile storage with the same PPID as the PPID in the SECS-II message, then the existing process program shall be replaced. The PPID in the SECS-II message shall identify the process program in nonvolatile storage.

7.7.4.7 For scenarios that use the Process Program Load Inquire/Grant transaction (S7F1/F2), its use is recommended prior to all host-initiated transfers to the equipment. This provides a means of verifying process program size before the transfer request is approved. Note that S7F1 should not be used to select a process program.

7.7.5 *Process Program Management Options*

7.7.5.1 There are several options available to implement Process Program Management. In order to comply with Process Program Management requirements, at least one option shall be supported. However, multiple or all of the options can be supported. Each option is listed separately in the GEM Compliance Statement.

Table 7 Process Program Management Options

Process Program Management Option	*Description*
Stream 7 Unformatted	The process program structure is determined by the equipment supplier and can be transferred as text or binary data using S7F3/F4 and S7F5/F6. When using HSMS, process program content is in a single data item and limited to 16777215 bytes.
Stream 21 Single Message Unformatted	The process program structure is determined by the equipment supplier and can be transferred as text or binary data. Process program size is limited by the maximum size of the ITEMLENGTH data item. Because the process program body is sent and received in a single SECS-II message, this technique is ideal for small to medium sized process programs. Process programs in Stream 21 also include a version property. This capability is an alternative to Stream 7 Unformatted and Stream 21 Multiple Message. Using this technique and HSMS communication, process programs can be up to 4294967295 bytes.
Stream 21 Multiple Message Unformatted	The process program structure is determined by the equipment supplier and can be transferred as text or binary data. Process program size is limited by the maximum size of the ITEMLENGTH data item. Because only a part of process program body is sent and received in each SECS-II message, the technique can be used for process programs of any size and may be the only alternative for large to very large process programs. Process programs in Stream 21 also include a version property. This capability is an alternative to Stream 7 Unformatted and Stream 21 Single Message. Using this technique, process programs size is not limited by the HSMS or SECS-II standard. Additionally, by sending each process program part in separate messages, this technique allows for other messages to be interleaved while process program transfer occurs.
Stream 7 Formatted	The process program structure is organized into a series of CCODE and PPARM name/value(s) argument pairs. Formatted process programs are transferred using messages S7F23/F24 and S7F25/F26. Each PPARM value is limited to 16777215 bytes.

Table 8 SECS-II Message Summary for Each Process Program Management Option

Scenario	*Stream 7 Unformatted*	*Stream 21 Single Message Unformatted*	*Stream 21 Multiple Message Unformatted*	*Stream 7 Formatted*
Host queries the list of process programs on the equipment	S7F19/F20	S21F7/F8	S21F7/F8	S7F19/F20
Host requests a process program from the equipment	S7F5/F6	S21F5/6	S21F15/16 followed by one or more S21F17/18	S7F25/F26
Host sends a process program to the equipment	S7F1/F2 followed by S7F3/F4	S21F1/F2 followed by S21F3/F4	S21F13/14 followed by one or more S21F17/18	S7F1/F2 followed by S7F23/F24, S7F29/F30 and S7F27/F28

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Scenario	*Stream 7 Unformatted*	*Stream 21 Single Message Unformatted*	*Stream 21 Multiple Message Unformatted*	*Stream 7 Formatted*
Equipment queries the list of available process programs on the host	Not supported	S21F7/F8	S21F7/F8	Not supported
Equipment sends a process program to the host	S7F1/F2 followed by S7F3/F4	S21F1/F2 followed by S21F3/F4	S21F13/14 followed by one or more S21F17/18	S7F1/F2 followed by S7F23/F24
Equipment requests a process program from the host	S7F5/F6	S21F5/6	S21F15/16 followed by one or more S21F17/18	S7F25/F26 followed by S7F29/F30, S7F27/F28
Host deletes one or more process programs on the equipment	S7F17/F18	S21F11/F12	S21F11/F12	S7F17/F18

7.7.5.2 Additional process program implementation options from previous revisions of this Specification are included in the Appendices for backward compatibility with equipment that might have implemented an option before it was removed from the main body of this Specification. Upgraded and new implementations of this specification are recommended to restrict implementations to the current options list in Table 7, Process Program Management Options.

7.7.5.3 There shall be one process program repository on the equipment in nonvolatile storage even when multiple Process Program Management options are supported, as illustrated in the Figure 11 below. The options provide support for different messages to access to the same process programs. For example, the list of process programs returned by S7F20 would be the same as the list returned by S21F8 assuming that no process programs have been deleted or created in between the request messages. As another example, when the host sends a new process program to the equipment using S7F3, the new process program would be listed by subsequent S21F8.

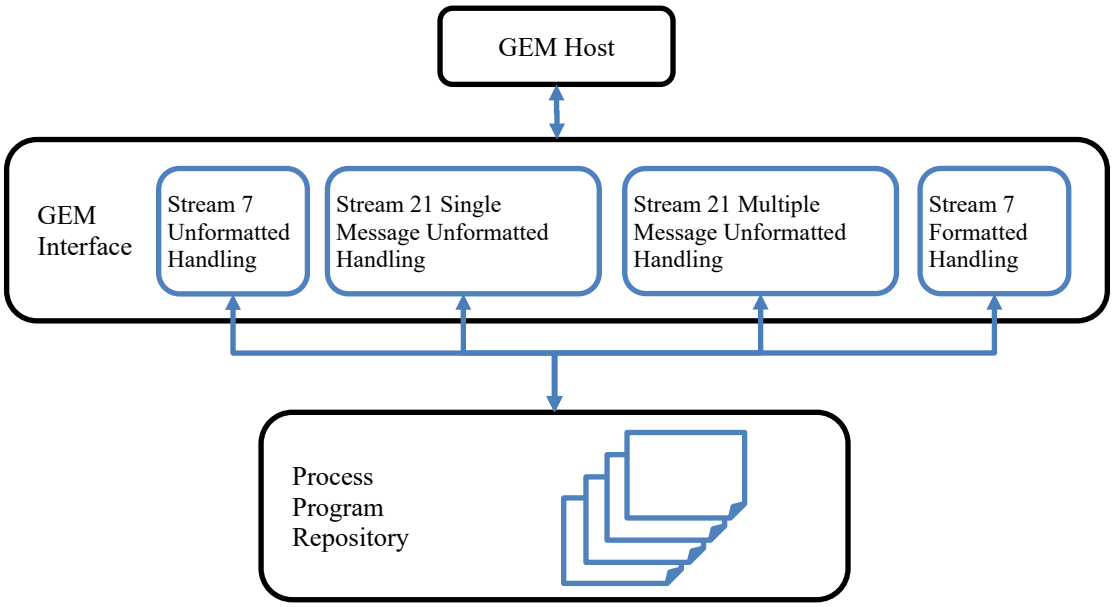

Figure 11
Single Repository of Process Programs

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.5.4 When both S7F17 and S21F11 are supported, deleting a process program with either S7F17 or S21F11 shall have the same result regardless of the message used.

7.7.5.5 Even when a process program is listed as 'available' by message S7F20 or S21F8, this does not mean that each process program shall be available through all implemented Process Program Management options. For example, a process program too large for Stream 7 transfer would be limited to Stream 21 transfer. Because S21F8 returns the size of each process program a host can predict when Stream 7 cannot be used.

7.7.5.6 When implementing the Stream 7 Formatted Process Program Management option, if a Parameter is too large to fit in a single PPARM data item (larger than 16777215 bytes), then the Parameter can be split into multiple PPARM values when transferred over S7F23 or S7F26 similar to how Stream 21 splits an item in S21F3. When this technique is used, the equipment shall describe the implementation in the GEM documentation.

7.7.6 *General Scenarios for Process Program Management*

7.7.6.1 The following scenarios apply to all implementations of Process Program Management, regardless of the Process Program Management option(s) implemented.

7.7.6.2 Process Program Created, Edited, or Deleted by Operator:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

                                                                            Process program
                                                                            created, edited, or
                                                                            deleted by operator at
                                                                            equipment.
                                                                            PPChangeName = PPID
                                                                            PPChangeStatus = 1
                                                                            (Created)
                                                                                           = 2
                                                                            (Edited)
                                                                                           = 3
                                                                            (Deleted)
                                                                            [IF] CEID for Process
                                                                            Program Change Event
                                                                            enabled
                                                <--S6F11                    [THEN]
                                                                            Send Event Report
Event Report                          S6F12-->
Acknowledge
                                                                            [END_IF]
```

7.7.6.3 *Equipment Constant E30EquipmentInitiatedPPOption*

7.7.6.3.1 The equipment shall include an equipment constant called E30EquipmentInitiatedPPOption with SECS-II format 51 and type PPOPTION.

7.7.6.3.2 The value of E30EquipmentInitiatedPPOption shall determine the process program option used when the equipment initiates sending process programs to the host and when the equipment is requesting process programs from the host.

7.7.6.3.3 If the host attempts to set the value of E30EquipmentInitiatedPPOption to a value not supported by the equipment, then the equipment shall return with an EAC value of 3 (refer to EAC data item and S2F16 in SEMI E5).

7.7.6.3.4 E30EquipmentInitiatedPPOption shall have a value of 0 (None) when the equipment does not support any option for equipment-initiated transfer of process programs.

7.7.6.4 *Status Variable E30SupportedPPOptions*

7.7.6.4.1 The equipment shall include a status variable called 'E30SupportedPPOptions' with SECS-II Format 0.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.6.4.2 Status variable E30SupportedPPOptions shall identify the process program management options supported by the equipment.

Table 9 E30SupportedPPOptions Value Structure

Value	*Description*
L, n <PPOPTION$_1$> … <PPOPTION$_n$>	List of type PPOPTION, where n is the number of supported process program management options. PPOPTION shall have SECS-II format 51. E30SupportedPPOptions shall be an empty list when no process program management option is supported.

Table 10 PPOPTION Enumeration

Value	*Description*
0	None
1	Stream 7 Unformatted
2	Stream 21Single Message Unformatted
3	Stream 21 Multiple Message Unformatted
4	Stream 7 Formatted
5	Stream 13 Large Formatted
6	Stream 13 Large Unformatted
7	SEMI E42 Recipe Management Standard (RMS)
8	SEMI E139 Recipe and Parameter Management (RaP)

7.7.6.4.3 Note that PPOPTION values 5, 6, 7, and 8 are included for backward compatibility.

7.7.7 *Scenarios for Process Program Management Option: Stream 7 Unformatted*

7.7.7.1 Host queries the list of process programs on the equipment:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT
Current EPPD Request                     S7F19-->
                                                  <--S7F20                Current EPPD Data
```

7.7.7.2 Host requests a process program from the equipment:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT
Process Program Request                  S7F5-->
                                                  <--S7F6                 Process Program Data
                                                                          Note that if the process
                                                                          program does not exist,
                                                                          a zero-length list will
                                                                          be sent.
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.7.3 Host sends a process program to the equipment:

```
COMMENT                     HOST                    EQUIPMENT                COMMENT

[IF] Process program is
multi-block
[THEN]

Process Program Load                     S7F1-->
Inquire
                                                <--S7F2                  Process Program Load
                                                                         Grant
[END_IF]
Process Program Send                     S7F3-->
                                                <--S7F4                  Process Program
                                                                         Acknowledge
```

7.7.7.4 Equipment sends a process program to the host:

```
COMMENT                     HOST                    EQUIPMENT                COMMENT

                                                                         [IF] Process program is
                                                                         multi-block
                                                                         [THEN]
                                                <--S7F1                  Process Program Load
                                                                         Inquire
Process Program Load                     S7F2-->
Grant
                                                                         [END_IF]
                                                <--S7F3                  Process Program Send
Process Program                          S7F4-->
Acknowledge
```

7.7.7.5 Equipment requests a process program from the host:

```
COMMENT                     HOST                    EQUIPMENT                COMMENT

                                                <--S7F5                  Process Program Request
Process Program Send                     S7F6-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.7.6 Host deletes one or more process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the equipment to delete one or more process programs.	S7F17-->		
			Equipment deletes the requested process program(s) from nonvolatile storage.
		<--S7F18	Equipment acknowledges the request.

7.7.8 *Scenarios for Process Program Management Option: Stream 21 Single Message Unformatted*

7.7.8.1 *Requirements for Stream 21 Single Message Unformatted*

7.7.8.1.1 The value of data item ITEMTYPE shall be ‘SEMI:UNFORMATTEDPROCESSPROGRAM’ when using Stream 21 Single Message Unformatted.

7.7.8.2 Host queries the list of process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the current list of available process programs.	S21F7-->		
		<--S21F8	Equipment returns the current list of available process programs.

7.7.8.3 Host requests a process program from the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests process program data	S21F5-->		
		<--S21F6	Equipment returns the process program data

7.7.8.4 Host sends a process program to the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests permission to send the process program.	S21F1-->		
		<--S21F2	Equipment grants permission.
Host sends the process program.	S21F3-->		
		<--S21F4	Equipment acknowledges receipt of the process program.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.8.5 Equipment sends a process program to the host:

```
COMMENT                     HOST                EQUIPMENT              COMMENT
                                                <--S21F1               Equipment requests
                                                                       permission to send the
                                                                       process program.
Host grants permission.           S21F2-->
                                                <--S21F3               Equipment sends the
                                                                       process program.
Host acknowledges                 S21F4-->
receipt of the process
program.
```

7.7.8.6 Equipment queries the list of available process programs on the host:

```
COMMENT                     HOST                EQUIPMENT              COMMENT
                                                <--S21F7               Equipment requests the
                                                                       current list of
                                                                       available process
                                                                       programs.
Host returns the list             S21F8-->
of process programs
currently available for
this Equipment.
```

7.7.8.7 Equipment requests a process program from the host:

```
COMMENT                     HOST                EQUIPMENT              COMMENT
                                                <--S21F5               Equipment requests
                                                                       process program.
Host returns the                  S21F6-->
process program.
```

7.7.8.8 Host deletes one or more process programs on the equipment:

```
COMMENT                     HOST                EQUIPMENT              COMMENT
Host requests the                S21F11-->
equipment to delete one
or more process
programs.
                                                                       Equipment deletes the
                                                                       requested process
                                                                       program(s) from
                                                                       nonvolatile storage.
                                                <--S21F12              Equipment acknowledges
                                                                       the request.
```

7.7.9 *Scenarios for Process Program Management Option: Stream 21 Multiple Message Unformatted*

7.7.9.1 *Requirements for Stream 21 Multiple Message Unformatted*

7.7.9.1.1 The value of data item ITEMTYPE shall be ‘SEMI:UNFORMATTEDPROCESSPROGRAM’ when using Stream 21 Multiple Message Unformatted.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.9.2 Host queries the list of process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the current list of available process programs.	S21F7-->		
		<--S21F8	Equipment returns the current list of available process programs.

7.7.9.3 Host requests a process program from the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the process program.	S21F15-->		
		<--S21F16	Equipment accepts the request.
			[START]
			Repeat until all item parts are sent or the transfer is aborted.
		<--S21F17	Equipment sends one item part.
Host acknowledges receipt of an item part.	S21F18-->		
			[END]
		<--S21F17	Equipment sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.
Host acknowledges receipt of the final message in the sequence.	S21F18-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.9.4 Host sends process program to the equipment:

COMMENT	HOST		EQUIPMENT	COMMENT
Host requests permission to send the process program.		S21F13-->		
			<--S21F14	Equipment grants permission.
[START]				
Repeat until all item parts are sent.				
Host sends one item part.		S21F17-->		
			<--S21F18	Equipment acknowledges receipt of an item part.
[END]				
Host sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.		S21F17-->		
			<--S21F18	Equipment acknowledges receipt of the final message in the sequence.

7.7.9.5 Equipment queries the list of available process programs on the host:

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S21F7	Equipment requests the current list of available process programs.
Host returns the list of process programs currently available for this Equipment.		S21F8-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.9.6 Equipment sends a process program to the host:

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S21F13	Equipment requests permission to send the process program.
Host grants permission.		S21F14-->		
				[START] Repeat until all item parts are sent.
			<--S21F17	Equipment sends one item part.
Host acknowledges receipt of an item part.		S21F18-->		
				[END]
			<--S21F17	Equipment sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.
Host acknowledges receipt of the final message in the sequence.		S21F18-->		

7.7.9.7 Equipment requests a process program from the host:

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S21F15	Equipment requests the process program.
Host accepts the request.		S21F16-->		
[START] Repeat until all item parts are sent.				
Host sends one item part.		S21F17-->		
			<--S21F18	Equipment acknowledges receipt of an item part.
[END]				
Host sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.		S21F17-->		
			<--S21F18	Equipment acknowledges receipt of the final message in the sequence.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.9.8 Host deletes one or more process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the equipment to delete one or more process programs.	S21F11-->		
			Equipment deletes the requested process program(s) from nonvolatile storage.
		<--S21F12	Equipment acknowledges the request.

7.7.10 *Scenarios for Process Program Management Option: Stream 7 Formatted*

7.7.10.1 *Requirements for Stream 7 Formatted*

7.7.10.1.1 Formatted process programs shall be verified immediately following any transfer to the equipment.

7.7.10.2 Host queries the list of process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Current EPPD Request	S7F19-->		
		<--S7F20	Current EPPD Data

7.7.10.3 Host requests a process program from the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Formatted Process Program Request	S7F25-->		
		<--S7F26	Formatted Process Program Data

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.10.4 Host sends process program to the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
[IF] Process program is multi-block			
[THEN]			
Process Program Load Inquire	S7F1-->		
		<--S7F2	Process Program Load Grant
[ENDIF]			
Formatted Process Program Send	S7F23-->		
		<--S7F24	Formatted Process Program Acknowledge Verify process program
			[IF] S7F27 is multi-block
			[THEN]
		<--S7F29	Process Program Verification Inquire
Process Program Verification Grant	S7F30-->		
			[END IF]
		<--S7F27	Process Program Verification Send
Process Program Verification Acknowledge	S7F28-->		

7.7.10.5 Equipment sends a process program to the host:

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF] Process program is multi-block
			[THEN]
		<--S7F1	Process Program Load Inquire
Process Program Load Grant	S7F2-->		
			[END_IF]
		<--S7F23	Formatted Process Program Send
Formatted Process Program Acknowledge	S7F24-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.10.6 Equipment requests a process program from the host:

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S7F25	Formatted Process Program Request
Formatted Process Program Data		S7F26-->		
				[IF] S7F27 is multi-block
				[THEN]
			<--S7F29	Process Program Verification Inquire
Process Program Verification Grant		S7F30-->		
				[END IF]
			<--S7F27	Process Program Verification Send
Process Program Verification Acknowledge		S7F28-->		

7.7.10.7 Host deletes one or more process programs on the equipment:

COMMENT	HOST		EQUIPMENT	COMMENT
Host requests the equipment to delete one or more process programs.		S7F17-->		
				Equipment deletes the requested process program(s) from nonvolatile storage.
			<--S7F18	Equipment acknowledges the request.

7.7.11 *Scenarios for Process Program Management Option: Stream 21 Single Message Unformatted*

7.7.11.1 *Requirements for Stream 21 Single Message Unformatted*

7.7.11.1.1 The value of data item ITEMTYPE shall be ‘SEMI:UNFORMATTEDPROCESSPROGRAM’ when using Stream 21 Single Message Unformatted.

7.7.11.2 Host queries the list of process programs on the equipment:

Comment	Host		Equipment	Comment
Host requests the current list of available process programs.		S21F7-->		
			<--S21F8	Equipment returns the current list of available process programs.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.11.3 Host requests a process program from the equipment:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT
Host requests process                    S21F5-->
program data
                                                 <--S21F6                Equipment returns the
                                                                         process program data
```

7.7.11.4 Host sends a process program to the equipment:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

Host requests                            S21F1-->
permission to send the
process program.

                                                 <--S21F2                Equipment grants
                                                                         permission.
Host sends the process                   S21F3-->
program.
                                                 <--S21F4                Equipment acknowledges
                                                                         receipt of the process
                                                                         program.
```

7.7.11.5 Equipment sends a process program to the host:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT
                                                 <--S21F1                Equipment requests
                                                                         permission to send the
                                                                         process program.
Host grants permission.                  S21F2-->
                                                 <--S21F3                Equipment sends the
                                                                         process program.
Host acknowledges                        S21F4-->
receipt of the process
program.
```

7.7.11.6 Equipment queries the list of available process programs on the host:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT
                                                 <--S21F7                Equipment requests the
                                                                         current list of
                                                                         available process
                                                                         programs.
Host returns the list                    S21F8-->
of process programs
currently available for
this Equipment.
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.11.7 Equipment requests a process program from the host:

COMMENT	HOST	EQUIPMENT	COMMENT
		<--S21F5	Equipment requests process program.
Host returns the process program.	S21F6-->		

7.7.11.8 Host deletes one or more process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the equipment to delete one or more process programs.	S21F11-->		
			Equipment deletes the requested process program(s) from nonvolatile storage.
		<--S21F12	Equipment acknowledges the request.

7.7.12 *Scenarios for Process Program Management Option: Stream 21 Multiple Message Unformatted*

7.7.12.1 *Requirements for Stream 21 Multiple Message Unformatted*

7.7.12.1.1 The value of data item ITEMTYPE shall be ‘SEMI:UNFORMATTEDPROCESSPROGRAM’ when using Stream 21 Multiple Message Unformatted.

7.7.12.2 Host queries the list of process programs on the equipment:

Comment	Host	Equipment	Comment
Host requests the current list of available process programs.	S21F7-->		
		<--S21F8	Equipment returns the current list of available process programs.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.12.3 Host requests a process program from the equipment:

COMMENT	HOST		EQUIPMENT	COMMENT
Host requests the process program.		S21F15-->		
			<--S21F16	Equipment accepts the request.
				[START]
				Repeat until all item parts are sent or the transfer is aborted.
			<--S21F17	Equipment sends one item part.
Host acknowledges receipt of an item part.		S21F18-->		
				[END]
			<--S21F17	Equipment sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.
Host acknowledges receipt of the final message in the sequence.		S21F18-->		

7.7.12.4 Host sends process program to the equipment:

COMMENT	HOST		EQUIPMENT	COMMENT
Host requests permission to send the process program.		S21F13-->		
			<--S21F14	Equipment grants permission.
[START]				
Repeat until all item parts are sent.				
Host sends one item part.		S21F17-->		
			<--S21F18	Equipment acknowledges receipt of an item part.
[END]				
Host sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.		S21F17-->		
			<--S21F18	Equipment acknowledges receipt of the final message in the sequence.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.12.5 Equipment queries the list of available process programs on the host:

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S21F7	Equipment requests the current list of available process programs.
Host returns the list of process programs currently available for this Equipment.		S21F8-->		

7.7.12.6 Equipment sends a process program to the host:

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S21F13	Equipment requests permission to send the process program.
Host grants permission.		S21F14-->		
				[START]
				Repeat until all item parts are sent.
			<--S21F17	Equipment sends one item part.
Host acknowledges receipt of an item part.		S21F18-->		
				[END]
			<--S21F17	Equipment sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.
Host acknowledges receipt of the final message in the sequence.		S21F18-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.12.7 Equipment requests a process program from the host:

COMMENT	HOST	EQUIPMENT	COMMENT
		<--S21F15	Equipment requests the process program.
Host accepts the request.	S21F16-->		
[START]			
Repeat until all item parts are sent.			
Host sends one item part.	S21F17-->		
		<--S21F18	Equipment acknowledges receipt of an item part.
[END]			
Host sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.	S21F17-->		
		<--S21F18	Equipment acknowledges receipt of the final message in the sequence.

7.7.12.8 Host deletes one or more process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the equipment to delete one or more process programs.	S21F11-->		
			Equipment deletes the requested process program(s) from nonvolatile storage.
		<--S21F12	Equipment acknowledges the request.

7.7.13 *Scenarios for Process Program Management Option: Stream 7 Formatted*

7.7.13.1 *Requirements for Stream 7 Formatted*

7.7.13.1.1 Formatted process programs shall be verified immediately following any transfer to the equipment.

7.7.13.2 Host queries the list of process programs on the equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Current EPPD Request	S7F19-->		
		<--S7F20	Current EPPD Data

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.13.3 Host requests a process program from the equipment:

COMMENT	HOST		EQUIPMENT	COMMENT
Formatted Process Program Request		S7F25-->		
			<--S7F26	Formatted Process Program Data

7.7.13.4 Host sends process program to the equipment:

COMMENT	HOST		EQUIPMENT	COMMENT
[IF] Process program is multi-block				
[THEN]				
Process Program Load Inquire		S7F1-->		
			<--S7F2	Process Program Load Grant
[ENDIF]				
Formatted Process Program Send		S7F23-->		
			<--S7F24	Formatted Process Program Acknowledge Verify process program
				[IF] S7F27 is multi-block
				[THEN]
			<--S7F29	Process Program Verification Inquire
Process Program Verification Grant		S7F30->		
				[END IF]
			<--S7F27	Process Program Verification Send
Process Program Verification Acknowledge		S7F28-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.7.13.5 Equipment sends a process program to the host:

```
COMMENT                          HOST                    EQUIPMENT                COMMENT
                                                                                  [IF] Process program is
                                                                                  multi-block
                                                                                  [THEN]
                                                <--S7F1                           Process Program Load
                                                                                  Inquire
Process Program Load                      S7F2-->
Grant
                                                                                  [END_IF]
                                                <--S7F23                          Formatted Process
                                                                                  Program Send
Formatted Process                        S7F24-->
Program Acknowledge
```

7.7.13.6 Equipment requests a process program from the host:

```
COMMENT                          HOST                    EQUIPMENT                COMMENT
                                                <--S7F25                          Formatted Process
                                                                                  Program Request
Formatted Process                        S7F26-->
Program Data
                                                                                  [IF] S7F27 is multi-
                                                                                  block
                                                                                  [THEN]
                                                <--S7F29                          Process Program
                                                                                  Verification Inquire
Process Program                           S7F30->
Verification Grant
                                                                                  [END IF]
                                                <--S7F27                          Process Program
                                                                                  Verification Send
Process Program                          S7F28-->
Verification Acknowledge
```

7.7.13.7 Host deletes one or more process programs on the equipment:

```
COMMENT                          HOST                    EQUIPMENT                COMMENT
Host requests the                        S7F17-->
equipment to delete one
or more process
programs.
                                                                                  Equipment deletes the
                                                                                  requested process
                                                                                  program(s) from
                                                                                  nonvolatile storage.
                                                <--S7F18                          Equipment acknowledges
                                                                                  the request.
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.8 *Material Movement* — The material movement capability includes the physical transfer of material (WIP, tools, expendable materials, etc.) between equipment, buffers, and storage facilities. The transfer of material can be performed by operators, AGV robots, tracks, or dedicated fixed material handling equipment.

7.8.1 *Purpose* — This capability is limited in implementation, serving to notify the host of the appearance or removal of material at the equipment's ports.

7.8.2 *Definitions*

7.8.2.1 *Port* — A point or area on the equipment at which a change in equipment ownership of material may occur.

7.8.3 *Description* — This capability consists of alerting the host whenever material is sent or received from any of the ports on the equipment. Event-specific information, such as port identification and material identification, also may be useful, but definition of these and other related DVVALs are left to the implementation.

7.8.4 *Requirements* — The equipment shall supply two CEIDs, one to report when material is sent from any port and the other to report when material is received at any port.

7.8.5 *Scenario*

```
COMMENT                    HOST                    EQUIPMENT                COMMENT
                                                                            Material is sent or
                                                                            received at an
                                                                            equipment port.
                                                   <--S6F11                 Send collection event
                                                                            to host.
Host acknowledges.                    S6F12-->
```

7.9 *Equipment Terminal Services* — Equipment Terminal Services allows the host to display information on the equipment's display device or the operator of the equipment to send information to the host.

7.9.1 *Purpose* — Equipment Terminal Services allows the factory operators to exchange information with the host from their equipment workstations.

7.9.2 *Detailed Description* — The equipment shall be capable of displaying information passed to it by the host for the operator's attention. The information, or an indication of a message, shall remain on the equipment's display until the operator indicates message recognition (refer to ¶ 7.9.2.3). Message recognition results in a collection event that informs the host that the operator has actually viewed the information.

7.9.2.1 The equipment shall be capable of passing information to the host that has been entered from the operator's equipment console. This information is intended for host applications and is not processed by the equipment.

7.9.2.2 The equipment has no responsibility for interpreting any of the data passed to or from the host using this method.

7.9.2.3 *Message Recognition* — A positive action by the equipment operator indicating the operator has viewed the text of a host-initiated message.

7.9.3 *Requirements*

- Any new Terminal Display message sent by the host shall overwrite an unrecognized message at the same equipment terminal.
- The equipment shall provide a display device capable of displaying at least 160 characters to the operator.
- The equipment shall provide a mechanism for displaying information sent to it by the host.
- The equipment shall provide an indicator to notify the operator when an unrecognized message is present.
- The equipment shall provide a mechanism for the operator to indicate message recognition (e.g., push button, terminal function).

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

- The equipment shall provide a means for alpha numeric data entry that can be used by the operator.
- The equipment shall support operator entry of at least 160 characters per message.
- The equipment shall have a mechanism to send operator-entered messages to the host.
- The equipment shall support single-block messages as a minimum. Support of multi-block messages is optional.
- A Terminal Display message received by the equipment with a zero length TEXT data item shall be accepted and replace any previous unrecognized message, but shall not itself be considered an unrecognized message. This provides a method of clearing an unrecognized message and turning off the unrecognized message indicator.

7.9.4 *Scenarios*

7.9.4.1 Host Sends Information to an Equipment's Display Device:

COMMENT	HOST		EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal x.		S10F3-->		
			<--S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
				Operator indicates message recognition (Equipment clears unrecognized message indicator.)
			<--S6F11	Message recognition event. (Refer to § 7.3.1, Event Data Collection, for details.)
Host acknowledges Optional:		S6F12-->		
			<--S10F1	Operator responds with text via terminal x.
Host acknowledges receipt of operator text.		S10F2-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.9.4.2 Host Sends Information to an Equipment's Display Device and then Overwrites the Information Before Operator Recognizes Message:

COMMENT	HOST		EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal x.		S10F3-->		
			<--S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
Host sends textual information to equipment for display to the operator on terminal x. This message overwrites the first one sent by the host since it is still unrecognized.		S10F3-->		
			<--S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
				Operator indicates message recognition (Equipment clears unrecognized message indicator.)
			<--S6F11	Message recognition event. (Refer to § 7.3.1, Event Data Collection, for details.)
Host acknowledges		S6F12-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.9.4.3 Operator Sends Information to the Host:

COMMENT	HOST	EQUIPMENT	COMMENT
		<--S10F1	Operator sends textual information via equipment terminal x.
Host acknowledges receipt of operator initiated message.	S10F2-->		
Optional: Host responds with information for display to the operator on terminal x.	S10F3-->		
		<--S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
		<--S6F11	Operator indicates message recognition (Equipment clears unrecognized message indicator.)
Host acknowledges	S6F12-->		Message recognition collection event. (Refer to § 7.3.1, Event Data Collection, for details.)

7.9.4.4 Host Sends a Multi-Block Display Message:

COMMENT	HOST	EQUIPMENT	COMMENT
Send information	S10F5-->		
		<--S10F6	Accepted or denied. [IF] Message from host is multi-block and multi-block is not supported by the equipment, [THEN]
		<--S10F7	Send Multi-block Not Allowed. [END_IF]

7.10 *Error Messages*

7.10.1 *Purpose* — Error messages provide the host with information describing the reason for a particular message or communication fault detected by the equipment.

7.10.2 *Definitions*

7.10.2.1 *Communication Fault* — Refer to § 5.2 for the definition.

7.10.2.2 *Message Fault* — Refer to § 5.2 for the definition.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.10.3 *Detailed Description* — The equipment shall inform the host that it cannot process a message due to an incorrect:

- device ID,
- message stream type,
- message function,
- message format, or
- data format.

7.10.3.1 The equipment shall inform the host if the message has more data than it can handle.

7.10.3.2 The equipment shall inform the host if the equipment's transaction timer expires.

7.10.3.3 The equipment shall treat the above conditions as application-level errors and shall not take any further action on any message in error.

7.10.3.4 Error messages are invoked whenever the equipment detects communication or message faults.

7.10.4 *Requirements* — Support of all Stream 9 messages.

7.10.5 *Scenarios*

7.10.5.1 Message Fault Due to Unrecognized Device ID:

COMMENT	HOST		EQUIPMENT	COMMENT
Host sends a message.		SxFy-->		
				Equipment detects an unrecognized device ID within the message from the host.
			<--S9F1	Equipment reports to the host that an "unrecognized device ID" was detected in the received message.

7.10.5.2 Message Fault Due to Unrecognized Stream Type:

COMMENT	HOST		EQUIPMENT	COMMENT
Host sends a message.		SxFy-->		
				Equipment detects an unrecognized stream type within the message from the host.
			<--S9F3	Equipment reports to the host that an "unrecognized stream type" was detected in the received message.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.10.5.3 Message Fault Due to Unrecognized Function Type:

COMMENT	HOST		EQUIPMENT	COMMENT
Host sends a message.		SxFy-->		
				Equipment detects an unrecognized function type within the message from the host.
			<--S9F5	Equipment reports to the host that an "unrecognized function type" was detected in the received message.

7.10.5.4 Message Fault Due to Illegal Data Format:

COMMENT	HOST		EQUIPMENT	COMMENT
Host sends a message.		SxFy-->		
				Equipment detects illegal data format within the message from the host.
			<--S9F7	Equipment reports to the host that an "illegal data format" was detected in the received message.

7.10.5.5 Communication Fault Due to Transaction Timer Timeout:

COMMENT	HOST		EQUIPMENT	COMMENT
				Equipment does not receive an expected reply message from the host and a transaction timer timeout occurs.
			<--S9F9	Equipment reports to the host that a transaction timer timeout occurred.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.10.5.6 Message Fault Due to Data Too Long:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy-->		
			Equipment detects that the message from the host contains more data than it can handle.
		<--S9F11	Equipment reports to the host that an "data too long" was detected in the received message.

7.10.5.7 Communication Fault Due to Conversation Timeout:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy-->		
		<--SxFy+1	Equipment sends reply.
			Equipment is now expecting a specific message from the host as a result of the previous transaction. Equipment has not received the expected message from the host and a conversation timeout occurs.
		<--S9F13	Equipment reports to the host that a conversation timeout occurred.

7.11 *Clock* — The clock capability enables host management of time-related activities and occurrences associated with the equipment and across multiple pieces of equipment.

7.11.1 *Purpose* — The purpose of the clock capability is to enable time stamping of collection event and alarm reports. Time stamping is useful for resolving relative order of collection event/alarm occurrences and scheduling of equipment activities by the host.

7.11.1.1 The ability for the host to instruct the equipment to set an internal clock to a specified time value, and for the equipment to request the current date and time, is needed for effective time management and synchronization between host and equipment.

7.11.2 *Definitions*

7.11.2.1 *Clock* — Clock is a status variable containing the current value of time at the equipment. Clock may be included in report definitions and/or queried separately by the host. Refer to SEMI E5 for a full definition of this variable data item and its required formatting.

7.11.2.2 *TIME* — TIME is a data item contained in messages used by the host to set time at the equipment and by the equipment or host to request the current time from the other. Refer to SEMI E5 for a full definition of this data item.

7.11.3 *Detailed Description* — The clock capability assumes the existence of a relative time reference on the equipment. This time reference is used as a basis for updating the time value of an equipment status variable called 'Clock'. The time reference shall reflect the current time to within a resolution range of seconds to centiseconds (refer to the format for Clock in SEMI E5). The purpose of time stamping with centiseconds is to resolve the order in which

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

nearly simultaneous events occur rather than to provide a more precise record of the time of day at which they occurred. Where more than one event occurs within a given period of clock resolution, the centiseconds reported in the time stamps for these events shall reflect the actual order in which the events were detected. Equipment with a clock resolution of less than a second should report centiseconds. Otherwise, centiseconds should be assigned to reflect the relative order in which events were detected. Equipment unable to resolve time to less than a second and unable to reflect the relative order in which events were detected may report centiseconds as ‘00’.

7.11.3.1 The host employs the ‘Date and Time Set Request’ message (S2F31) to initialize the value of Clock to the value contained in the TIME data item. Similarly, the equipment may employ the ‘Date and Time Request’ message (S2F17) to obtain a new initialization time for Clock. As before, the value of TIME returned by the host is used to set Clock. Note that, in the event that the precision of TIME is seconds and that for Clock is centiseconds, in both cases the initial value of Clock shall contain ‘00’ for its Centisecond digits upon initialization. Additionally, for any field in TIME that is not supported by the equipment, the local value of this field is equipment dependent. For example, Equipment that cannot resolve time to less than a second might round or ignore centiseconds and always set the Centisecond field to ‘00’.

7.11.4 *Requirements*

- The resolution and update rate of the internal time reference shall be sufficient to distinguish between two nearly simultaneous collection events and/or alarms.
- The equipment supplier shall provide documentation describing the resolution of the internal time reference.
- The equipment supplier shall provide documentation describing how centisecond values are assigned, including the case of unresolvable simultaneous events.

7.11.5 *Scenarios*

7.11.5.1 Equipment Requests TIME (Optional Scenario):

COMMENT	HOST		EQUIPMENT	COMMENT
			<--S2F17	Equipment requests a time value from the host.
Host responds with a TIME value		S2F18-->		Equipment sets its internal time reference to the value of TIME received from the host.

7.11.5.2 Host Instructs Equipment to Set Time:

COMMENT	HOST		EQUIPMENT	COMMENT
Host instructs equipment to set its time.		S2F31-->		
			<--S2F32	Equipment sets its internal time reference to the value of TIME received from the host and acknowledges completion.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.11.5.3 Host Requests Equipment's Current Time Value:

```
COMMENT                    HOST                EQUIPMENT              COMMENT

Host requests equipment          S2F17-->
time.

                                            <--S2F18              Equipment returns its
                                                                  internal time reference
                                                                  value to the host.
```

7.12 *Spooling* — Spooling is a capability whereby the equipment can queue messages intended for the host during times of communication failure and subsequently deliver these messages when communication is restored. Spooling is limited to primary messages of user-selected streams.

7.12.1 *Purpose* — The purpose of spooling is to provide a method for retaining equipment message data that might otherwise be lost due to communication failure. The motive for producing this functionality is to retain valuable data used to track material and to improve product quality. The spooling capability fills a gap in the SEMI E5 Standard. In the past, without a spooling capability, the equipment has typically discarded messages that could not be delivered, or turned messaging off altogether. It is intended that the host initiate the spool unload process immediately following the reestablishment of communications.

7.12.2 *Definitions*

7.12.2.1 *MaxSpoolTransmit* — An equipment constant containing the maximum number of messages that the equipment shall transmit from the spool in response to an S6F23 'Transmit Spooled Messages' request. If MaxSpoolTransmit is set to zero, no limit is placed on the messages sent from the spool. Multi-block inquire/grant messages are not counted in this total.

7.12.2.2 *OverWriteSpool* — A Boolean equipment constant used to indicate to the equipment whether to overwrite data in the spool area or to discard further messages whenever the spool area limits are exceeded.

7.12.2.3 *Send Queue* — Refers to the queue into which equipment generated SECS messages are placed in preparation for transmission to the host.

7.12.2.4 *Spool* — The spool is an area of nonvolatile storage in which the equipment stores certain messages that cannot be delivered to the host (when the equipment is in the NOT COMMUNICATING substate of COMMUNICATIONS ENABLED). The spool area can be thought of as a sequential 'ring' buffer. The term spool is also used to denote the action of placing messages into the spool area.

7.12.2.5 *SpoolCountActual* — A status variable used to keep a count of the messages actually stored in the equipment's spool area. Multi-block inquire/grant messages are not spooled and not included in this count.

7.12.2.6 *SpoolCountTotal* — A status variable used to keep a count of the total number of primary messages directed to the spool, regardless of whether placed or currently retained in the spool. Multi-block inquire/grant messages are not spooled and not included in this count.

7.12.2.7 *SpoolFullTime* — A status variable containing the timestamp when the spool last became full. If the spool was not filled during the last spooling period, this will contain a time value prior to the current SpoolStartTime.

7.12.2.8 *SpoolStartTime* — A status variable containing the timestamp from when spooling was last activated.

7.12.3 *Description*

7.12.3.1 *Spooling State Model* — There are two major states of spooling: SPOOL INACTIVE and SPOOL ACTIVE. SPOOL ACTIVE has two components: SPOOL UNLOAD and SPOOL LOAD. These are each broken into substates. A description of all spooling states, substates, and applicable state transitions follows. The POWER OFF and POWER ON parent states are common to all equipment subsystems. They are shown here to illustrate the retention of spooling context during a power down situation.

NOTE 12: Disabling SECS communications does not affect the current spooling state since no messages are generated until communications are subsequently enabled. Spooling is effectively frozen in this case.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.12.3.2 *POWER OFF*

7.12.3.2.1 The equipment has lost power for any reason (e.g., power failure, power switch set to off).

7.12.3.3 *POWER ON*

7.12.3.3.1 The equipment is powered up.

7.12.3.4 *SPOOL INACTIVE*

7.12.3.4.1 This is the normal operating mode. No spooling occurs. The spool area is empty. Primary SECS-II messages are transmitted normally.

7.12.3.5 *SPOOL ACTIVE*

7.12.3.5.1 All primary SECS-II messages ready for sending and for which spooling is enabled (refer to S2F43) are directed to the spool area. All other primary messages, except Stream 1, are discarded. The equipment shall attempt to send any secondary messages that are generated and discard these messages should the attempt to send fail.

7.12.3.6 Spool state transitions from SPOOL INACTIVE to SPOOL ACTIVE if the communications state changes from COMMUNICATING to NOT COMMUNICATING (Communication State Transition Table, #14) or from WAIT CRA to WAIT DELAY (Communication STATE Transition Table, #6).

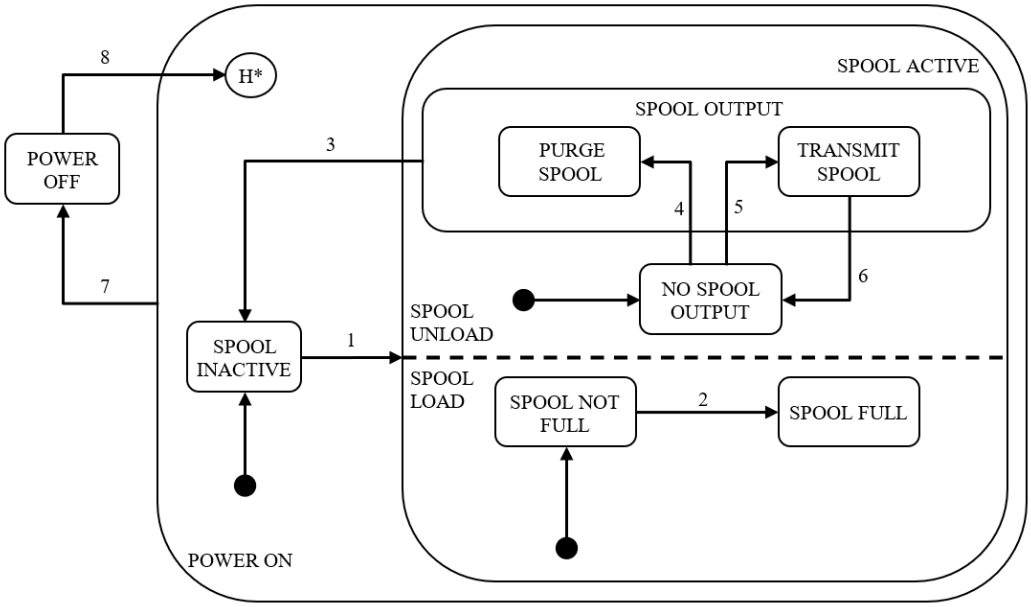

Figure 12
Spooling State Model Diagram

7.12.3.7 Once communications are established, the host shall initiate the spool unload sequence to restore full functionality (refer to below). Since the equipment will deliver secondary messages, the host may inquire for information or send commands as needed.

7.12.3.8 The SPOOL ACTIVE state has two AND states: SPOOL LOAD and SPOOL UNLOAD. This means that they operate independently, though sharing data and some state change stimuli. Refer to § 6.3 for explanation of state model terms and notation.

7.12.3.9 *SPOOL LOAD*

7.12.3.9.1 The SPOOL LOAD component enters messages into the spool area. It is divided into two substates: SPOOL NOT FULL and SPOOL FULL. SPOOL NOT FULL is the default entry substate of the parent state SPOOL LOAD.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.12.3.10 *SPOOL NOT FULL*

7.12.3.10.1 As primary SECS-II messages are directed to the spool area, the equipment shall 'write' the SECS-II message to the end of the spool. Status variables SpoolCountTotal and SpoolCountActual shall be incremented each time a message is placed in the spool area.

7.12.3.11 *SPOOL FULL*

7.12.3.11.1 In this state, all of the allocated spooling area is filled. Choice of the following options shall be controlled by the setting of a Boolean equipment constant called 'OverWriteSpool'. The first message to be dealt with is that which could not be fit into the spool prior to transition from SPOOL NOT FULL (refer to transition table below).

7.12.3.12 *OverWriteSpool is True* — The equipment deletes as many of the 'oldest' records (e.g., SECS-II messages) contained in the spool area necessary to make space for the new message and then adds the message. Status variable SpoolCountTotal shall be incremented whenever a message is submitted to the spool area. Status variable SpoolCountActual shall be manipulated to keep an accurate count of the number of messages contained in the spool area. For example, if it is necessary to delete three messages in the spool area to spool one message, SpoolCountActual would have three subtracted and then one added to the total.

7.12.3.13 *OverWriteSpool is False* — Any subsequent primary messages shall be discarded. When such a message is discarded, SpoolCountTotal is incremented.

7.12.3.14 *SPOOL UNLOAD*

7.12.3.14.1 The SPOOL UNLOAD component of SPOOLACTIVE deals with movement of messages out of the spool. It has an active substate (SPOOL OUTPUT) and a passive substate (NO SPOOL OUTPUT). NO SPOOL OUTPUT is the default entry substate, since the equipment is NOT COMMUNICATING at the time spooling is initiated. When communications between equipment and host are restored, there is an opportunity for the host to recover spooled messages. No action is taken until the host initiates the spool output process via the S6F23 (Request Spooled Data). The host has the option to either receive the spooled messages (refer to substate TRANSMIT SPOOL) or discard all messages in the spool (refer to substate PURGE SPOOL).

7.12.3.15 *NO SPOOL OUTPUT*

7.12.3.15.1 In this state, no messages are removed from the spool.

7.12.3.16 *SPOOL OUTPUT*

7.12.3.16.1 The SPOOL OUTPUT state encompasses the removal of messages from the spool. Its substates are TRANSMIT SPOOL and PURGE SPOOL.

7.12.3.17 *TRANSMIT SPOOL*

7.12.3.17.1 The host elects to receive all messages contained in the spool area. The equipment is expected to keep track of the oldest record (i.e., message) within the spool area. When communications are re-established with the host and transmission of the spool area is started, the oldest record shall be the first record transmitted, then the next oldest record, etc. There is no prioritization of messages to be sent from the spool.

7.12.3.17.2 As each spooled message is successfully transmitted to the host, it is removed from the spool area upon successful completion of the transaction. SpoolCountActual is decremented as each message is removed from the spool. The equipment shall transmit messages only from the spool area until all spooled messages have been completely transmitted to the host.

7.12.3.17.3 Flow control of the spool transmit process is achieved in two ways. First, only one open transaction on the equipment is allowed during spool unload. Thus, if a message requires a reply, the equipment shall wait for that reply before transmitting the next spooled message. Messages which require no reply may be transmitted sequentially as rapidly as the message transfer mechanism will allow.

7.12.3.17.4 The second flow control method is to allow the host to limit the maximum number of messages sent from the spool in response to the S6F23 request. An equipment constant named MaxSpoolTransmit may be set by the host to achieve this behavior. If MaxSpoolTransmit is set to five, for example, the equipment will send the first five messages from the spool and then transition to the NO SPOOL OUTPUT state, awaiting the next S6F23 request. There is no event report generated when MaxSpoolTransmit is reached. The host is responsible for determining this situation by (a) counting the messages received, (b) timing out waiting for the next message, (c) inquiring to the equipment for

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

the curent value of the SpoolCountActual status variable, or (d) some combination of the above. If MaxSpoolTransmit is set to zero, the spool shall be transmitted completely in response to S6F23.

7.12.3.17.5 Normal spooling continues during the spool transmit process. If the SPOOL LOAD component should transition to SPOOL FULL, it shall not have any effect on the SPOOL UNLOAD component. Once full, the spool cannot make the transition back to SPOOL NOT FULL except via the SPOOL INACTIVE state. Space made available due to the spool unload process shall not be used in this case.

7.12.3.17.6 When a multi-block message is to be transmitted from the spool, any required inquire/grant transaction shall be initiated. If the host's response denies permission to send the multi-block message, the equipment shall discard that message and continue with the transmit process. This sequence shall count as one message in the MaxSpoolTransmit count.

7.12.3.17.7 There is one area where SPOOL LOAD and SPOOL UNLOAD may interact: When the spool is full and OverWriteSpool is True. During the spool transmit process, spooled messages are being removed and new primary messages are being written to the spool. These new messages are overwriting the oldest messages available, unless the unload process has freed sufficient spool space. There is a possibility that the unload and overwrite processes may compete for control of the same message area. For example, if the spool holds messages ABCDE, with A oldest and E newest, A might be sent to the host, B (and the space from A) overwritten by the new message F, C sent to the host, D and E (and the space from message C) overwritten by G, etc. The loss of continuity may be 'disorienting' to the host program receiving the messages. It is expected that the unload process will be fast relative to the generation of new messages, so that this occurrence will be rare.

7.12.3.17.8 Should a communication failure occur during the spool transmit process, spooling shall continue as before the transmit process began. However, the spool unload sequence shall halt (i.e., transition to NO SPOOL OUTPUT will occur—refer to transition table below).

7.12.3.18 *PURGE SPOOL*

7.12.3.18.1 The equipment shall discard all messages in the spool and, when the spool is empty, zero SpoolCountActual.

7.12.3.19 *Spooling State Transitions*

7.12.3.19.1 A table follows detailing all spooling state transitions as presented in the state transition diagram.

Table 11 Spooling State Transition

#	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	SPOOL INACTIVE	Communication state changes from COMMUNICAT-ING to NOT COMMUNICAT-ING or from WAIT CRA to WAIT DELAY and Enable Spool is true.	SPOOL ACTIVE	SpoolCountActual and SpoolCountTotal are initialized to zero. Any open transactions with the host are aborted. SpoolStart-Time (SV) is set to current time. Alert the operator that spooling is active.	The default state in each AND substate is entered. The message which could not be sent remains in the send queue and is dealt with in Spool Active state. The collection event Spooling Activated has occurred.
2	SPOOL NOT FULL	Message generate which will not fit into spool area.	SPOOL FULL	SpoolFullTime (SV) is set to current time. Alert the operator that spool is full.	The message which would not fit into the spooling area is dealt with after the transition. No collection event is generated.
3	SPOOL OUTPUT	Spool area emptied.	SPOOL INACTIVE	Spooling process disabled. Alert the operator that spooling has been halted.	The collection event Spooling Deactivated has occurred. Transition from the AND substate Spool Unload component occurs.
4	NO SPOOL OUTPUT	S6F23 received w/RSDC = 1.	PURGE SPOOL	No action.	Initiates purging process. No collection event is generated since this is based on host request.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

#	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
5	NO SPOOL OUTPUT	S6F23 received w/RSDC = 0.	TRANSMIT SPOOL	No action.	Initiates message transmission from spool. No collection event is generated since this is based on host request.
6	TRANSMIT SPOOL	Communication failure or MaxSpoolTransmit reached.	NO SPOOL OUTPUT	Spool transmission process suspended.	If communications failure, the event Spool Transmit Failure has occurred. No collection event generated for MaxSpoolTransmit reached.
7	POWER ON	Equipment power source discontinued.	POWER OFF	No action.	Spooling context has been maintained in nonvolatile storage prior to this transition.
8	POWER OFF	Equipment power source restored.	POWER ON	Spooling context restored from nonvolatile memory.	If spooling were active prior to power down, it shall continue. If TRANSMIT SPOOL were active at powerdown, transition #6 is expected to follow since communications state is initially NOT COMMUNICATING.

7.12.3.20 *Enabling Spooling* — The equipment shall provide the host with the ability to enable and disable Spooling for any message (except Stream 1 messages [i.e., S1F1, S1F13]) via the S2F43/F44 transaction. Spooling may be enabled for an entire Stream, for individual messages within a stream, or for any combination of the two. Streams and Functions not referenced in this message are not spooled. Spooling can be totally disabled by sending an S2F43 with a zero-length list for the first item (refer to S2F43 definition). In addition, the equipment shall provide an equipment constant, EnableSpooling, to allow setting the enable or disable of spooling.

7.12.3.21 When EnableSpooling is false, the SPOOL state cannot transition from SPOOL INACTIVE to SPOOL ACTIVE. Changing EnableSpooling does not change the spool state, purge the spool or change streams and functions enabled for spooling. Once the equipment is 'SPOOL ACTIVE', the host shall initiate the spool unload sequence to restore full functionality even though 'Enable Spooling' has changed to false.

7.12.4 *Requirements* — The following items are required to support the spooling capability:

- At a minimum, the equipment shall reserve for spooling nonvolatile storage with sufficient capacity to store all of the primary SECS-II messages that would occur during a normal processing cycle.
- While spooling is enabled, the equipment shall reply to primary messages sent by the host with the appropriate secondary message.
- Secondary messages which cannot be delivered shall be discarded, never spooled.
- All spooling-related status variables and setup information (as per S2F43) shall be stored in nonvolatile memory along with any other information required for the potential unloading of the spool area after a power loss.
- Upon power-up, the equipment shall retain all spooling context from the time the equipment was last shutdown or reset. This means that spooling, if previously active, continues upon system power up.
- The Equipment shall reject any message that attempts to set 'Spooling' for Stream 1.
- If a multi-block primary message need for the inquire/grant is to be sent during SPOOL ACTIVE, the message should be placed in the spool and the grant resolved during spool transmit.

7.12.5 *Scenarios*

7.12.5.1 Define the Set of Messages to be Spooled:

7.12.5.1.1 This Scenario is used to set up the list of messages that the equipment should spool (or by defining none, to disable spooling).

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

COMMENT	HOST	EQUIPMENT	COMMENT
Host defines messages to be spooled in case of communications failure.	S2F43-->		
		<--S2F44	Equipment acknowledges setup.

7.12.5.2 Define the Maximum Number of Messages to Send in Response to S6F23:

7.12.5.2.1 This Scenario sets the value of the equipment constant MaxSpoolTransmit.

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends value for equipment constant MaxSpoolTransmit.	S2F15-->		
		<--S2F16	Equipment acknowledges equipment constant change.

7.12.5.3 Request or Delete Spooled Data (MaxSpoolTransmit = 0):

7.12.5.3.1 This Scenario is used to initiate the transfer of the spooled data from the equipment to the host or to purge the spool.

COMMENT	HOST	EQUIPMENT	COMMENT
			Communications were lost and then re-established.
Host requests data that includes spool-related status variables.	S1F3-->		
		<--S1F4	Send status data.
NOTE: S1F3 is one of various methods that could be used. Request or delete spooled data.	S6F23-->		
		<--S6F24	Request spooled data acknowledgement. [IF] RSDC = 0 (Spool data requested.) [THEN] The appropriate Streams and Functions are used to transmit the spooled data to the host. [ELSE_IF] RSDC = 1 [THEN] Spool data discarded. [END_IF]
		<--S6F11	Spooling Deactivated event report sent.
Acknowledge	S6F12-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.12.5.4 Request or Delete Spooled Data (MaxSpoolTransmit > 0):

7.12.5.4.1 This Scenario shows the effect of MaxSpoolTransmit < SpoolCountActual on the Spool Transmit process. For the purpose of illustration, the value of MaxSpoolTransmit is 5 and the SpoolCountActual is 8 (at the time communications are re-established). No messages are added to the Spool during the transmit process.

```
COMMENT                     HOST                    EQUIPMENT                COMMENT

                                                                             Communications were
                                                                             lost and then re-
                                                                             established.

Host requests data that                  S1F3-->
includes spool-related
status variables.

                                                 <--S1F4                     Send status data (e.g.,
                                                                             SpoolCountActual = 8,
                                                                             MaxSpoolTransmit = 5).

Host requests spooled                   S6F23-->
data (RSDC=0).

                                                 <--S6F24                    Request spooled data
                                                                             acknowledgement.

                                                                             The five oldest
                                                                             messages in the Spool
                                                                             are transmitted to the
                                                                             host. Spooling remains
                                                                             active.

Host recognizes that
MaxSpoolTransmit is
reached.

Host requests                           S6F23-->
additional spooled data
(RSDC = 0).

                                                 <--S6F24                    Request spooled data
                                                                             acknowledgement.

                                                                             The three remaining
                                                                             messages are
                                                                             transmitted from the
                                                                             spool.

                                                 <--S6F11                    Spooling Deactivated
                                                                             event report sent.

Acknowledge                             S6F12-->
```

7.12.6 *Spooling Information* — This section describes a method for the host to query the current spooling setup.

7.12.6.1 *Purpose* — Spooling Information allows the host to discover the current streams and functions configured for spooling. For example, when a host first connects to an equipment, the host can discover if previously configured spooling configuration is still active in the equipment. Without these features, the host cannot know whether or not the spooling setup was changed. Additionally, this Spooling Information can be used when testing a GEM interface to confirm that the spooling configuration was correctly received.

7.12.6.2 *Detailed Description* — The equipment shall support the following requests through the SECS-II interface:

- Request Selected Spooling Stream and Functions (RQSSF)

7.12.6.3 *Requirements*

7.12.6.3.1 The equipment shall support the message S2F59 Request Selected Spooling Stream and Functions (RQSSF) to allow the host to query spooling setup information from the equipment.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.12.6.4 *Scenarios*

7.12.6.4.1 Host Initiates: Request Selected Spooling Stream and Functions (RQSSF):

```
        COMMENT                     HOST                    EQUIPMENT                COMMENT

Host initiates RQSSF                    S2F59-->

                                                  <--S2F60                Return Selected Spooling
                                                                          Stream and Functions
                                                                          (RTSSF)
```

7.13 *Control* — The control-related capabilities allow for configuration and manipulation of the control state model. In this way the host and/or user may modify the equipment's control-related behavior.

7.13.1 *Purpose* — This section complements the CONTROL state model description found in § 6.5. It defines the requirements for implementation of this model.

7.13.2 *Definitions* — None.

7.13.3 *Description*

7.13.3.1 *Control Configuration* — The control state model has two areas of configuration. The first area is related to default entry states of the state model. Upon system initialization, the system shall activate either the ON-LINE or OFF-LINE state. Upon entry to OFF-LINE, the system shall in turn activate one of the substates of OFF-LINE (EQUIPMENT OFF-LINE, ATTEMPT ON-LINE, or HOST OFF-LINE). In both these cases, the user shall configure the equipment to make the choices appropriate to that factory. Entry to the ON-LINE state also involves a choice of substates. In this case, the equipment reads the front panel REMOTE/LOCAL switch to determine the appropriate state.

7.13.3.1.1 The second area of configuration involves the transition to be made if the ON-LINE attempt should fail. The model may be set to transition to either HOST OFF-LINE or to EQUIPMENT OFF-LINE should the S1F1 transaction be halted unsuccessfully. Choosing HOST OFF-LINE allows the host to cause the equipment to transition to ON-LINE when the host becomes ready. This is accomplished via the message S1F17 (per below).

7.13.3.2 *Changing Control State* — In the control state model, both the operator and the host can affect the control state. The operator retains ultimate authority to set the equipment OFF-LINE by means of an OFF-LINE switch mechanism. The operator also can cause the equipment to attempt to go ON-LINE. Under some circumstances, the host can initiate the transition to ON-LINE.

7.13.3.2.1 If the operator requests ON-LINE, the equipment will send an S1F1 to the host. The host may confirm ON-LINE with an S1F2 or deny ON-LINE by sending an S1F0. If there is no host response (i.e., reply timeout), the equipment shall treat it as a denial.

7.13.3.2.2 When the equipment is ON-LINE, the host may request that it transition to OFF-LINE. It will transition into the HOST OFF-LINE substate. When the equipment HOST OFF-LINE state is active, the host may request that it transition to ON-LINE. The combination of these two allow the host to cycle the equipment between ON-LINE and OFF-LINE.

7.13.3.2.3 Only the operator may change the ON-LINE substate (REMOTE or LOCAL).

7.13.4 *Requirements*

- The equipment shall supply a method for configuring the default CONTROL state to be activated upon system initialization. The choice of states shall be among ATTEMPT ON-LINE, EQUIPMENT OFF-LINE, HOST OFF-LINE, and ON-LINE.
- The equipment shall supply a method for configuring which state should be activated when the attempt to go ON-LINE fails. The option is a transition to either the HOST OFF-LINE state or the EQUIPMENT OFF-LINE state.
- The equipment shall supply a momentary switch which will initiate the transition to OFF-LINE and another which will begin the process to go ON-LINE. Discrete position switches shall not be used. These should be designed so

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

that they may not be actuated simultaneously. The switch may be mounted on the front panel or be available via keyboard input at the operator console.

- The equipment shall supply a discrete two-position switch which the operator may use to indicate the desired substate for ON-LINE (i.e., REMOTE or LOCAL). The switch may be mounted on the front panel or be available via keyboard input at the operator console. If implemented in software, this setting shall be retained in non-volatile storage.
- The equipment shall supply an indicator on the front panel which displays the full identification of the current CONTROL state/substate (e.g., OFF-LINE/ATTEMPT ON-LINE). This may be accomplished either with labelled display lights or via the operator console display. It is recommended that this indicator be visible at all times.
- The equipment shall supply a status variable that contains the current state/substate of the CONTROL state model.
- Whenever the ON-LINE/REMOTE state is active and the operator issues a command to the equipment, the equipment shall cause an ‘operator command issued’ event.

7.13.5 *Scenarios*

7.13.5.1 *Operator-Initiated Scenarios*

7.13.5.1.1.1 Host Accepts ON-LINE:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

                                                                            Operator actuates ON-
                                                                            LINE switch when
                                                                            equipment OFF-LINE
                                                                            state is active.

                                          <--S1F1                           Equipment requests ON-
                                                                            LINE.

Host grants ON-LINE.                   S1F2-->

                                          <--S6F11                          "Control State LOCAL
                                                                            (or REMOTE)" event.

Acknowledge.                          S6F12-->
```

7.13.5.1.1.2 Host Denies ON-LINE:

```
COMMENT                    HOST                    EQUIPMENT                COMMENT

                                                                            Operator actuates ON-
                                                                            LINE switch when
                                                                            equipment OFF-LINE
                                                                            state is active.

                                          <--S1F1                           Equipment requests ON-
                                                                            LINE.

Host denies ON-LINE.                   S1F0-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.13.5.1.1.3 Operator Sets OFF-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
			Operator actuates OFF-LINE switch when equipment ON-LINE state is active.
		<--S6F11	"Equipment requests OFF-LINE" event.
Acknowledge.	S6F12-->		

7.13.5.1.1.4 Operator Sets REMOTE:

COMMENT	HOST	EQUIPMENT	COMMENT
			Operator sets switch from LOCAL to REMOTE.
		<--S6F11	"Control State REMOTE" event.
Acknowledge.	S6F12-->		

7.13.5.1.1.5 Operator Sets LOCAL:

COMMENT	HOST	EQUIPMENT	COMMENT
			Operator sets switch from REMOTE to LOCAL.
		<--S6F11	"Control State LOCAL" event.
Acknowledge.	S6F12-->		

7.13.5.2 *Host-Initiated Scenarios*

7.13.5.2.1 Host Sets OFF-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests OFF-LINE.	S1F15-->		
		<--S1F0	[IF] Equipment is OFF-LINE [THEN]: Equipment does not process requests.
		<--S1F16	[ELSE] Equipment ON-LINE Equipment acknowledges request and transitions to OFF-LINE.
		<--S6F11	"Equipment OFF-LINE" event.
Acknowledge	S6F12-->		
			[END_IF]

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.13.5.2.2 Host Sets ON-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests ON-LINE.	S1F17-->		
		<--S1F18	[IF] Equipment is ON-LINE [THEN] Equipment denies request (ONLACK = 2).
		<--S1F18	[ELSE IF] Equipment is not Host OFF-LINE Equipment denies request (ONLACK = 1).
		<--S1F18	[ELSE] Equipment is Host OFF-LINE Equipment acknowledges request (ONLACK = 0).
		<--S6F11	"Control state LOCAL (or REMOTE)" event.
Acknowledge	S6F12-->		
			[END_IF]

7.14 *Documentation File Transfer* — Transfer GEM documentation in a standard portable document format (PDF) or comma separated values (CSV) format through the GEM interface.

7.14.1 *Purpose* — This section complements the GEM documentation requirements by making a view friendly and print friendly file format of the documentation available through the GEM interface. This is intended to make it easier both for the equipment supplier to distribute the GEM documentation with the equipment and for the end user to gain access to the current GEM documentation.

7.14.2 *Definition*

7.14.2.1 *GEM Documentation* — Structured data that contains documentation defined in § 11.4, of the GEM capabilities implemented on the equipment.

7.14.3 *Requirements*

7.14.3.1 The equipment shall support Stream 21 Single Message and/or Multiple Message technique to allow the host to request equipment to send its GEM documentation. The equipment shall not request the host to send the GEM documentation.

NOTE 13: Some equipment suppliers control who should have access to the GEM documentation, and they do not distribute it through the GEM interface. In these situations, the equipment does not support the Documentation File Transfer GEM capability.

7.14.3.2 If the host attempts to transfer GEM documentation to the equipment it shall be rejected.

7.14.3.3 The document shall be transferred using SECS-II format 10 as the ITEMDATA in Stream 21 messages.

7.14.3.4 The ITEMID shall include a '.pdf' or '.csv' file extension. For example, 'GEMManual.TSSupplier.TrackSys.V3.2.pdf' or 'GEMManual.TSSupplier.TrackSys.V3.2.csv'.

7.14.3.5 If the ITEMID file extension is '.pdf', then the document shall be formatted according to ISO 32000 (standard portable document format, also known as PDF).

7.14.3.6 If the ITEMID file extension is '.csv', then the document shall be formatted according to IETF RFC 4180.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.14.3.7 The ITEMTYPE in the Stream 21 messages for this feature shall be ‘SEMI:GEMDOCUMENTATION’.

7.14.3.8 The ITEMVERSION shall be the file’s last modification date.

7.14.3.9 The GEM documentation can be provided in a single or multiple PDF files.

7.14.3.10 The GEM documentation can also include one or more CSV files. For example, an equipment might provide a list of alarm information in CSV format.

7.14.3.11 Equipment implementing this feature should avoid including equipment sensitive information and intellectual property within the GEM documentation transferred through the GEM interface.

7.14.3.12 Documentation files returned by the equipment shall correspond to the current software of the equipment implementing the GEM interface.

7.14.4 *Scenarios*

7.14.4.1 Host queries the GEM documentation from the equipment using Stream 21 Single Message technique:

```
        Comment                    Host                  Equipment                Comment
Host requests the                        S21F7-->
current list of
available GEM
documentation files.
                                                 <--S21F8                Equipment returns the
                                                                         current list of GEM
                                                                         documentation files.

                                                                         The following exchange
                                                                         would occur for each
                                                                         file.
Host requests GEM                        S21F5-->
documentation file.
                                                 <--S21F6                Equipment returns the
                                                                         GEM documentation file.
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.14.4.2 Host queries the GEM documentation from the equipment using Stream 21 Multiple Message technique:

Comment	Host	Equipment	Comment
Host requests the current list of available GEM documentation files.	S21F7-->		
		<--S21F8	Equipment returns the current list of GEM documentation files. The following exchanges would occur for each file.
Host requests the GEM documentation.	S21F15-->		
		<--S21F16	Equipment accepts the request. [START] Repeat until all item parts are sent or the transfer is aborted.
		<--S21F17	Equipment sends one item part.
Host acknowledges receipt of an item part.	S21F18-->		
			[END]
		<--S21F17	Equipment sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.
Host acknowledges receipt of the final message in the sequence.	S21F18-->		

7.15 *SEDD File Transfer* — Provide GEM documentation in SEMI E172 SEDD format through the GEM interface.

7.15.1 *Purpose* — This section complements the GEM documentation requirements by making the SEMI E172 SEDD file format of the documentation available through the GEM interface. This intends to make it easier both for the equipment supplier to distribute the GEM documentation for the equipment and for the end user to get access to the GEM documentation.

7.15.2 *Requirements*

7.15.2.1 The equipment shall support Stream 21 Single Message and/or Multiple Message technique to allow the host to request SEDD GEM documentation files. The equipment shall not request SEDD GEM documentation files from the host.

7.15.2.2 If the host attempts to transfer a SEDD file to the equipment it shall be rejected.

7.15.2.3 The document shall be transferred using SECS-II format 10 as the ITEMDATA in Stream 21 messages.

7.15.2.4 The document shall comply with SEMI E172 SEDD.

7.15.2.5 The ITEMID shall include ‘.xml’ file extension. For example, ‘TrackSysSupplier.TrackSys.GEMSEDD.V3.2.xml’.

7.15.2.6 The ITEMTYPE in the Stream 21 messages for this feature shall be ‘SEMI:SEDD’.

7.15.2.7 The ITEMVERSION shall be the file’s last modification date.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

7.15.2.8 The equipment is expected to have one SEDD file.

7.15.2.9 A SEDD file returned by the equipment shall correspond to the current software of the equipment implementing the GEM interface.

7.15.3 *Scenarios*

7.15.3.1 Host queries the SEDD documentation from the equipment using Stream 21 Single Message technique:

Comment	Host		Equipment	Comment
Host requests the current list of available SEDD GEM documentation files (expected to be a list of one).		S21F7-->		
			<--S21F8	Equipment returns the current list of SEDD GEM documentation files (expected to be a list of one).
Host requests SEDD GEM documentation file.		S21F5-->		
			<--S21F6	Equipment returns the SEDD GEM documentation file.

7.15.3.2 Host queries the SEDD documentation from the equipment using Stream 21 Multiple Message technique:

Comment	Host		Equipment	Comment
Host requests the current list of available SEDD GEM documentation files (expected to be a list of one).		S21F7-->		
			<--S21F8	Equipment returns the current list of SEDD GEM documentation files (expected to be a list of one).
Host requests the SEDD GEM documentation.		S21F15-->		
			<--S21F16	Equipment accepts the request.
				[START] Repeat until all item parts are sent or the transfer is aborted.
			<--S21F17	Equipment sends one item part.
Host acknowledges receipt of an item part.		S21F18-->		
				[END]
			<--S21F17	Equipment sends a final S21F17 message with an ITEMINDEX value of 0 and zero-length ITEMPART.
Host acknowledges receipt of the final		S21F18-->		

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
message in the
sequence.
```

8 Data Items

8.1 The following sections specify which data items and variable data items are required.

8.2 Except for the specified format restrictions, all data items and variable data items follow the definitions contained in SEMI E5.

8.3 *Data Item Restrictions* — The following is a subset of the data items used by SECS-II messages specified in this Standard. Each data item listed in this section is restricted in its SEMI E5 defined usage. Most are limited to a single format from their standard list of formats. Data items used by SECS-II messages contained in this Standard, but which have no restrictions are not duplicated here.

NOTE 14: One data item, ALCD, is restricted in other than its format. Take note of this restriction as described below.

8.3.1 The equipment supplier shall document any restrictions on length or format of CPNAME. Suppliers shall document the behavior of spaces in a CPNAME. The maximum length of CPNAME shall be 40.

ACKC7A	Format: 51
ALCD	Only bit 8 (alarm set/cleared) of the binary byte is used. Bits 1–7, denoting alarm category, are not used.
ALID	Format: 5()
CCODE	Format: 20, 32, 34, 52, 54
CEID	Format: 5()
CPNAME	Format: 20
DATAID	Format: 5()
DATALENGTH	Format: 5()
ECID	Format: 5()
LENGTH	Format: 5()
PPID	Format: 20
RCMD	Format: 20
REPGSZ	Format: 5()
RPTID	Format: 5()
SEQNUM	Format: 52
SMPLN	Format: 5()
SVID	Format: 5()
TEXT	Format: 20
TOTSMP	Format: 5()
TRID	Format: 5()
VID	Format: 5()

8.4 *Variable Item List* — The following variable data items (several from the Variable Item Dictionary in SEMI E5) shall be implemented when the associated GEM feature is implemented. Format restrictions are noted. The Well-Known Name is a case-sensitive, standardized name for use in other specifications that reference the variable.

8.4.1 *Well-Known Names*

8.4.1.1 Implementers of this Specification may choose any names for variables, collection events and alarms. For example, SEMI E116 defines a collection event for all Equipment Performance Tracking transitions. Implementers

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

might define the CENAME for this collection event as EPTStateChange, E116_EPTStateChange, EPTTransition, or CollectionEvent116. All of these CENAME are valid.

8.4.1.2 For end users, it can be useful to map the names for variables, collection events and alarms in a specific implementation to the variables, collection events and alarms defined in this Specification. Well-known names are standardized names for the variables, collection events and alarms defined in this Specification.

8.4.1.3 Equipment implementing this Specification can publish a map between the names in their implementation and the well-known names defined in this Appendix using a SEMI E172 SEDD file. The map can also be published in the SEMI E30 documentation.

8.4.1.4 New implementations can use well-known names as the SECS item name (CENAME, SVNAME, etc.).

8.4.1.5 The lists of well-known variables, collection events and alarms do not intend to imply that implementations of this Specification shall include all the listed variables, collection events and alarms.

8.4.1.6 For completeness, well-known tables for variables, collection events and alarms included yet deliberately left empty when no applicable variable, collection event or alarm is defined in this Specification.

8.4.1.7 Well-known names are case-sensitive.

8.4.2 *Format Restrictions and Well-Known Names for Variables*

8.4.2.1 This section defines well-known names for variables currently defined in this Specification.

8.4.2.2 Refer to SEMI E5 for more information on the variable classes.

Table 12 Format Restrictions and Well-Known Names for Variables

Variable	*Variable Class*	*Format Restrictions*	*Well-Known Name*
AlarmID	DV	5()	E30:AlarmID
EventLimit	DV		E30:EventLimit
LimitVariable	DV		E30:LimitVariable
PPChangeName	DV	20	E30:PPChangeName
PPChangeStatus	DV		E30:PPChangeStatus
PPError	DV		E30:PPError
PPErrorPPID	DV		E30:PPErrorPPID
TransitionType	DV		E30:TransitionType
ECID	DV		E30:EquipmentConstant
OperatorCommand	DV		E30:OperatorCommand
AlarmsEnabled	SV		E30:AlarmsEnabled
AlarmsSet	SV		E30:AlarmsSet
Clock	SV		E30:Clock
ControlState	SV		E30:ControlState
EqpSerialNum	SV		E30:EquipmentSerialNumber
E30EquipmentSupplier	SV		E30:EquipmentSupplier
E30SupportedPPOptions	SV		E30:SupportedPPOptions
EventsEnabled	SV		E30:EventsEnabled
PPExecName	SV	0, 20	E30:PPExecName
PreviousProcessState	SV		E30:PreviousProcessState
ProcessState	SV		E30:ProcessState
SpoolCountActual	SV		E30:SpoolCountActual
SpoolCountTotal	SV		E30:SpoolCountTotal
SpoolFullTime	SV		E30:SpoolFullTime

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Variable	*Variable Class*	*Format Restrictions*	*Well-Known Name*
SpoolStartTime	SV		E30:SpoolStartTime
E30EquipmentInitiatedPPOption	EC		E30:EquipmentInitiatedPPOption
EqpName	EC		E30:EquipmentName
EnableSpooling	EC		E30:EnableSpooling
EstablishCommunicationsTimeout	EC		E30:EstablishCommunicationsTimeout
MaxSpoolTransmit	EC		E30:MaxSpoolTransmit
OverWriteSpool	EC		E30:OverWriteSpool
TimeFormat	EC		E30:TimeFormat

8.5 *Data Collection Item Properties* — The properties of data collection items are listed below.

8.5.1 Properties of an alarm include the alarm identifier (ALID).

8.5.2 Properties of a collection event include a collection event name (CENAME), and collection event identifier (CEID).

8.5.3 Properties of a data variable include a data variable name (DVVALNAME), data variable value (DVVAL), and data variable identifier (DVID).

8.5.4 Properties of an equipment constant include an equipment constant name (ECNAME), equipment constant value (ECV), and equipment constant identifier (ECID).

8.5.5 Properties of a status variable include a status variable name (SVNAME), status variable value (SV), and status variable identifier (SVID).

9 Collection Events

9.1 Table 8 provides the list of collection events required to support the capabilities addressed within this Standard. Also shown are typical variable data that would most likely be included in the associated collection event report and a reference to the event trigger and to the appropriate section of the Standard.

9.2 This list does not represent all events that might be needed to properly monitor/control equipment. Many events are unique to the specific equipment characteristics. The needed additions are a matter for other standards and for collaboration between equipment supplier and user.

9.3 Refer to § 8.4 for further detail of variable data items.

9.4 The Well-Known Name is a case-sensitive, standardized name for use in other specifications that reference the collection event.

9.5 When collection events correspond to the Processing state model, the well-known name uses the naming convention ‘E30:Processing:<transition number>:<from state>-<to state>’ where ‘<transition number>’ is the respective state model transition number using 3 digits and leading zeros when necessary, ‘<from state>’ is the name of the initial state without spaces, and “<to state>” is the name of the ending state without spaces.

9.6 When an alarm has a ‘well known’ name, the alarm’s corresponding Alarm Set collection event uses the convention ‘Alarm:001:<alarm well-known name>:Set’. The corresponding Alarm Clear collection event uses the convention ‘Alarm:002:<alarm well-known name>:Clear’.

9.7 When a variable is a assigned a limits monitoring collection event, the well-known name for that collection event uses the convention ‘E30:LimitsMonitoring:<variable name>’.

Table 13 GEM-Defined Collection Events

Event Designation	*Typical Variable Data*	*Reference*	*Well-Known Name*
Control-Related Events:		§ 6.5	
Equipment OFF-LINE	ControlState, Clock	ON-LINE->OFF-LINE	E30:Control:Online-OffLine

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Event Designation	*Typical Variable Data*	*Reference*	*Well-Known Name*
Control State LOCAL	ControlState, Clock	REMOTE->LOCAL or OFF-LINE->LOCAL	E30:Control:Local
Control State REMOTE	ControlState, Clock	LOCAL->REMOTE or OFF-LINE->REMOTE	E30:Control:Remote
Operator Command Issued	OperatorCommand	Operator Activity while REMOTE state is active.	E30:OperatorCommandIssued
Processing-Related Events:	Any transition in the implemented processing state model shall have a corresponding collection event.	§ 6.6	Refer to § 9.5
Processing Started	Clock, PreviousProcessState	Entry into EXECUTING state.	E30:ProcessingStarted
Processing Completed	Clock, PreviousProcessState	Normal exit of EXECUTING state.	E30:ProcessingCompleted
Processing Stopped	Clock, PreviousProcessState	Result of STOP command from host or operator.	E30:ProcessingStopped
Processing State Change	Clock, ProcessState, PreviousProcessState	Any processing state transition.	E30:ProcessingStateChange
Alarm Management Events:		§ 6.4	
$Alarm_n$Detected	Clock, AlarmID, AlarmsSet, Associated variable data	$ALARM_n$CLEAR->$ALARM_n$SET	Refer to § 9.6
$Alarm_n$Cleared	Clock, AlarmID, AlarmsSet	$ALARM_n$SET->$ALARM_n$CLEAR	Refer to § 9.6
Equipment Constant Events:		§ 7.6	
Operator Equipment Constant Change	ECID	Operator activity	E30:OperatorEquipmentConstantChange
Limits Monitoring:		§ 7.3.4	
Limit Zone $Transition_n$ (separate CEID per variable)	Clock, LimitVariable, EventLimit, Transition Type	Entry into BELOW LIMIT or ABOVE LIMIT states.	Refer to § 9.7
Process Program Management Events:		§ 7.7	
Process Program Change (required only if process programs are implemented)	PPChangeName, PPChangeStatus	Operator activity	E30:ProcessProgramChange
Process Program(s) Selected	PPExecName	Operator/Host activity	E30:ProcessRecipeSelected
Process Program Verification Failed	PPError, PPErrorPPID		E30:ProcessProgramVerificationFailed
Process Program Validation Failed	PPError, PPErrorPPID		E30:ProcessProgramValidationFailed
Material Movement Events:		§ 7.8	
Material Received	Clock		E30:MaterialReceived
Material Removed	Clock		E30:MaterialRemoved
Spooling Events:		§ 7.12	

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Event Designation	*Typical Variable Data*	*Reference*	*Well-Known Name*
Spooling Activated	SpoolStartTime	SPOOL INACTIVE->SPOOL ACTIVE	E30:Spooling:Activated
Spooling Deactivated	SpoolCountTotal	SPOOL OUTPUT->SPOOL INACTIVE	E30:Spooling:Deactivated
Spool Transmit Failure	Clock, SpoolCountActual SpoolCountTotal	TRANSMIT SPOOL->NO SPOOL OUTPUT	E30:Spooling:TransmitFailure
Terminal Services Events:		§ 7.9	
Message Recognition	Clock	Operator	E30:MessageRecognition

10 SECS-II Message Subset

10.1 This section lists the required set of SECS-II messages as referenced in this Standard. Definitions for these messages can be found in SEMI E5. All primary messages (for which SEMI E5 defines replies) should have replies available. Replies are required or optional as specified in SEMI E5.

STREAM 1: Equipment Status

S1F1 Are You There Request (R)	S,H<->E
S1F2 On-Line Data (D)	S,H<->E
S1F3 Selected Equipment Status Request (SSR)	S,H->E
S1F4 Selected Equipment Status Data (SSD)	M,H<-E
S1F11 Status Variable Namelist Request (SVNR)	S,H->E
S1F12 Status Variable Namelist Reply (SVNRR)	M,H<-E
S1F13 Establish Communications Request (CR)	S,H<->E
S1F14 Establish Communications Request Acknowledge (CRA)	S,H->E
S1F15 Request OFF-LINE (ROFL)	S,H<-E
S1F16 OFF-LINE Acknowledge (OFLA)	S,H->E
S1F17 Request ON-LINE (RONL)	S,H<-E
S1F18 ON-LINE Acknowledge (ONLA)	S,H<-E
S1F21 Data Variable Namelist Request (DVNR)	S,H->E
S1F22 Data Variable Namelist (DVN)	M,H<-E
S1F23 Collection Event Namelist Request (CENR)	S,H->E
S1F24 Collection Event Namelist (CEN)	M,H<-E

STREAM 2: Equipment Control and Diagnostics

S2F13 Equipment Constant Request (ECR)	S,H->E
S2F14 Equipment Constant Data (ECD)	M,H<-E
S2F15 New Equipment Constant Send (ECS)	S,H->E
S2F16 New Equipment Constant Acknowledge (ECA)	S,H<-E
S2F17 Date and Time Request (DTR)	S,H<->E
S2F18 Date and Time Data (DTD)	S,H<->E
S2F23 Trace Initialize Send (TIS)	S,H->E
S2F24 Trace Initialize Acknowledge (TIA)	S,H<-E
S2F29 Equipment Constant Namelist Request (ECNR)	S,H->E
S2F30 Equipment Constant Namelist (ECN)	M,H<-E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

S2F31 Date and Time Send (DTS)	S,H->E
S2F32 Date and Time Acknowledge (DTA)	S,H<-E
S2F33 Define Report (DR)	M,H->E
S2F34 Define-Report Acknowledge (DRA)	S,H<-E
S2F35 Link Event Report (LER)	M,H->E
S2F36 Link Event Report Acknowledge (LERA)	S,H<-E
S2F37 Enable/Disable Event Report (EDER)	S,H->E
S2F38 Enable/Disable Event Report Acknowledge (EDEA)	S,H<-E
S2F39 Multi-Block Inquire (DMBI)	S,H->E
S2F40 Multi-Block Grant (DMBG)	S,H<-E
S2F41 Host Command Send (HCS)	S,H->E
S2F42 Host Command Acknowledge (HCA)	S,H<-E
S2F43 Reset Spooling Streams and Functions (RSSF)	S,H->E
S2F44 Reset Spooling Acknowledge (RSA)	M,H<-E
S2F45 Define Variable Limit Attributes (DVLA)	M,H->E
S2F46 Variable Limit Attribute Acknowledge (VLAA)	M,H<-E
S2F47 Variable Limit Attribute Request (VLAR)	S,H->E
S2F48 Variable Limit Attributes Send (VLAS)	M,H<-E
S2F49 Enhanced Remote Command	M,H->E
S2F50 Enhanced Remote Command Acknowledge	M,H<-E
S2F51 Request List of Report Identifiers (RQRPTID)	S,H->E
S2F52 Return List of Report Identifiers (RTRPTID)	M,H<-E
S2F53 Request List of Report Definitions (RQLRD)	M,H<-E
S2F54 Return List of Report Definitions (RTLRD)	M,H<-E
S2F55 Request Event Report Links (RQERL)	M,H<-E
S2F56 Return Event Report Links (RTERL)	M,H<-E
S2F57 Request List of Enabled Events (RQEE)	S,H->E
S2F58 Return List of Enabled Events (RTEE)	M,H<-E
S2F59 Request Selected Spooling Stream and Functions (RQSSF)	S,H->E
S2F60 Return Selected Spooling Stream and Functions (RTSSF)	M,H<-E
S2F61 Request List of Trace Identifiers (RQTRID)	S,H->E
S2F62 Return List of Trace Identifiers (RTTRID)	M,H<-E
S2F63 Request Trace Definitions (RQTD)	M,H<-E
S2F64 Return Trace Definitions (RTTD)	M,H<-E

STREAM 5: Exception (Alarm) Reporting

S5F1 Alarm Report Send (ARS)	S,H<-E
S5F2 Alarm Report Acknowledge (ARA)	S,H->E
S5F3 Enable/Disable Alarm Send (EAS)	S,H->E
S5F4 Enable/Disable Alarm Acknowledge (EAA)	S,H<-E
S5F5 List Alarms Request (LAR)	S,H->E
S5F6 List Alarm Data (LAD)	M,H<-E
S5F7 List Enabled Alarm Request (LEAR)	S,H->E
S5F8 List Enabled Alarm Data (LEAD)	S,H<-E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

STREAM 6: Data Collection

S6F1 Trace Data Send (TDS)	S,H<-E
S6F2 Trace Data Acknowledge (TDA)	S,H->E
S6F5 Multi-block Data Send Inquire (MBI)	S,H<-E
S6F6 Multi-block Grant (MBG)	S,H->E
S6F11 Event Report Send (ERS)	M,H<-E
S6F12 Event Report Acknowledge (ERA)	S,H->E
S6F15 Event Report Request (ERR)	S,H->E
S6F16 Event Report Data (ERD)	M,H<-E
S6F19 Individual Report Request (IRR)	S,H->E
S6F20 Individual Report Data (IRD)	M,H<-E
S6F23 Request Spooled Data (RSD)	S,H->E
S6F24 Request Spooled Data Acknowledgement Send (RSDAS)	S,H<-E

STREAM 7: Process Program Load – Required only if equipment implements process programs

S7F1 Process Program Load Inquire (PPI)	S,H<->E
S7F2 Process Program Load Grant (PPG)	S,H<->E
S7F3 Process Program Send (PPS)	M,H<->E
S7F4 Process Program Acknowledge (PPA)	S,H<->E
S7F5 Process Program Request (PPR)	S,H<->E
S7F6 Process Program Data (PPD)	M,H<->E
S7F17 Delete Process Program Send (DPS)	S,H->E
S7F18 Delete Process Program Acknowledge (DPA)	S,H<-E
S7F19 Current EPPD Request (RER)	S,H->E
S7F20 Current EPPD Data (RED)	M,H<-E
S7F23 Formatted Process Program Send (FPS)	M,H<->E
S7F24 Formatted Process Program Acknowledge (FPA)	S,H<->E
S7F25 Formatted Process Program Request (FPR)	S,H<->E
S7F26 Formatted Process Program Data (FPD)	M,H<->E
S7F27 Process Program Verification Send (PVS)	S,H<-E
S7F28 Process Program Verification Acknowledge (PVA)	S,H->E
S7F29 Process Program Verification Inquire (PVA)	S,H<-E
S7F30 Process Program Verification Grant (PVG)	S,H->E

STREAM 9: System Errors

S9F1 Unrecognized Device ID (UDN)	S,H<-E
S9F3 Unrecognized Stream Type (USN)	S,H<-E
S9F5 Unrecognized Function Type (UFN)	S,H<-E
S9F7 Illegal Data (IDN)	S,H<-E
S9F9 Transaction Timer Timeout (TTN)	S,H<-E
S9F11 Data Too Long (DLN)	S,H<-E
S9F13 Conversation Timeout (CTN)	S,H<-E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

STREAM 10: Terminal Services

S10F1 Terminal Request (TRN)	S,H<-E
S10F2 Terminal Request Acknowledge (TRA)	S,H->E
S10F3 Terminal Display, Single (VTN)	S,H->E
S10F4 Terminal Display, Single Acknowledge (VTA)	S,H<-E
S10F5 Terminal Display, Multi-block (VMN)	M,H->E
S10F6 Terminal Display, Multi-block Acknowledge (VMA)	S,H<-E
S10F7 Multi-block Not Allowed (MNN)	S,H<-E

STREAM 21: Item Transfer

S21F1 Item Load Inquire (ITEMLI)	S,H<->E
S21F2 Item Load Grant (ITEMLG)	S,H<->E
S21F3 Item Send (ITEMS)	M,H<->E
S21F4 Item Send Acknowledge (ITEMSA)	S,H<->E
S21F5 Item Request (ITEMR)	S,H<->E
S21F6 Item Data (ITEMD)	M,H<->E
S21F7 Item Type List Request (ITEMTLRQ)	S,H<->E
S21F8 Item Type List Results (ITEMTLRS)	M,H<->E
S21F9 Supported Item Type List Request (ITEMTYPERQ)	S,H<->E
S21F10 Supported Item Type List Results (ITEMTYPERS)	M,H<->E
S21F11 Item Delete (ITEMDEL)	M,H<->E
S21F12 Item Delete Acknowledge (ITEMDELACK)	M,H<->E
S21F13 Request Permission To Send Item Using S21F17 (RPTSIF17)	S,H<->E
S21F14 Grant Permission To Send Item Using S21F17 (GPTSIF17)	S,H<->E
S21F15 Item Request Using S21F17 (ITEMRF17)	S,H<->E
S21F16 Item Request Grant Using S21F17 (ITEMRGF17)	M,H<->E
S21F17 Send Item Part (SIP)	M,H<->E
S21F18 Send Item Part Acknowledge (IPACK)	S,H<->E
S21F19 Item Type Feature Support (ITEMFS)	S,H<->E
S21F20 Item Type Feature Support Results (ITEMFSR)	M,H<->E

11 GEM Compliance

11.1 This section defines compliance to the GEM Standard. It describes the fundamental GEM requirements and additional GEM capabilities. It provides references to other sections of the Standard where detailed requirements are located. This section also defines standard terminology and documentation that can be used by equipment suppliers and device manufacturers to describe compliance with this Standard.

11.1.1 The GEM Standard contains two types of specifications:

- fundamental GEM requirements and
- requirements pertaining to additional GEM capabilities.

11.1.2 The fundamental GEM requirements form the foundation of the GEM Standard. The additional GEM capabilities provide functionality required for some types of factory automation or functionality applicable to specific types of equipment.

11.2 *Fundamental GEM Requirements* — All equipment shall comply with the fundamental GEM requirements listed in Table 14. Compliance to these requirements involves precise and complete adherence to all sections of the GEM Standard referenced in Table 14.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

11.2.1 *Fundamental GEM Requirements*

Table 14 References for Fundamental GEM Requirements

Requirement	*Section References*
State Models	6.2, 6.4
Equipment Processing States	6.6
Host-Initiated S1F13/F14 Scenario	7.2.5.1
Event Notification	7.3.1.2
On-line Identification	7.3.6
Error Messages	7.10
Control (Operator-Initiated)	7.13
Documentation	11.4

11.2.2 In addition, compliance requires adherence to the portions of the following sections that are applicable to the fundamental GEM requirements:

- Variable data items (GEM, § 8)
- SECS-II data item restrictions (GEM, § 8)
- Collection events (GEM, § 9)

11.3 *GEM Compliance* — Equipment is GEM-compliant for a specific GEM capability if, and only if, the following three criteria are met:

- The fundamental GEM requirements are satisfied.
- The capability is implemented to conform with all applicable definitions, descriptions, and requirements defined for the capability in this Standard.
- The equipment does not exhibit behavior related to this capability that conflicts with the GEM behavior defined for the capability.

11.3.1 For example, equipment that provides SECS-II messages for management of process programs shall precisely implement the GEM Process Program Management capability to be 'GEM-Compliant for Process Program Management.'

11.3.2 Equipment may supply additional functionality not specified in the GEM Standard by using any messages defined in the SECS-II Standard as long as the additional functionality does not conflict with compliance to GEM capabilities.

11.3.3 The GEM capabilities are built upon the fundamental GEM requirements and present GEM-compliant behavior to the host when they are not obstructed by conflicting functionality. Additional non-GEM capabilities and non-obstructing extensions to GEM capabilities provide additional functionality while maintaining GEM behavior from the host view.

11.3.4 One additional term is defined to facilitate discussion of GEM capability. Equipment is 'Fully GEM Capable' if and only if it meets the following two criteria:

- The equipment supplies all the GEM capabilities.
- Every implemented GEM capability is GEM-Compliant.

11.4 *Documentation* — This section describes documentation requirements in addition to those specified in § 6 and § 7 of this Standard. All documentation of the SECS-II interface shall be supplied as a single volume, including Message Documentation, a Compliance Statement and the documentation required by § 6 and § 7.

11.4.1 *Message Documentation* — The equipment supplier shall provide message documentation in conformance with § 11 (Message Documentation) of SEMI E5.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

11.4.2 *GEM Compliance Statement* — The SECS-II interface documentation provided by an equipment supplier shall address GEM compliance. This documentation shall include a GEM Compliance Statement that accurately indicates for each capability whether it has been implemented and whether it has been implemented in a GEM-compliant manner. The format for this statement is supplied as Table 15.

11.4.2.1 The table consists of three columns. The first column lists the requirements and capabilities. The second column identifies the section(s) that contain the detailed requirements for implementing a GEM capability. The other two columns pose questions to the supplier:

- *Implemented* — Does the equipment provide functionality that is similar to that defined for the GEM requirement or capability?
- *GEM-Compliant* — Has that requirement or capability been implemented in a GEM-compliant manner?

11.4.2.2 Legacy Capabilities are considered optional and less important to implement than Additional Capabilities.

Table 15 GEM Compliance Statement

E30 Version: _____	*GEM COMPLIANCE STATEMENT*		
FUNDAMENTAL GEM REQUIREMENTS	*Section Reference*	*IMPLEMENTED*	*GEM-COMPLIANT*
State Models	6.2, 6.4	□ Yes □ No	□ Yes □ No
Equipment Processing States	6.6	□ Yes □ No	□ Yes □ No
Host-Initiated S1F13/F14 Scenario	7.2.5.1	□ Yes □ No	□ Yes □ No
Event Notification	7.3.1.2	□ Yes □ No	□ Yes □ No
On-Line Identification	7.3.6	□ Yes □ No	□ Yes □ No
Error Messages	7.10	□ Yes □ No	□ Yes □ No
Control (Operator Initiated)	7.13	□ Yes □ No	□ Yes □ No
Documentation	11.4	□ Yes □ No	□ Yes □ No
ADDITIONAL CAPABILITIES		*IMPLEMENTED*	*GEM-COMPLIANT*
Establish Communications	7.2, 6.4	□ Yes □ No	□ Yes □ No
Dynamic Event Report Configuration	7.3.1.2	□ Yes □ No	□ Yes □ No
Data Variable and Collection Event Namelist Requests	7.3.1.4	□ Yes □ No	□ Yes □ No
Event Report Information	7.3.1.5	□ Yes □ No	□ Yes □ No
Variable Data Collection	7.3.2	□ Yes □ No	□ Yes □ No
Trace Data Collection	7.3.3	□ Yes □ No	□ Yes □ No
Trace Information	7.3.3.6	□ Yes □ No	□ Yes □ No
Status Data Collection	7.3.5	□ Yes □ No	□ Yes □ No
Unique Identification	7.3.7	□ Yes □ No	□ Yes □ No
Alarm Management	7.4	□ Yes □ No	□ Yes □ No
Remote Control	7.5	□ Yes □ No	□ Yes □ No
Equipment Constants	7.6	□ Yes □ No	□ Yes □ No
Process Program Management	7.7	□ Yes □ No	□ Yes □ No
Stream 7 Unformatted	7.7.7	□ Yes □ No	□ Yes □ No
Stream 21 Single Message Unformatted	7,7.8	□ Yes □ No	□ Yes □ No
Stream 21 Multiple Message Unformatted	7.7.9	□ Yes □ No	□ Yes □ No
Stream 7 Formatted	7.7.10	□ Yes □ No	□ Yes □ No
Material Movement	7.8	□ Yes □ No	□ Yes □ No
Equipment Terminal Services	7.9	□ Yes □ No	□ Yes □ No

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Clock	7.11	□ Yes □ No	□ Yes □ No
Spooling	7.12	□ Yes □ No	□ Yes □ No
Spooling Information	7.12.6	□ Yes □ No	□ Yes □ No
Control (Host-Initiated)	7.13.5.1	□ Yes □ No	□ Yes □ No
Documentation File Transfer	7.14	□ Yes □ No	□ Yes □ No
SEDD File Transfer	7.15	□ Yes □ No	□ Yes □ No
LEGACY CAPABILITIES		*IMPLEMENTED*	*GEM-COMPLIANT*
Limits Monitoring	7.3.4	□ Yes □ No	□ Yes □ No

11.4.3 The equipment supplier may provide documentation on the format of required data items (refer to § 8) using SMN (refer to SEMI E173) or SECS Message Language™ Notation (SML®). The SMN element names and SML formats are provided in Table 16.

Table 16 SMN Elements and SML Notation

Item Format	*SECS-II Format Code*		*SMN Format*	*SML Item Format Mnemonic*
	Binary	*Octal*		
LIST	000000	00	LST	L [length]
Binary	001000	10	BIN	B
Boolean	001001	11	BOO	BOOLEAN
ASCII	010000	20	ASC	A [length] or A [min., max.]
JIS-8	010001	21	JIS	J [length] or J [min., max.]
2-byte character	010010	22	MBC	
8-byte integer (signed)	011000	30	SI8	I8
1-byte integer (signed)	011001	31	SI1	I1
2-byte integer (signed)	011010	32	SI2	I2
4-byte integer (signed)	011100	34	SI4	I4
8-byte floating point	100000	40	FP8	F8
4-byte floating point	100100	44	FP4	F4
8-byte integer (unsigned)	101000	50	UI8	U8
1-byte integer (unsigned)	101001	51	UI1	U1
2-byte integer (unsigned)	101010	52	UI2	U2
4-byte integer (unsigned)	101100	54	UI4	U4

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

APPENDIX 1
ALTERNATIVE SCENARIOS FOR LARGE FORMATTED PROCESS PROGRAMS

NOTICE: The material in this Appendix is an official part of SEMI E30 and was approved by full letter ballot procedures on December 21, 2022.

A1-1 Large Formatted Process Programs

A1-1.1 Although large formatted process programs are removed from the main SEMI E30 Specification, they are retained in this Appendix for backward compatibility.

A1-1.2 Formatted process programs should be verified immediately after transferring a process program to the equipment.

A1-2 Scenarios for Large Formatted Process Programs

A1-2.1 Host Requests a Process Program From the Equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Large Formatted Process Program Request	S7F43-->		
		<--S7F44	Large Formatted Process Program Acknowledge Acknowledge may be "accepted and will be performed later with completion signaled". When the equipment is ready to send:
		<--S13F1	Send Data Set Send
Send Data Set Send Acknowledge	S13F2-->		
Open Data Set Request	S13F3-->		
		<--S13F4	Open Data Set Data
[START] Repeat until ACKC13 indicates error:			
Read Data Set Request	S13F5-->		
		<--S13F6	Read Data Set Data
[END]			
			[IF] CEID for upload event enabled [THEN] [IF] ACKC13 is "ERROR: END OF DATA" [THEN]
		<--S6F11	Send Event Report "successful upload"
Event Report Acknowledge	S6F12-->		
			[ELSE]
		<--S6F11	Send Event Report "bad upload"
Event Report Acknowledge	S6F12-->		
			[END IF]

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
Close Data Set Send                     S13F7-->
                                                <--S13F8          Close Data Set Acknowledge
```

A1-2.2 Equipment Sends a Process Program to the Host:

```
          COMMENT                   HOST              EQUIPMENT            COMMENT
                                                  <--S7F39          Large Formatted Process
                                                                    Program Send
Large Formatted Process                 S7F40-->
Program Acknowledge
                                                  <--S13F1          Send Data Set Send
Send Data Set Send                      S13F2-->
Acknowledge
Open Data Set request                   S13F3-->
                                                  <--S13F4          Open Data Set Data
[START] repeat until
ACKC13 is error:
Read Data Set Request                   S13F5-->

                                                  <--S13F6          Read Data Set Data
[END]
                                                                    [IF]
                                                                    CEID for upload event enabled
                                                                    [THEN]
                                                                    [IF]
                                                                    ACKC13 is "ERROR: END OF
                                                                       DATA"
                                                                    [THEN]
                                                  <--S6F11          Send Event Report
                                                                    “successful upload”
Event Report Acknowledge                S6F12-->
                                                                    [ELSE]
                                                  <--S6F11          Send Event Report “bad
                                                                       upload”
Event Report Acknowledge                S6F12-->
                                                                    [END IF]
Close Data Set Send                     S13F7-->
                                                  <--S13F8          Close Data Set Acknowledge
```

A1-2.3 Host Sends a Process Program to the Equipment:

```
          COMMENT                   HOST              EQUIPMENT            COMMENT
Large Formatted Process                 S7F39-->
Program Send
                                                  <--S7F40          Large Formatted Process
                                                                    Program Acknowledge
                                                  <--S13F3          Open Data Set Request
Open Data Set Data                      S13F4-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                                  [START] repeat until ACKC13
                                                                     is
                                                                  error:
                                          <--S13F5                Read Data Set Request

Read Data Set Data                     S13F6-->

                                                                  [END]

                                          <--S13F7                Close Data Set Send

Close Data Set Acknowledge             S13F8-->

                                                                  [IF]
                                                                  S7F27 is multi-block
                                                                  [THEN]
                                          <--S7F29                Process Program Verification
                                                                  Inquire

Process Program Verification           S7F30-->
Grant

                                                                  [END IF]

                                          <--S7F27                Process Program
                                                                     Verification Send

Process Program Verification           S7F28-->
Acknowledge
```

A1-2.4 Equipment Requests a Process Program from the Host:

```
COMMENT                                HOST                EQUIPMENT                COMMENT

                                                           <--S7F43                Large Formatted Process
                                                                                   Program Request

Large Formatted Process                S7F44-->
Program Acknowledge

                                                           <--S13F3                Open Data Set Request

Open Data Set Data                     S13F4-->

                                                                                   [START] repeat until ACK13 is
                                                                                   error:
                                                           <--S13F5                Read Data Set Request

Read Data Set Data                     S13F6-->

                                                                                   [END]

                                                           <--S13F7                Close Data Set Send

Close Data Set Acknowledge             S13F8-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                        [IF]
                                                        S7F27 is multi-block
                                                        [THEN]
                                   <--S7F29             Process  Program  Verification
                                                           Inquire
Process Program Verification
   Grant                       S7F30-->

                                                        [END IF]
                                   <--S7F27             Process Program Verification

                                   Send
                               S7F28-->
Process Program Verification
   Acknowledge
```

A1-3 SECS-II Message Subset for Large Formatted Process Programs

STREAM 7:

S7F39 Large Formatted Process Program Send	S,H<->E
S7F40 Large Formatted Process Program Acknowledge	S,H<->E
S7F43 Large Formatted Process Program Request	S,H<->E
S7F44 Large Formatted Process Program Acknowledge	S,H<->E

STREAM 13:

S13F1 Send Data Set Send (DSSS)	S,H<->E
S13F2 Send Data Set Acknowledge (DSSA)	S,H<->E
S13F3 Open Data Set Request (DSOR)	S,H<->E
S13F4 Open Data Set Data (DSOD)	S,H<->E
S13F5 Read Data Set Request (DSSR)	S,H<->E
S13F6 Read Data Set Data (DSRD)	M,H<->E
S13F7 Close Data Set Send (DSCS)	S,H<->E
S13F8 Close Data Set Acknowledge (DSCA)	S,H<->E
S13F9 Reset Data Set Send (DSRS)	S,H<->E
S13F10 Reset Data Set Acknowledge (DSRA)	S,H<->E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

APPENDIX 2
ALTERNATIVE SCENARIOS FOR LARGE UNFORMATTED PROCESS PROGRAMS

NOTICE: The material in this Appendix is an official part of SEMI E30 and was approved by full letter ballot procedures on December 21, 2022.

A2-1 Large Unformatted Process Programs

A2-1.1 Although large unformatted process programs have been removed from the main SEMI E30 Specification, they are retained in this Appendix for backward compatibility.

A2-1.2 The terms upload and download are retained as originally used in SEMI E30.

A2-2 Scenarios for Large Unformatted Process Programs

A2-2.1 Host Requests a Process Program from the Equipment:

```
COMMENT                          HOST          EQUIPMENT          COMMENT

Large Process Program Request        S7F41-->
                                             <--S7F42          Large Process Program
                                                               Acknowledge

                                                               Acknowledge may be "accepted
                                                               and will be performed later
                                                               with completion signaled".
                                                               When the equipment is ready
                                                               to send:
                                             <--S13F1          Send Data Set Send

Send Data Set Send                   S13F2-->
Acknowledge

Open Data Set request                S13F3-->
                                             <--S13F4          Open Data Set Data

[START]
Repeat until ACKC13 indicates
   error:
Read Data Set Request                S13F5-->
                                             <--S13F6          Read Data Set Data

[END]
                                                               [IF]
                                                               CEID for upload event enabled
                                                               THEN]
                                                               [IF]
                                                               ACKC13 is "ERROR: END OF DATA"
                                                               [Then]
                                             <--S6F11          Send Event Report

Event Report Acknowledge             S6F12-->                  "successful upload"

                                                               [ELSE]

                                             <--S6F11          Send Event Report "bad upload"

Event Report Acknowledge             S6F12-->

                                                               [END_IF]
                                                               [END_IF]
Close Data Set Send                  S13F7-->
                                             <--S13F8          Close Data Set Acknowledge
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

A2-2.2 Equipment Sends a Process Program to the Host:

```
COMMENT                          HOST          EQUIPMENT          COMMENT
                                                <--S7F37          Large Process Program Send

Large Process Program            S7F38-->
Acknowledge

                                                <--S13F1          Send Data set Send

Send Data Set Send               S13F2-->
Acknowledge

Open Data Set Request            S13F3-->

                                                <--S13F4          Open Data Set Data

[START] Repeat until
ACKC13 is error:
Read Data Set Request            S13F5-->

                                                <--S13F6          Read Data Set Data

[END]

                                                                  [IF]
                                                                  CEID for upload event enabled
                                                                  [THEN]
                                                                  [IF]
                                                                  ACKC13 is "ERROR: END OF DATA"
                                                                  [THEN]
                                                <--S6F11          Send Event Report

Event Report Acknowledge         S6F12-->                         "successful upload"

                                                                  [ELSE]

                                                <--S6F11          Send Event Report "bad upload"

Event Report Acknowledge         S6F12-->                         [END_IF]
                                                                  [END_IF]

Close Data Set Send              S13F7-->

                                                <--S13F8          Close Data Set Acknowledge
```

A2-2.3 Host Sends a Process Program to the Equipment:

```
COMMENT                          HOST          EQUIPMENT          COMMENT

Large Process Program Send       S7F37-->

                                                <--S7F38          Large Process Program
                                                                  Acknowledge
                                                <--S13F3          Open Data Set Request

Open Data Set Data               S13F4-->

                                                                  [START] repeat until ACKC13 is
                                                                  error:
                                                <--S13F5          Read Data Set Request

Read Data Set Data               S13F6-->

                                                                  [END]

                                                <--S13F7          Close Data Set Send

Close Data Set Acknowledge       S13F8-->

                                                                  [IF]
                                                                  S7F27 is multi-block
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                        [THEN]
                                                        Process Program Verification
                                        <--S7F29        Inquire
Process Program Verification        S7F30-->
Grant
                                                        [END_IF]
                                        <--S7F27        Process Program Verification
                                                           Send
Process Program Verification        S7F28-->
Acknowledge
```

A2-2.4 Equipment Requests a Process Program from the Host:

```
        COMMENT                 HOST          EQUIPMENT            COMMENT
                                                <--S7F41        Large Process Program Request
Large Process Program             S7F42-->
Program Acknowledge
                                                <--S13F3        Open Data Set Request
Open Data Set Data                S13F4-->
                                                                [START] repeat until ACKC13 is
                                                                error:
                                                <--S13F5        Read Data Set Request
Open Data Set Data                S13F6-->
                                                                [END]
                                                <--S13F7        Close Data Set Send
Close Data Set Acknowledge        S13F8-->
                                                                [IF]
                                                                S7F27 is multi-block
                                                                [THEN]
                                                <--S7F29        Process Program Verification
                                                                Inquire
Process Program Verification      S7F30-->
Grant
                                                                [END_IF]
                                                <--S7F27        Process Program
Process Program Verification      S7F28-->                      Verification Send
Acknowledge
```

A2-3 SECS-II Message Subset for Large Unformatted Process Programs

STREAM 7:

S7F37 Large Process Program Send	S,H<->E
S7F38 Large Process Program Acknowledge	S,H<->E
S7F41 Large Process Program Request	S,H<->E
S7F42 Large Process Program Acknowledge	

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

STREAM 13:

S13F1 Send Data Set Send (DSSS)	S,H<->E
S13F2 Send Data Set Acknowledge (DSSA)	S,H<->E
S13F3 Open Data Set Request (DSOR)	S,H<->E
S13F4 Open Data Set Data (DSOD)	S,H<->E
S13F5 Read Data Set Request (DSSR)	S,H<->E
S13F6 Read Data Set Data (DSRD)	M,H<->E
S13F7 Close Data Set Send (DSCS)	S,H<->E
S13F8 Close Data Set Acknowledge (DSCA)	S,H<->E
S13F9 Reset Data Set Send (DSRS)	S,H<->E
S13F10 Reset Data Set Acknowledge (DSRA)	S,H<->E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

APPENDIX 3
ALTERNATIVE SCENARIOS FOR SEMI E42 RECIPES

NOTICE: The material in this Appendix is an official part of SEMI E30 and was approved by full letter ballot procedures on December 21, 2022.

A3-1 SEMI E42 Recipes

A3-1.1 Although SEMI E42 recipes have been removed from the main SEMI E30 Specification, they are retained in this Appendix for backward compatibility.

A3-1.2 The terms upload and download are retained as originally used in SEMI E30.

A3-2 Definitions Applicable to SEMI E42 Recipes

A3-2.1 *RcpExecName* — The status variable containing the specifiers of the currently selected recipe(s). Refer to the SEMI E5 Standard for a full definition of this variable data item.

A3-3 Description

A3-3.1 *Process Program Description*

A3-3.1.1 Process programs allow the equipment's process, and/or the parameters used by that process, to be set and modified by the engineer to achieve different results. Different process programs may be required for different products, while often the same process program will be used for all lots of a given product. The engineer shall be able to create such programs, to modify current programs, and to delete programs from equipment storage.

A3-3.1.2 For the host to ensure that the proper process programs are in place at the equipment, there shall be a means of transferring them from equipment to host and from host to equipment. The host also may need to delete process programs from the equipment's storage to make room for a process program to be downloaded. In addition, the host shall be kept informed whenever a local change occurs in the contents or status of a process program.

A3-3.1.3 Both formatted and unformatted process programs may be uploaded and downloaded. This capability provides for both host- and equipment-initiated transfers. The equipment-initiated transfer may be used at the request of the process engineer or operator at the equipment.

A3-3.1.4 If a process program exists with the same PPID as the one given in the SECS-II message, the old process program shall be replaced. The PPID in the SECS-II message shall identify the process program in nonvolatile storage.

A3-3.2 *SEMI E42 Recipe Description*

A3-3.2.1 Specifications in the main standard apply to SEMI E42 recipes as well as process programs, with the following differences:

- A recipe contains a body corresponding to a process program. In addition, it contains attributes defined for execution recipes in SEMI E42, § 6. Recipe attributes are transferred whenever the recipe is downloaded or uploaded.
- The same SECS-II messages are used for all execution recipes, regardless of the internal structure of the recipe body.
- If an execution recipe already exists with the same identifier as the one given in the SECS-II message, the downloaded recipe shall be rejected (not stored) unless the host has specified a 'forced overwrite' in the data item RCPOWCODE.
- A recipe currently being edited shall be protected from inadvertent change or overwriting by a recipe with the same identifier that is downloaded during this time. If the downloaded recipe is accepted (stored), the equipment shall require the operator either to save the edited recipe to a new (unused) identifier or to discard it.
- For the equipment to initiate either an upload or download of a recipe, it shall request the host to initiate an upload or download procedure. In addition, it may be necessary to also specify the name of the repository (recipe namespace) at the host.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

A3-3.2.2 *Large Process Programs and SEMI E42 Recipes*

A3-3.2.2.1 Process programs and recipes for certain types of equipment, such as metrology or inspection equipment, contain images and are, therefore, very large. A process program or recipe cannot be sent as a single item in a SECS-II message if its size exceeds SECS-II data item size limits as defined in SEMI E5, or if the total message size would exceed the SECS message size limits defined in transport protocol Standards such as SEMI E4 and SEMI E37. Furthermore, process programs and recipes may also require some preparation prior to transfer.

A3-3.2.2.2 The commonly used Stream 7 and Stream 15 transfer functions require that the entire process program be sent in a single message. Thus, it is not possible to send a large process program or recipe with such messages. However, there is an alternative set of Stream 7 and Stream 15 functions that supports large transfers. These messages invoke Stream 13 Data Set Transfer Protocol messages to transfer large process programs (or recipes) by using a sequence of read messages. The completion of such a read transaction is indicated by 'ERROR: End of Data'. The Data Set Transfer Protocol does not set any limit on the size of the data set.

A3-4 Requirements

- The equipment manufacturer shall provide a method to create, modify, and delete Process Recipes. This method shall exist on either the equipment or on a separate computing system.
- If the equipment manufacturer has implemented process programs, a CEID shall be defined for a collection event for the creation, the deletion, or the modification (completion of an editing session) of a process program (Process Program Change Event).
- If the equipment manufacturer has implemented SEMI E42 recipes, there are two separate collection events with corresponding CEIDs, one for the creation of a new recipe (New Execution Recipe Event) and one when a recipe is changed or deleted (Execution Recipe Change Event). A New Execution Recipe Event shall occur whenever a new recipe identifier is created through download, edit, copy, or rename operations. An Execution Recipe Change Event shall occur whenever the body of an existing recipe is modified.
- If the equipment manufacturer has implemented process programs or SEMI E42 recipes, the name (identifier) that the engineer or operator uses to refer to the process program or recipe is the same as the identifier used by the host.
- Upon request from the host or operator, the equipment shall perform the following actions with regard to Process Recipes stored in nonvolatile storage: upload, download, delete, and list current equipment Process Recipe directory.
- The equipment shall be able to store in nonvolatile memory the number of Process Recipes sufficient to execute three unique process cycles. For example, if a wire-bonder requires both an 'ALIGN' process program and a 'BOND' process program for a full process cycle, then it shall provide nonvolatile storage for at least three pairs of process programs. These stored Process Recipes may not be modified in any way by the execution process, nor may the execution process be affected by the modification of any Process Recipes in storage, either by downloading or by local editing or by deletion, while that Process Recipe is being executed.
- The equipment shall provide verification and validation of all downloaded Process Recipes.
- The equipment shall support at least one of the defined alternatives for Process Recipe management:
  - Stream 7 facilitates Process Recipe management for formatted and unformatted process programs.
  - Stream 15 facilitates Process Recipe management for SEMI E42 recipes.
  - Stream 13 is also utilized for Process Recipe management for large process programs and large SEMI E42 recipes.
- The equipment supplier shall document any restrictions on the length or text format of PPID. The maximum length allowed by equipment may be less than that allowed by SECS-II.
- Where SEMI E42 recipes are supported, the following requirements also apply:
- The variable PPFormat shall be provided to indicate to the user the messages supported by the equipment.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

The recipe and its attributes shall comply to the requirements for execution recipes as defined in SEMI E42, § 6.

A3-4.1 Recipe Deletion by the Host:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

Recipe Delete Request                  S15F35-->

                                                 <--S15F36                Recipe Delete
                                                                          Acknowledge.
                                                                          The recipe is removed
                                                                          from nonvolatile
                                                                          storage.
```

A3-4.2 *Recipe Directory Request*

A3-4.2.1 Host Requests a List of Identifiers of Currently Stored Recipes:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

GetAttr Request                         S14F1-->

                                                 <--S14F2                 GetAttr Data
```

A3-4.3 *Recipe Upload*

A3-4.3.1 Host Requests a Recipe from the Equipment:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

Recipe Upload Request                  S15F31-->

                                                 <--S15F32                Recipe Upload Data
```

A3-4.3.2 Host Requests a Large Recipe from the Equipment:

```
        COMMENT                    HOST                  EQUIPMENT                COMMENT

Large Recipe Upload                    S15F51-->
Request

                                                 <--S15F52                Large Recipe Upload
                                                                          Acknowledge

                                                                          Acknowledge may be
                                                                          "accepted and will be
                                                                          performed later with
                                                                          completion signaled".
                                                                          When the equipment is
                                                                          ready to send:

                                                 <--S13F1                 Send Data Set Send

Send Data Set                           S13F2-->
Acknowledge


Open Data Set request                   S13F3-->

                                                 <--S13F4                 Open Data Set Data

[START] repeat until
ACKC13 is error:

Read Data Set Request                   S13F5-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
[END]                                   <--S13F6        Read Data Set Data

                                                        [IF]
                                                        CEID for upload event
                                                        enabled
                                                        [THEN]
                                                        [IF]
                                                        ACKC13 is "ERROR: END OF
                                                        DATA"
                                                        [Then]

                                        <--S6F11        Send Event Report
                                                        "successful upload"

Event Report Acknowledge        S6F12-->

                                                        [ELSE]

                                        <--S6F11        Send Event Report "bad
                                                        upload"

Event Report Acknowledge        S6F12-->                [END_IF]
                                                        [END_IF]

Close Data Set Send             S13F7-->

                                        <--S13F8        Close Data Set
                                                        Acknowledge
```

A3-4.3.3 Equipment Sends a Recipe to the Host:

```
COMMENT                 HOST            EQUIPMENT       COMMENT

                                                        RCPCMD = Upload
                                                        [IF] multi-block request

                                        <--S15F1        [THEN] Recipe Management
                                                        Multi-block inquire

Recipe Management Multi-        S15F2-->
block Grant

                                                        [END_IF]

                                        <--S15F21       Recipe Action Request

Recipe Action                   S15F22-->
Acknowledge

Host requests upload            S15F31-->

                                        <--S15F32       Recipe Upload Data
```

A3-4.3.4 Equipment Sends a Large Recipe to the Host:

```
COMMENT                 HOST            EQUIPMENT       COMMENT

                                                        RCPCMD = Upload
                                                        [IF] multi-block request

                                        <--S15F1        [THEN]
                                                        Recipe Management

Recipe Management               S15F2-->                Multi-block Inquire

Multi-block Grant                                       [END_IF]

                                        <--S15F21       Recipe Action Request

Recipe Action                   S15F22-->
Acknowledge
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                        S15F51-->

Large Recipe Upload
Request

                                                <--S15F52               Large Recipe Upload
                                                                        Acknowledge

                                                <--S13F1                Send Data Set Send

Send Data Set                           S13F2-->
Acknowledge

                                                <--S13F3                Open Data Set Request

Open Data Set Data                      S13F4-->

[START] repeat until
ACKC13 is error:

Read Data Set Request                   S13F5-->

[END]                                           <--S13F6                Read Data Set Data

                                                                        [IF]
                                                                        CEID for upload event
                                                                        enabled
                                                                        [THEN]
                                                                        [IF]
                                                                        ACKC13 is "ERROR: END OF
                                                                        DATA"
                                                                        [Then]

                                                <--S6F11                Send Event Report
                                                                        “successful upload”

Event Report Acknowledge                S6F12-->                        [ELSE]

                                                <--S6F11                Send Event Report “bad
                                                                        upload”

Event Report Acknowledge                S6F12-->                        [END_IF]
                                                                        [END_IF]

Close Data Set Send                     S13F7-->

                                                <--S13F8                Close Data Set
                                                                        Acknowledge
```

A3-4.4 *Recipe Download*

A3-4.4.1 The Recipe Management Multi-Block Inquire/Grant transaction (S15F1/F2) is required for the transfer of multi-block recipes. However, its use is recommended prior to all downloads, as it provides recipe size to the equipment. If the data item RCPOWCODE is TRUE in S15F27, then a pre-existing recipe with the same identifier shall be overwritten.

A3-4.4.2 Host Sends a Recipe to the Equipment:

```
        COMMENT                     HOST                    EQUIPMENT                COMMENT

[IF]
Recipe is multi-block
[THEN]

Recipe Management Multi-                S15F1-->
block Inquire

                                                <--S15F2                Recipe Management Multi-
                                                                        block
                                                                        Grant

[END IF]

Recipe Download Request                 S15F27-->
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                        [IF] RCPOWCODE = TRUE
                                                        delete any pre-existing
                                                        recipe
                                                        with the same identifier
                                                        before storing new
                                                        recipe
                                                        [END_IF]
                                  <--S15F28             Recipe Download
                                                        Acknowledge
```

A3-4.4.3 Host Sends a Large Recipe to the Equipment:

```
COMMENT                    HOST                EQUIPMENT              COMMENT
Large Recipe Download            S15F49-->
Request
                                                        [IF] RCPOWCODE = TRUE
                                                        delete any pre-existing
                                                        recipe
                                                        with the same identifier
                                                        before storing new
                                                        recipe
                                                        [END_IF]
                                  <--S15F50             Large Recipe Download
                                                        Acknowledge
                                  <--S13F3              Open Data Set Request
Open Data Set Data                S13F4-->
                                                        [START] repeat until
                                                        ACKC13 is error:
                                  <--S13F5              Read Data Set Request
Read Data Set Data                S13F6-->
                                                        [END]
                                  <--S13F7              Close Data Set Send
Close Data Set                    S13F8-->
Acknowledge
                                                        [IF] multi-block request
                                  <--S15F1              [THEN]
Recipe Management                 S15F2-->              Recipe Management
Multi-block Grant                                       Multi-block Inquire
                                                        [END_IF]
                                  <--S15F53
                                                        Recipe Verification Send
Recipe Verification              S15F54-->
Acknowledge
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

A3-4.4.4 Equipment Requests a Recipe from the Host:

```
COMMENT                          HOST                    EQUIPMENT                COMMENT

                                                                                  RCPCMD = Download
                                                                                  [IF] multi-block request

                                                <--S15F1                          [THEN] Recipe Management
                                                                                  Multi-block inquire

Recipe Management Multi-                 S15F2-->
block

Grant                                                                             [END_IF]

                                                <--S15F21                         Recipe Action Request

Recipe Action                           S15F22-->
Acknowledge

[IF]
Recipe is multi-block
[THEN]

Recipe Management Multi-                 S15F1-->
block

Inquire

                                                <--S15F2                          Recipe Management Multi-
                                                                                  block

                                                                                  Grant

[END IF]

Recipe Download Request                 S15F27-->

                                                                                  [IF] RCPOWCODE = TRUE
                                                                                  delete any pre-existing
                                                                                  recipe
                                                                                  with the same identifier
                                                                                  before storing new
                                                                                  recipe
                                                                                  [END_IF]

                                                <--S15F28                         Recipe Download
                                                                                  Acknowledge
```

A3-4.4.5 Equipment Requests a Large Recipe from the Host:

```
COMMENT                          HOST                    EQUIPMENT                COMMENT

                                                                                  RCPCMD = Download
                                                                                  [IF] multi-block request

                                                <--S15F1                          [THEN]

Recipe Management                        S15F2-->                                 Recipe Management

Multi-block Grant                                                                 Multi-block Inquire
                                                                                  [END_IF]

                                                <--S15F21                         Recipe Action Request

Recipe Action                           S15F22-->
Acknowledge

Large Recipe Download                   S15F49-->

Requests                                                                          [IF] RCPOWCODE = TRUE
                                                                                  delete any pre-existing
                                                                                  recipe
                                                                                  with the same identifier
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

```
                                                                    before storing new
                                                                    recipe
                                                                    [END_IF]
                                        <--S15F50                   Large Recipe Download
                                                                    Acknowledge
                                        <--S13F3                    Open Data Set Request
Open Data Set Data                  S13F4
                                                                    [START] repeat until
                                                                    ACKC13 is error:
                                        <--S13F5                    Read Data Set Request
Read Data Set Data                  S13F6-->
                                                                    [END]
                                        <--S13F7                    Close Data Set Send
Close Data Set                      S13F8-->
Acknowledge
                                                                    [IF] multi-block request
                                                                    [THEN]
                                        <--S15F1                    Recipe Management
Recipe Management                   S15F2-->                        Multi-block Inquire
Multi-block Grant                                                   [END_IF]
                                        <--S15F53                   Recipe Verification Send
Recipe Verification                 S15F54-->
Acknowledge
```

A3-4.5 *Recipe Verification*

A3-4.5.1 Host Requests Equipment to Verify a Recipe that it Has Stored:

```
        COMMENT                 HOST                EQUIPMENT               COMMENT
Recipe Verify Request               S15F29-->
                                                                    Equipment verifies
                                                                    recipe
                                        <--S15F30                   Recipe Verify Data
```

A3-5 SECS-II Message Subset for Large Unformatted Process Programs

STREAM 15: Recipe Management – Required only if equipment implements SEMI E42 recipes

S15F1 Recipe Management Multi-Block Inquire	S,H<->E
S15F2 Recipe Management Multi-Block Grant	S,H<->E
S15F21 Recipe Action Request	M,H<->E
S15F22 Recipe Action Acknowledge	M,H->E
S15F27 Recipe Download Request	M,H->E
S15F28 Recipe Download Acknowledge	M,H<-E
S15F29 Recipe Verify Request	M,H->E
S15F30 Recipe Verify Data	M,H<-E
S15F31 Recipe Upload Request	S,H->E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

S15F32 Recipe Upload Data	M,H<-E
S15F35 Recipe Delete Request	M,H->E
S15F36 Recipe Delete Acknowledge	M,H<-E
S15F49 Large Recipe Download Request	S,H->E
S15F50 Large Recipe Download Acknowledge	S,H<-E
S15F51 Large Recipe Upload Request	S,H->E
S15F52 Large Recipe Upload Acknowledge	S,H<-E
S15F53 Recipe Verification Send	M,H<-E
S15F54 Recipe Verification Acknowledge	S,H->E

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

APPENDIX 4
ADDITIONAL ITEMS

NOTICE: The material in this Appendix is an official part of SEMI E30 and was approved by full letter ballot procedures on December 21, 2022.

A4-1 PPFormat

A4-1.1 The definition of PPFormat remains in this Appendix for backward compatibility with implementations based on earlier versions of this Specification. When additional means of transferring process programs, the design of the PPFormat status variable became unsustainable when multiple combinations of transfer are supported.

A4-1.2 *PPFormat* — A status variable (SV) indicating the type or types of process programs and recipes that are supported.

1 = Unformatted process programs

2 = Formatted process programs

3 = Both unformatted and formatted process programs

4 = Execution recipes

5 = Large unformatted process programs

6 = Large formatted process programs

7 = Both large unformatted and large formatted process programs

8 = Large execution recipes and a combination of these formats. Refer to SEMI E5 for a complete list.

A4-1.3 The variable PPFormat shall be provided to indicate to the user the messages supported by the equipment.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

APPENDIX 5
WELL-KNOWN NAMES FOR RETIRED VARIABLES, COLLECTION EVENTS AND ALARMS

NOTICE: The material in this Appendix is an official part of SEMI E30 and was approved by full letter ballot procedures on December 14, 2023.

A5-1 Background

A5-1.1 This Appendix describes well-known names for items previously defined in this Standard that are not defined in the current revision. They are provided for backwards compatibility.

A5-1.2 For completeness, well-known tables for variables, collection events and alarms included yet deliberately left empty when no applicable variable, collection event or alarm is defined in this Specification.

A5-1.3 The Last Revision column captures the last revision of this Standard where the item was defined.

A5-2 Well-Known Names for Retired Variables

A5-2.1 This section describes well-known names for variables previously defined in the Standard.

A5-2.2 Refer to SEMI E5 for more information on the variable classes.

Table A5-1 Well-Known Names for Retired Variables

Variable	*Variable Class*	*Well-Known Name*	*Last Revision*
PPFormat	EC	E30:PPFormat	0520
RcpExecName	SV	E30:RcpExecName	0520

A5-3 Well-Known Names for Retired Collection Events

A5-3.1 This section describes well-known names for collection events previously defined in the Standard.

Table A5-2 Well-Known Names for Retired Collection Events

Collection Event	*Well-Known Name*	*Last Revision*

A5-4 Well-Known Names for Retired Alarms

A5-4.1 This section describes well-known names for alarms previously defined in the Standard.

Table A5-3 Well-Known Names for Retired Alarms

Alarm	*Well-Known Name*	*Last Revision*

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

RELATED INFORMATION 1

NOTICE: This Related Information is not an official part of SEMI E30 and was derived from the work of the Information & Control Global Technical Committee. This Related Information was approved for publication by full letter ballot procedures on May 16, 2017.

R1-1 Factory Operational Script

An Operational Script is a series of capabilities arranged in a typical factory operation sequence. The intent of having an Operational Script is to help put the SECS-II message Scenarios into a context. Although this context will vary, it represents a typical operational sequence found in most semiconductor device manufacturers' applications.

- System Initialization
- Synchronization
- Machine Setup
- Production Setup
- Processing
- Post-Processing
- Shutdown

The following script is not intended to be complete, but to serve as an example to be further developed on an implementation basis.

R1-1.1 *Anytime Capabilities* — All capabilities can generally occur at any time during the operational script sequence.

R1-1.2 *System Initialization and Synchronization* — Upon system initialization, the default setting for communication (enabled or disabled) becomes effective, as well as any equipment constants or other information retained in non-volatile storage. The initial communication status is displayed at the equipment.

Assuming the communication state is enabled, the equipment will attempt to establish communication with a host computer. Refer to § 7.2 for a description of the scenario for establishing communications.

Upon receiving an indication that the equipment was previously not communicating, the host would typically perform synchronization activities including setting the equipment's clock and requesting selected status information. Note that synchronization activity is host application-dependent and may be implemented using various scenarios.

R1-1.3 *Production Set-Up* — The host typically has the following information:

- what material
- what process step
- what process program to use (PPID)
- current equipment status, VIDs, SVIDs
- data collection requirements (trace data and event data)
- VIDs needed
- Equipment constants (ECIDs)

Based upon the above information, the host will perform setup activities as required. It should be verified that the correct process program is available and selected at the equipment.

R1-1.3.1 *Auxiliary Material and Manual Set-Up* — Auxiliary material can be checked and verified at this point. If status variables exist for auxiliary material, they may be requested by the host.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Any other manual, nonprocess, and/or nonproduct specific set-up also may take place at this point. The operator may interact with the equipment and the host. If the operator interacts with the equipment, the equipment communications link with the host should stay operational.

The operator and the host may exchange information via equipment terminal services.

R1-1.3.2 *Product/Process Set-Up* — Specific product and/or process information is communicated to the equipment prior to processing material.

R1-1.3.3 *Material Load* — The host may instruct an operator or a material handling system to deliver material to the equipment.

Once the material has arrived at the equipment, the equipment or the operator will notify the host.

R1-1.3.4 *Production Data Collection Set-Up* — The host instructs the equipment to collect event-based data. Reports are defined and linked to events. Event reports can be enabled or disabled.

The host instructs the equipment to collect data from the equipment based on time intervals.

The host configures the equipment to monitor specific variables and to send event reports when variables transition between monitoring zones.

R1-1.4 *Processing*

R1-1.4.1 *Start Process Executing* — The host or operator issues a command to start.

R1-1.4.2 *Equipment Signals End of Run* — When process execution is completed, the equipment generates events. If any of the events are enabled, they will be sent as event reports.

R1-1.5 *Post-Processing* — The equipment has completed processing material. It now makes the material available to the operator or material handling system for removal. The equipment signals the host that it is available for more work.

R1-1.5.1 *Material Unload* — Material is unloaded from the equipment by an operator or material handling system.

R1-2 Equipment Front Panel

In the GEM Standard, several requirements are stated that involve the display or input of information at the equipment front panel. The ‘equipment front panel’ refers to an area on the equipment that is available to the operator under normal use (i.e., without removing maintenance access panels). This may include a CRT display, keyboard, switches, and lights.

This application note provides some guidance for implementation of the GEM front panel capabilities. All of these requirements map directly to state models and capabilities defined in §§ 6 and 7. All capabilities may be implemented in either hardware (buttons, switches, lights) or in a software/CRT equivalent.

R1-2.1 *Displays and Indicators* — The intent of various displays is to inform the operator of either the current state of the equipment or of a recent change of state (or both). Therefore, it is most useful if these displays are continuously visible and easily recognized at a distance. Required displays/indicators include:

Communications State — This means that three distinct states should be represented: DISABLED, ENABLED/NOT COMMUNICATING, and ENABLED/COMMUNICATING.

Terminal Services — An ‘New Host Message’ indicator should be supplied.

R1-2.2 *Switches/Buttons* — Note that discrete switches also contain information for the user. However, these tend to represent the desired states of the operator/user. The equipment’s response to a change of a switch may not be instantaneous. Still, the current position of switches should be available to the operator.

It may be appropriate to limit the access to some switches and buttons. This might be done via any of the standard methods, keys, passwords, combinations, etc. This is especially true for system default switches that would not often be changed. Required switches/buttons include:

Communications State System Default — In what communications state should the equipment be when system initialization is complete? The choices are DISABLED and ENABLED.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Communications State Selector — This is a toggle or button that will initiate a transition from ENABLED to DISABLED or vice versa.

Message Recognition Button — This button is used to initiate an event message to the host which indicates that the 'New Host Message' has been read. This button should function only when the New Host Message Indicator is activated and when the received message is displayed in the terminal display.

R1-3 Examples of Equipment Alarms

Table R1-1 provides alarm examples pertaining to various configurational aspects of equipment.

NOTE 15: It is important to stress that these are just examples intended to illustrate that alarms pertain to situations in which there exists a potential for exceeding physical safety limits associated with people, equipment, and material being processed as per the GEM definition of an alarm.

NOTE 16: The alarm capability is intended as an addition to standard safety alarms (e.g., lights, horns). There is no intent to replace direct operator reaction to such problems. Nor is there the expectation that the host can necessarily prevent or directly address such alarms.

An actual machine shall have an associated set of alarms defined by the manufacturer that pertains to its specific configuration and design. The equipment manufacturer is responsible for supplying documentation associated with these alarm definitions.

Table R1-1 Alarm Examples per Equipment Configuration

Subsystem	*Alarm Description*	*ALID*	*Trigger*	*Reset*	*Operator*	*Equipment*	*Material*
Mainframe Power Supply	Overvoltage		Voltage supply over maximum limit			X	
	Undervoltage		Voltage supply under minimum limit			X	
Internal Power Distribution Bus	AC Low		AC under minimum limit			X	X
Cooling System	Overtemp		Temperature over maximum		X		X
	Pressure Low		Pressure below minimum				X

- *Subsystem* — The subsystem of the equipment to which the alarm is related.
- *Alarm Description* — Description of the alarm.
- *ALID* — The Alarm ID as specified by SECS-II.
- *Trigger* — Text description of what caused the alarm.
- *Reset* — Description of how to resolve the alarm condition.
- *Affected* — Who or what is affected by the alarm trigger: Operator, Equipment, and Material.

R1-4 Trace Data Collection Example

This example shows an implementation of the Trace Data Collection capability defined in § 7.3.3.

S2F23 sent by host:

```
TRID = ABCD
DSPER = 000100 (One minute per period)
TOTSMP = 9
REPGSZ = 3
SVID1 = Temperature
SVID2 = Relative humidity
```

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

S6F1 looks like this (starting at time 1 a.m.):

1st transmission <L,4>

```
1. ABCD (trace ID)
2. 3 (last sample of the transmission)
3.    88   5   01   01   03   00
    Year Month Day  Hour Min  Sec
4. <L, n> n = 2 SVIDs x REPGSZ of
   3 = 2 x 3 = 6
   72 (temperature)
   0.29 (relative humidity)
   73 (temp.)
   0.30 (r.h.)
   71 (temp.)
   0.30 (r.h.)
```

2nd transmission <L,4>

```
1. ABCD
2. 6
3. 88  05  01  01  06  00
                hr  min
4. <L,6>
      73
      0.31
      71
      0.32
      71
      0.31
```

3rd and last transmission <L,4>

```
1. ABCD
2. 9
3. 88  05  01  01  09  00
                hr  min
4. <L,6>
   71
   0.30
   72
   0.30
   71
   0.31
```

R1-5 Harel Notation

Harel's statecharts extend traditional state-transition diagrams with several additional concepts, most important of which are hierarchy and concurrence. Statecharts depict the behavior of a system by showing states it may take, events that prompt a change of state, and the composition of states. What follows is a very brief description of the symbols defined for use and how these are useful to describe a system. Refer to Figure R1-1 for the basic notational symbols.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

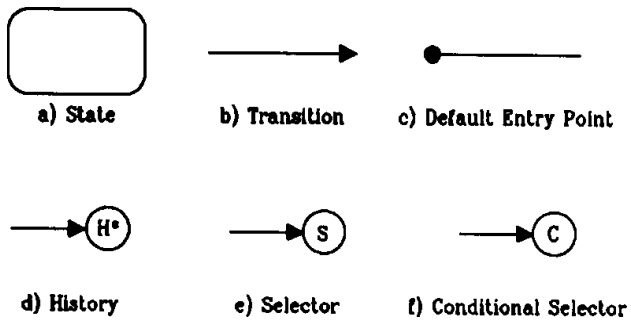

Figure R1-1
Harel Statechart Symbols

States are represented by rounded boxes. A state transition is shown graphically with a line from the old state terminating with the arrow symbol at the new state. Transitions are unidirectional-while the reverse transition may be possible, it is considered a different transition with different conditions for initiation and different resultant actions.

States may be subdivided into substates to facilitate more concise definition of behavior. Thus, a hierarchy is defined whereby any state may be a substate of some parent state and in turn be the parent of its own substates. Substates should be one of two types, termed AND substates and OR substates.

A parent may be divided into two or more OR substates of which one and only one is the active substate at any time. The accepted term for this exclusivity is XOR. Figure R1-2 gives an example of a simple case of OR substates. In this example, some system (perhaps a motor) has a state named FUNCTIONAL. When the motor is FUNCTIONAL, it may be either ON or OFF, but never both.

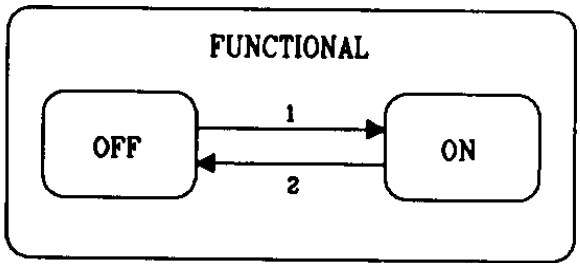

Figure R1-2
Example of OR Substates

Another way of dividing a parent state corresponds roughly to subsystems. These AND substates represent parallelism, such that every AND substate of an active parent state is considered active. Harel also uses the term 'Orthogonal Component' to refer to AND substates. However, these parallel substates tend to be highly interactive and interdependent. For this reason, the word orthogonal is considered confusing and has been excluded from use in this Standard. Figure R1-3 shows an example of AND substates representing (in part) an automobile. Note the convention of attaching the name of the parent state AUTOMOBILE to the outside of the state in a small box. The substates shown are independent components and may have their own substates (of either the AND or OR type):

- LIGHTS may be ON or OFF;
- DOOR may be OPEN or CLOSED;
- ENGINE is constructed of components such as pumps, pistons, carburetor, etc.

Exiting one of a set of AND substates requires the exit of all others. In some cases, a transition arrow will be shown from only one of the substates with the others implied.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

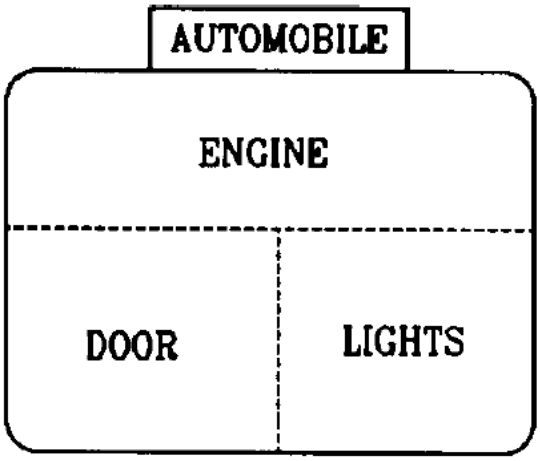

Figure R1-3
Example of AND Substates

A simplification that also helps to prevent in determinacy is implemented with the symbol for default entry point. This symbol will indicate which OR substate is initially active when there is not an explicit choice. This lack of specification is indicated by a transition arrow from one state to another that does not cross the boundary of the parent to point specifically to a substate.

An entrance to a state terminating in a history symbol (refer to Figure R1-1) indicates that the OR substate to be entered should be that which was active the last time the parent state was active (i.e., last time the car was running, the radio was on). The history symbol H refers to the choice of substates of the parent. The symbol H* extends further to the lowest level substates defined. In the absence of memory of a 'last time', the default entry is used.

The selector and conditional selector symbols serve to abbreviate complex entrances to states. Their meaning is similar and indicate that the choice of OR substate upon entry of a parent state depends on some condition that is not shown. The selector is usually used to combine several similar transition events, while the conditional selector will typically require some computation or test of conditions external to the stimulus for state transition. Please examine the referenced article for more detail.

NOTE 17: Within the body of this Standard, the term statechart is not used in favor of the more traditional term state model diagram.

R1-5.1 *State Definitions* — The state model diagram provides a concise description of the function of a system. However, a full definition requires detail that cannot be included on the diagram. A description of each state is required that covers the boundaries of the state and any responses that occur within that state to the environment. The convention in this Standard is to provide state names in ALL CAPS to help the reader identify where these are used. A sample state description of the ON state depicted in the Figure R1-2 might be:

ON

The switch is in the on position. Power is available to the motor. Speed of the motor will change in proportion to the speed knob adjustment.

R1-5.2 *Transition Table* — The last piece of the state model is the transition table. It consists of several columns that list the transition number from the diagram, the starting and ending state for the transition, and three columns titled trigger, action, and comment. The trigger column describes the combination of events and conditions that initiates the transition (e.g., message SxFy received). The trigger should be related to a single clearly defined event at the equipment. The action column identifies the activities associated directly with the transition. These activities may be of three types: (a) actions taken upon exit of the old state, (b) actions taken upon entry to the new state, and (c) actions not associated with either state. These are not differentiated in this Standard. The final column allows for additional comments that help to clarify the transition. Table R1-2, an example of transition table, illustrates the motor example in Figure R1-2.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

Table R1-2 Transition Table for Motor Example

#	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	OFF	Switch turned to on position.	ON	Power supplied to motor.	Power supply assumed available. Motor begins to turn.
2	ON	Switch turned to off position.	OFF	Power supply to motor disconnected.	Motor begins deceleration.

R1-6 Example Control Model Application

This section provides one example of a host's interaction with an equipment's control model. A host system should have a view of the control model to understand and predict equipment behavior. However, the implementor may simplify the host's view by assuming that some configuration settings are fixed and that the host-initiated features are not implemented. Applying these assumptions simplifies the behavior the host expects to see.

Figure R1-4 shows the effective control model (refer to § 6.5 for details of the control model) based on the following host assumptions:

- The fundamental requirements are met, but the additional host-initiated control capability is not implemented.
- The configuration for the default entry to CONTROL is set to an OFF-LINE substate (either ATTEMPT ON-LINE or EQUIPMENT OFF-LINE).
- The destination state for transition 4 (failure of S1F1 transaction) is configured to EQUIPMENT OFF-LINE.

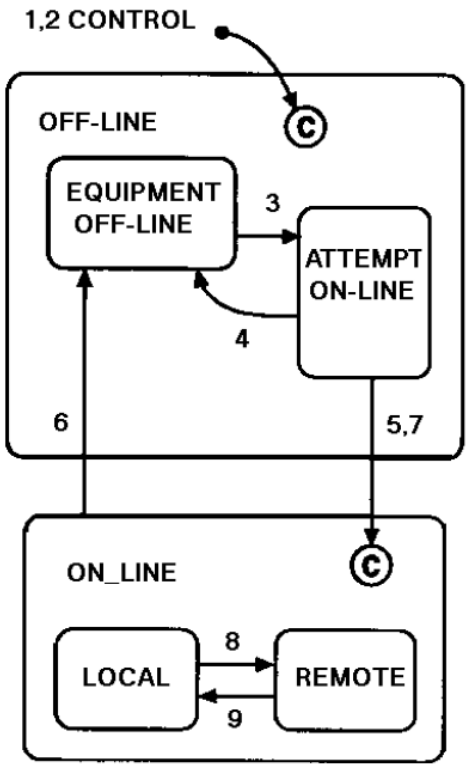

Figure R1-4
Example of the Simplified 'Effective' Control Model

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

This view of the model has two further settings that the host recognizes as changeable at the equipment. The first is the configuration of which substate of OFF-LINE to be activated upon system initialization. The second is the front panel switch that determines whether the active system substate is LOCAL or REMOTE when ON-LINE.

This application has the following implications:

- This application requires that the equipment begin with the OFF-LINE state active. Thus, an equipment initiated S1F1/F2 transaction should be completed before the equipment will begin sending all messages to the host.
- If a failed attempt to go ON-LINE is made by the equipment, it will not allow the host to complete the transition at a later time. An operator will be required to re-initiate the transition to ON-LINE when the host becomes ready.
- Once ON-LINE, the equipment will remain ON-LINE until an operator sets the equipment OFF-LINE at the equipment front panel.
- Since all transitions into the HOST OFF-LINE state are eliminated, this state is effectively eliminated from the host view of the control model.

This application retains the following features:

- The ON-LINE state is achieved only after the host acknowledges the equipment by replying to the S1F1 with and S1F2. This confirms to the operator attempting to put the equipment ON-LINE that the host application is ready for work to begin.
- It provides the operator the means to set the equipment OFF-LINE for non-host-related activities (e.g., maintenance, test lots). Which activities are 'non-host-related' varies from factory to factory. In general, fewer activities are 'non-host-related' as a factory's automation level increases.
- The operator has the ability to operate the equipment with either the REMOTE or LOCAL state active. As the equipment transitions to ON-LINE, the preferred substate is automatically chosen (based on a front panel switch).
- The user may configure which substate of OFF-LINE the equipment will initially activate at system initialization. If ATTEMPT ON-LINE is chosen, the equipment will automatically attempt the transition to the ON-LINE state as system initialization.

R1-7 Examples of Limits Monitoring

R1-7.1 *Introduction*

R1-7.1.1 Four limits monitoring examples are included below to help clarify the use of limits and to illustrate typical applications. The first example shows how to apply limits to Boolean values. The second illustrates application of several limits to a floating point variable in a classical control zone style. The third example shows an integer counter variable used to prompt for equipment maintenance.

R1-7.2 *Examples*

R1-7.2.1 *Example 1 — Valve Monitoring*

R1-7.2.1.1 The ACME Shine-Um-Rite Model 13 includes a sump which contains the chemical agent used to clean bare wafers. A chemical feeder system serves to refill the sump when the level drops below a certain level. The fill is accomplished via an on-off value driven by sensors in the sump. Facilities should be informed of the proportion of the time the valve is open (approximates usage) and any time the value remains open for more than 5 minutes (valve likely broken).

R1-7.2.1.2 To implement this requirement, a limit was defined for the Boolean status variable which contains the current state of the valve (i.e., 0 = Closed, 1 = Open). Refer to Figure R1-5 for illustration. LIMITID1 was defined with UPPERDB = LIMITMAX = 1 (Open) and LOWERDB = LIMITMIN = 0 (Closed). As a result, any time the valve opens, a collection event is generated with TransitionType = 0 and when the valve closes, a collection event is generated with TransitionType = 1. An event report containing the DV LimitVariable was attached to each collection event and reporting for the event was enabled.

NOTE 18: Boolean values are defined as 0 = False/Closed/Off and any value >0 = True/Open/On–never depend on a value of 1.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

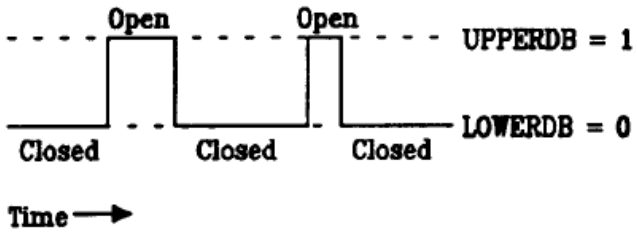

Figure R1-5
Valve Monitoring Example

R1-7.2.2 *Example 2 — Environment Monitoring*

R1-7.2.2.1 ACME also makes a Model 2 Stepper. The environmental control system of this equipment is designed to hold the internal temperature relatively constant, but is sensitive to large changes in the external environment, opening of access doors, etc. To ensure that processing conditions are appropriate, the internal stepper temperature is monitored to ensure it remains in a safe operating zone (within 'Shutdown' limits). In addition, a second set of limits are used within the Shutdown limits to bind the 'Normal' operating range. Frequent excursions from the normal range into the 'warning' range will prompt service on the environmental control system. The target temperature range is specified as 98° to 100°, the shutdown limits as 95° to 103°.

R1-7.2.2.2 Event reports are desired when the internal temperature moves outside of the normal operating zone into a warning zone (above or below), when the temperature moves back into the normal operating zone from the warning zones, and when the temperature moves out of the warning zones into the shutdown zones. Furthermore, temperature fluctuations of 0.5° should not trigger multiple event reports.

R1-7.2.2.3 Probably the most intuitive use of the limits monitoring capability is in establishing normal, warning, and shutdown zones for a particular equipment variable. Limits may be combined to provide such a scenario. The method is described below and illustrated in Figure R1-6. Please note that in the figure, limits are denoted as solid lines for simplicity, with deadbands indicated using the '±' notation.

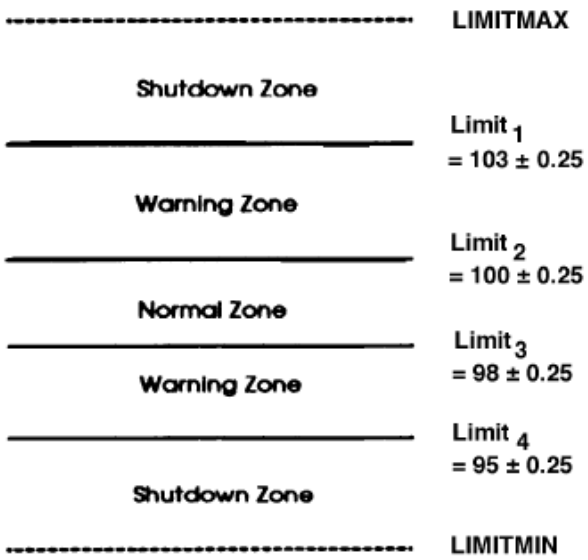

Figure R1-6
Environment Monitoring Example

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

R1-7.2.3 *Example 3 — Calibration Counter*

R1-7.2.3.1 Another ACME equipment is the multi-chamber Duz-It-All Model 7. This machine includes redundant chambers to increase throughput. One particular chamber on this equipment requires periodic calibration. The need for calibration is a nonlinear function of the number of wafers processed in that chamber. A status variable exists which contains the number of wafers processed since the last calibration was performed. Maintenance is definitely required after every 8 cycles, but the machine should be checked after 5 and 7 cycles to determine whether early calibration is necessary. This checking may be done by examining certain other equipment status variables.

R1-7.2.3.2 To meet this need, three limits are defined for the counter variable. Three limits were set, at 5, 7, and 8. Deadbands are set to zero, since chattering is not a problem. All the pertinent information is placed in an event report which is attached to the CEID for the limits of the counter to negate the need for further message exchange. Event reports are generated as each limit is reached (one zone transition each), and when the counter is reset following calibration (one, two, or three zone transitions referenced in one report). Figure R1-7 illustrates this example. Note that, disabling the report upon counter reset (downward transitions) is not possible.

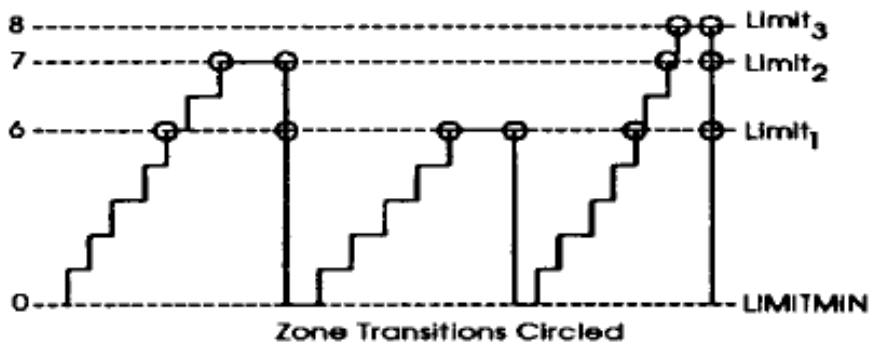

Figure R1-7
Calibration Counter Example

R1-7.2.4 *Example 4 — Derived Variables*

R1-7.2.4.1 The flagship of the ACME line is the new HotDog Furnace. This is a vertical furnace which exposes wafers to a variety of temperatures during processing. The temperature profile during the run is critical to the process and is typically contains a number of plateau's at different levels during the run. The owner wishes to monitor the temperature and be alerted whenever the actual temperature profile differs from the ideal by >0.5°. The derived variable was created to provide a steady target range during the run, no matter what the desired temperature range happened to be. Deviation from 'ideal temperature profile' was chosen as the new variable to be monitored. The equipment already had access to the profile for the run, which described the desired temperature at a given time into the process. The manufacturer added a calculation each time the actual temperature was sampled, subtracting the ideal temperature from the actual. They provided as status variables the actual temperature, the ideal temperature, and the new 'deviation from profile' variable. One limit was activated and set to 0.5° and a second set to −0.5° (each with a deadband width of 0.05). Thus, when the temperature deviation from setpoint exceeds ±0.5, an event is generated containing the current desired temperature and the actual temperature. For good measure, additional data was added, providing time since start of run to document the precise point in the process that the problem occurred.

R1-7.2.4.2 In order to achieve the desired behavior, the host defines four monitoring limits. Two of the limits establish the target zone. These are responsible for reporting transitions from normal to warning zones in either direction. The other two limits establish the transitions between the warning zones and the error zones. The difference between UPPERDB and LOWERDB for each limit is 0.5. This may also be expressed as limit ±0.25. Combining limits does not change the way the equipment treats limits monitoring, but rather builds a method of interpreting limits from the host's point of view.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

R1-8 Process Parameter Modification for Process and Equipment Control

R1-8.1 *Introduction*

R1-8.1.1 In many equipment control applications there is a need for a GEM host to modify one or a small set of process parameters associated with a process program. The number of parameters modified, frequency of modification (e.g., wafer-to-wafer, batch-to-batch, etc.), range of modification, etc., is largely a function of the equipment control application. Utilizing GEM, at least two methods are envisioned for modifying process parameters on a tool. With the first method, 'Equipment Constants' can be used to relate process parameters of the updated process program. Equipment Constants can also be used in a mode where they relate suggested modifications to process parameters from the stored process program; that is, the constants contain only the ± differential from a nominal value. The former mode is preferred because it better ensures data integrity between the controller and tool. With the second method the entire process program could be downloaded, but this results in an enormous amount of communication overhead. Note that, in all cases, the Equipment Constants do not replace the process parameters inside a process program, but are associated with (e.g., linked to) these parameters to relate modifications. The remainder of this application note provides a description of how process parameter modification can be implemented using existing GEM capabilities. The method may be used in a GEM compliant system provided that the specific GEM capabilities described are supported.

R1-8.2 *Equipment Constants*

R1-8.2.1 Incremental process parameter modification for process and equipment control can be supported over a GEM interface by using the *Equipment Constants* GEM capability (refer to § 7.6). With this capability, each process parameter (or process parameter at a step) that can be modified (e.g., for purposes of process control) is associated with an equipment constant. Using the Equipment Constant GEM scenarios (refer to § 7.6.5) the host can (1), send process parameters or parameter modifications, (2), request current values of modifiable process program parameters, (3), retrieve name lists of equipment constants associated with modifiable parameters, and (4), be informed by the equipment when an operator changes one of the modifiable process parameters.

R1-8.2.2 The equipment constants should represent the actual values of the process parameters with which they are associated. Depending on the equipment operation and control application, the equipment constant could represent the actual value of a process parameter at a process program step, or over the entire process program. The equipment constants could also be utilized to represent the differentials of process parameters from nominal values. However it is important to note that, when using differential values to relate process parameter modifications, any loss of synchronization between equipment and host could result in an incorrect assessment of the value of a process setpoint by the host. Note also that, upon system startup, and whenever the appropriate process parameters are modified, the equipment constants should also be modified as necessary to always reflect the (absolute or relative) values of the associated process parameters.

R1-8.2.3 In order to maintain synchronization between equipment and host, it is recommended that equipment constants associated with process program parameters be applied to only override the currently selected and active process program. A selected process program is considered to be active whenever the equipment is in the 'PROCESSING' state and the process program is the currently selected process program. If multiple process programs are utilized during one process event; for example, cluster tool scenario, it is recommended that separate equipment constants be utilized for each process program/process parameter pair.

R1-8.2.4 Note that, timing and traceability issues associated with utilizing the equipment constants capability (for process control) are application specific and beyond the scope of this application note. Equipment that provides for process program parameter overrides though setting of Equipment Constants should also provide additional Equipment Constants for the set that includes the name of the associated process program and a Boolean variable to enable and disable the override feature. In addition, the supplier should document for each parameter: (1), the associated Equipment Constant, and (2), any restrictions on the state (active or not) or the process program in which the parameter may be modified. Also, since override of the process setpoints may be provided by a Host controller application element, it is recommended that the equipment provide an event report whenever the associated process program has been modified.

R1-8.3 *Example*

R1-8.3.1 In the example of Figure R1-8, a chemical-mechanical planarizer (CMP) single wafer 'polishing' system includes a GEM compliant planarizer (equipment), a thickness metrology unit and a (host) controller. The tool polishes

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-17
Single user license only. Copying and Networking prohibited.

a wafer to a target thickness. The post-process thickness is measured by the metrology unit and reported to the host controller. The controller utilizes a feedback control algorithm to determine the appropriate polish 'time' process program parameter for the next wafer. This time should be reported to the CMP equipment utilizing the equipment's GEM interface so that the information can be utilized for the next wafer processing event.

R1-8.3.2 The mechanism described in this application note could be utilized to implement process control as follows. A settable equipment constant is associated or 'linked' with the 'time' parameter on the equipment. Equipment system documentation indicates the linkage and the conditions under which the linkage is valid. When the controller determines an appropriate 'time' parameter value for the next wafer to be polished, it sets the equipment constant to this value (refer to § 7.6). The equipment is configured to accept this equipment constant change and, if the equipment is in (or possibly when the equipment reaches) the appropriate state, the process program parameter is modified.

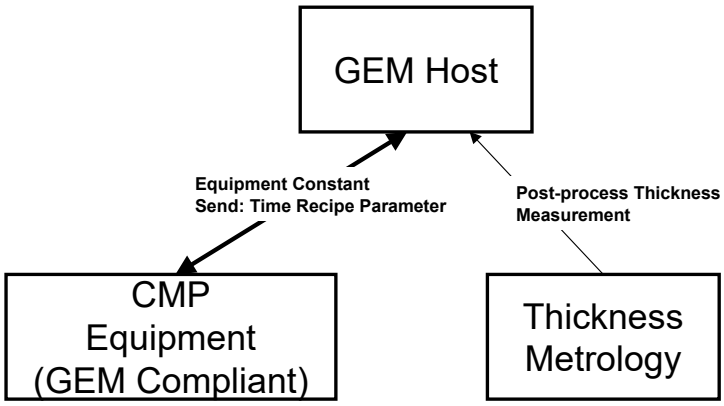

Figure R1-8
CMP Single Wafer 'Polishing' System with Host Process Program Parameter Modification Capability

NOTICE: SEMI makes no warranties or representations as to the suitability of the Standards and Safety Guidelines set forth herein for any particular application. The determination of the suitability of the Standard or Safety Guideline is solely the responsibility of the user. Users are cautioned to refer to manufacturer's instructions, product labels, product data sheets, and other relevant literature, respecting any materials or equipment mentioned herein. Standards and Safety Guidelines are subject to change without notice.

By publication of this Standard or Safety Guideline, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any items mentioned in this Standard or Safety Guideline. Users of this Standard or Safety Guideline are expressly advised that determination of any such patent rights or copyrights and the risk of infringement of such rights are entirely their own responsibility.

Copyright by SEMI®, 673 S. Milpitas Blvd., Milpitas, CA 95035. Reproduction of the contents in whole or in part is forbidden without express written consent of SEMI.